汽车电子与控制
Automotive Electronics and Control Systems (AECS)

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第三章：传感器、执行器及基本接口电路
Sensors, actuators and basic interface circuits

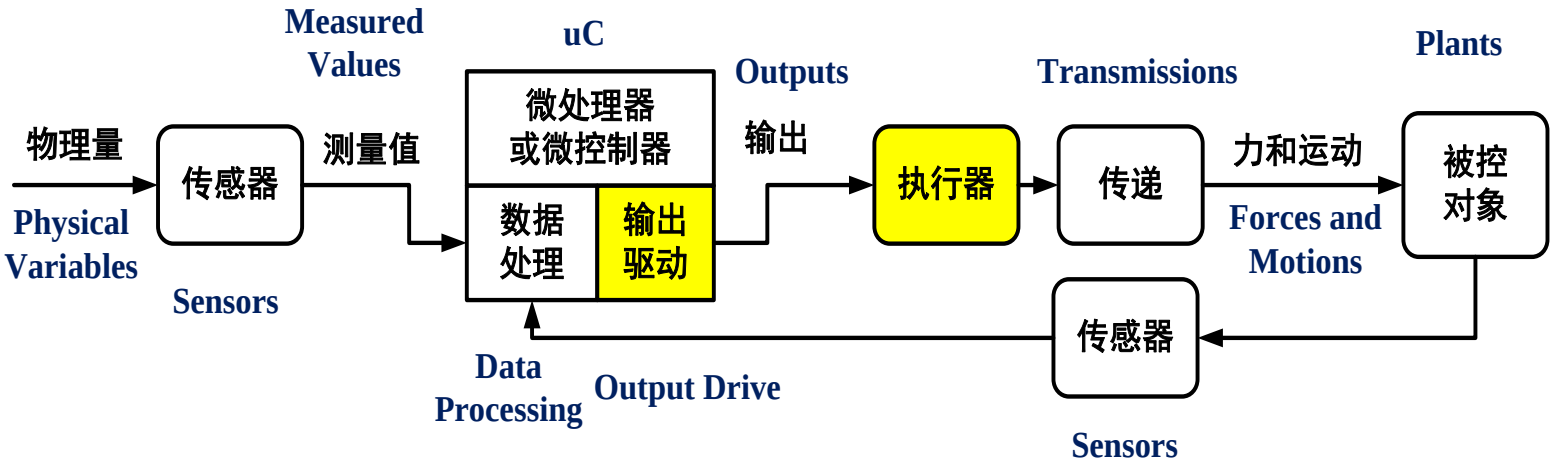
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§1 执行器及四种功率开关器件
Actuators and four power switches



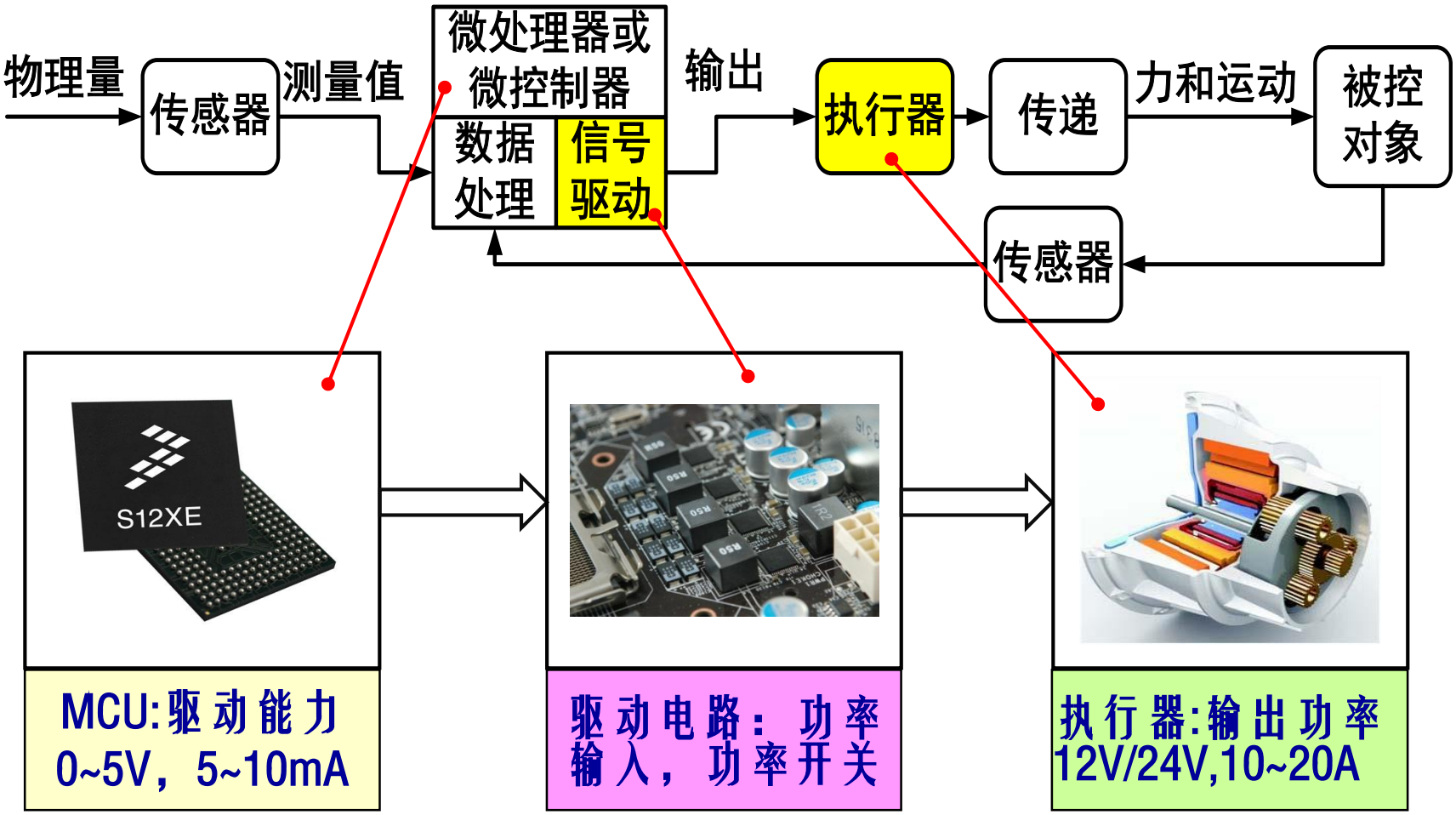
§1 执行器及四种功率开关器件
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§1 执行器及四种功率开关器件
Actuators and four power switches

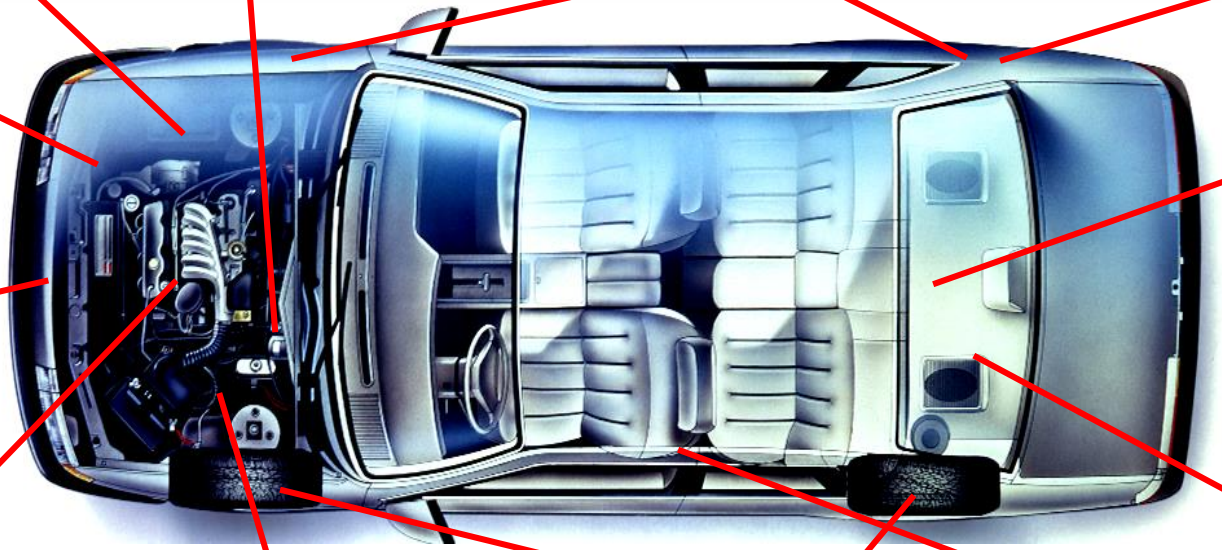
- 如何驱动一个执行器? /How to drive an actuator ?

属于控制系统中的前向通道（Feed forward channel)



§1 执行器及四种功率开关器件
Actuators and four power switches

●流体(液压和气动执行器)/Fluidic (Hydraulic and Pneumatic) Actuators



Secondary power pack

Power Steering

Shock absorbers, active spring suspension

Back wheel steering

Ventilation Heating

Friction type differential

cooling

Level regulation (automatic)

Motor, Hydraulic adjustable camshaft, pneumatic valve control

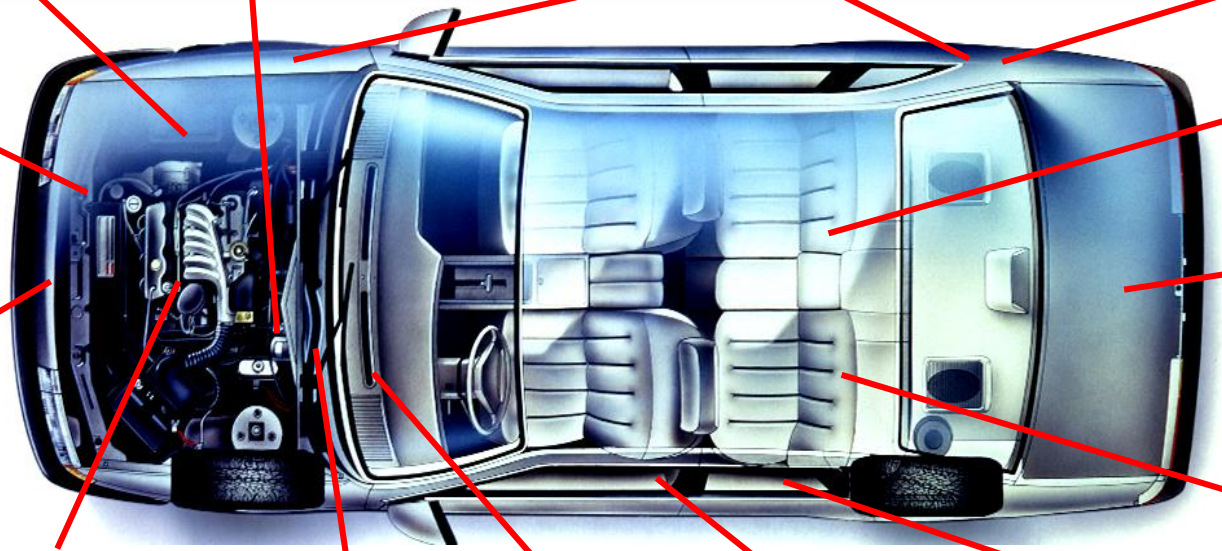
Gear box, automatic

Brake, ABS, ASR,…

Central locking system Comfort functions

§1 执行器及四种功率开关器件
Actuators and four power switches

●电执行器/ Electrical Actuators



- Air Conditioner
- Steering by wire Actuator
- Electronic Brake Actuators
- Motors ZV
- Cooler Fan
- Sunroof Drive
- Light Control Systems
- Water Pump
- Seat Motors
- Fuel Injection Valves, System, Fuel Pump, Electrical Valves, Starter,…
- Windshield Wipers
- Roll Bar Drive
- Mirror Drives
- Power window Motors

§1 执行器及四种功率开关器件
Actuators and four power switches

(1) 电磁阀：汽油机喷油器；柴油机喷油器（共轨系统）喷射控制用的高速电磁阀；ABS中的控制压力的电磁阀→感性负载/ Variant solenoid: Injector of gasoline, Injector of common rail–High speed solenoid valve, pressure control valve of ABS → Inductive loads

(2) 电机：直流电机，步进电机，交流电机；直线电机（线性电磁铁） →感性负载/ Variant motors： DC motor, step motor, AC motor, linear motor → Inductive loads

(3) 继电器：启动机，风扇，车灯… →感性负载 / Variant relay: starter, fans, lamps → Inductive loads

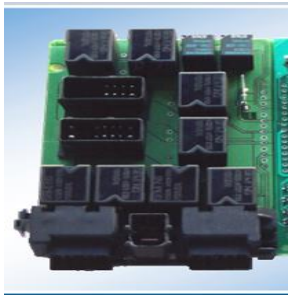
(4) 状态指示灯：档位指示，报警指示… →电阻负载 / Variant lamps: shift position, warning lamps.

绝大部分电气执行器的等效电路为感性负载（L/R）



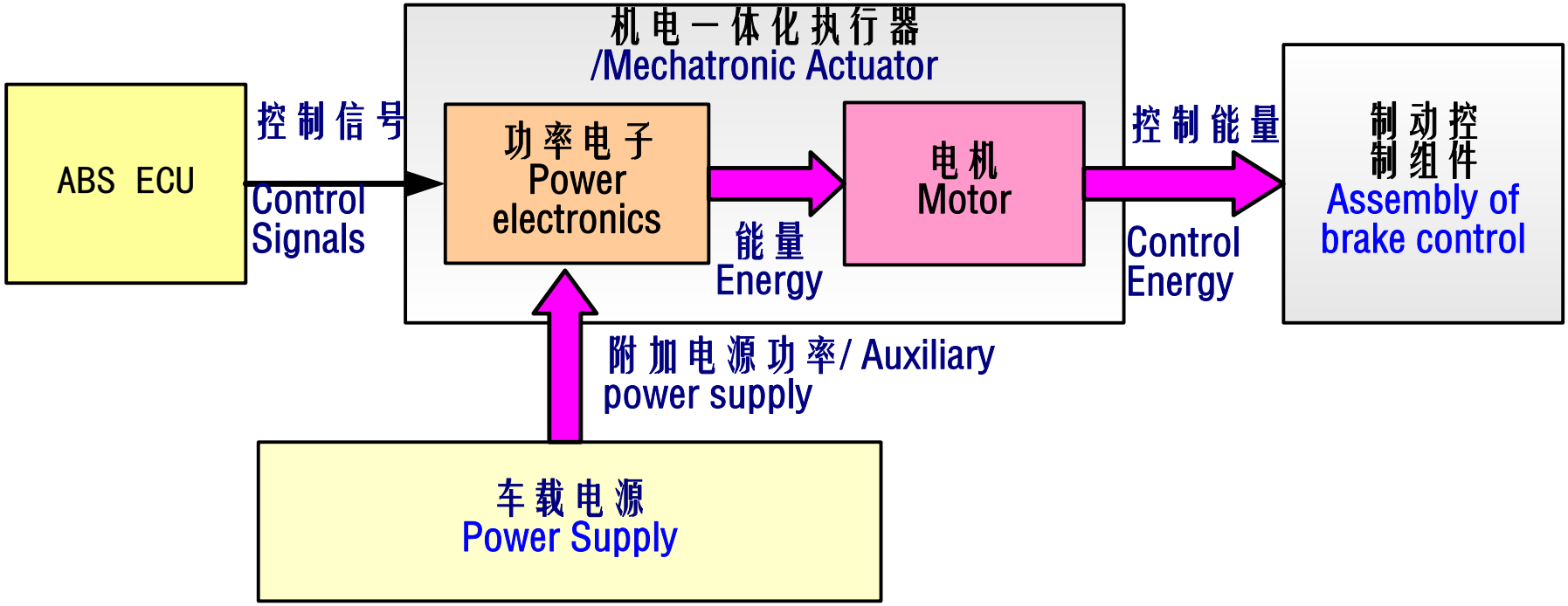






§1 执行器及四种功率开关器件
Actuators and four power switches

●执行器的驱动举例：ABS系统



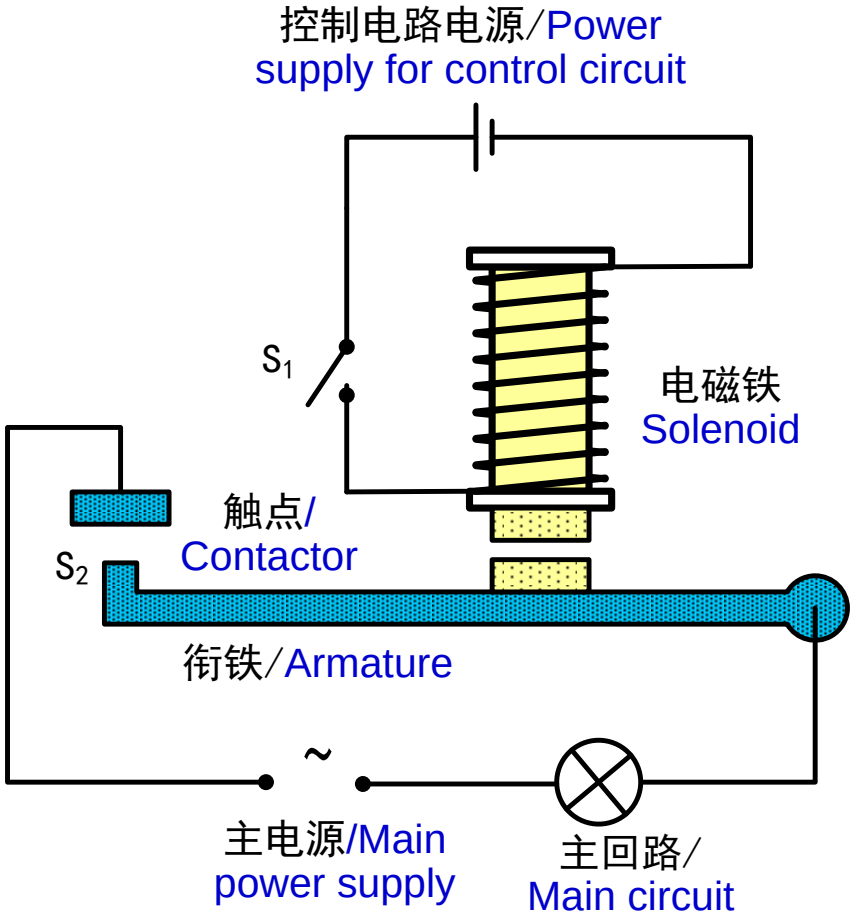
信号流-功率流-物质流

功率开关是驱动电执行器的基本器件

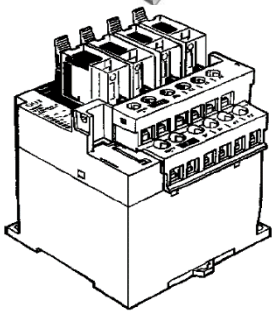
§1 执行器及四种功率开关器件
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● 功率开关1：继电器/Power switch 1: Relays

- ●开关继电器/Control relays
- ●辅助接触器/Auxiliary contactor
- ●电机接触器/Motor contactor
- ●特点：/Features
  - ➢双向；/Bidirectional
  - ➢电势独立；/Separate potential;
  - ➢频率慢；/Slow frequency
  - ➢电流大；/Large current
  - ➢寿命有限；/Limited lifecycle



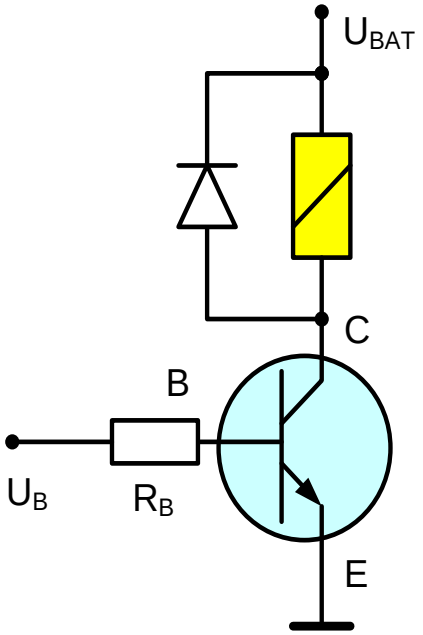




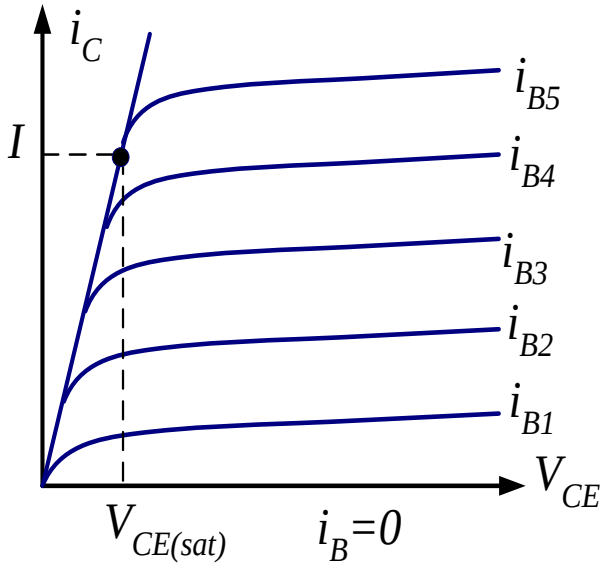


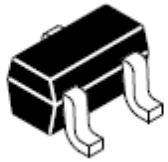


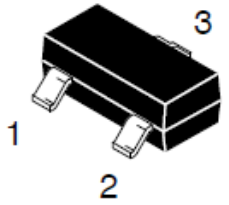
● 功率开关2：晶体管/Power switch 2: Transistor

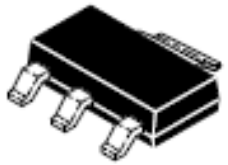


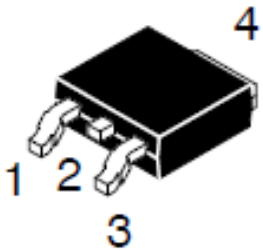
N型半导体、P型半导体、PN结

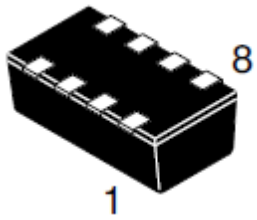






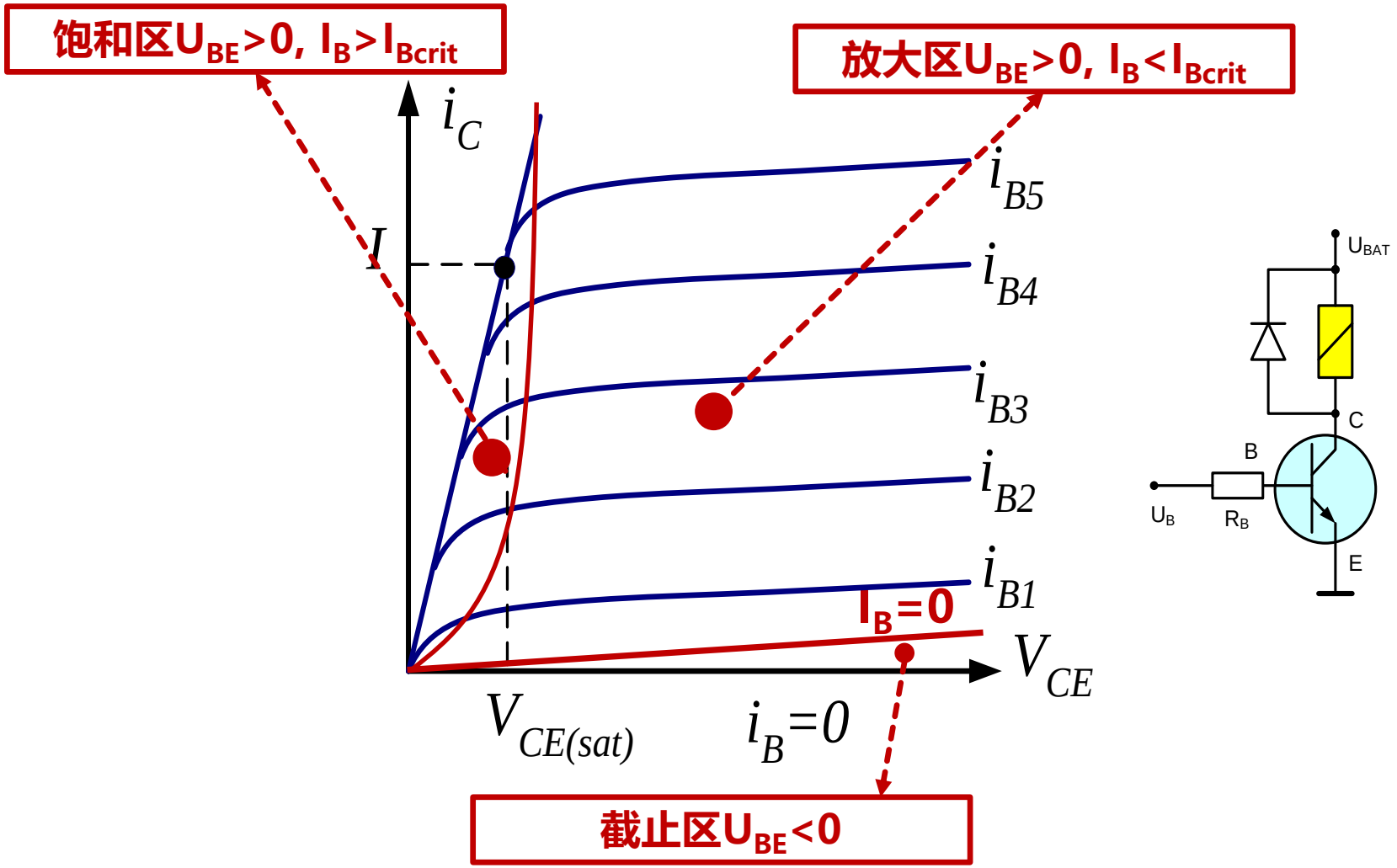




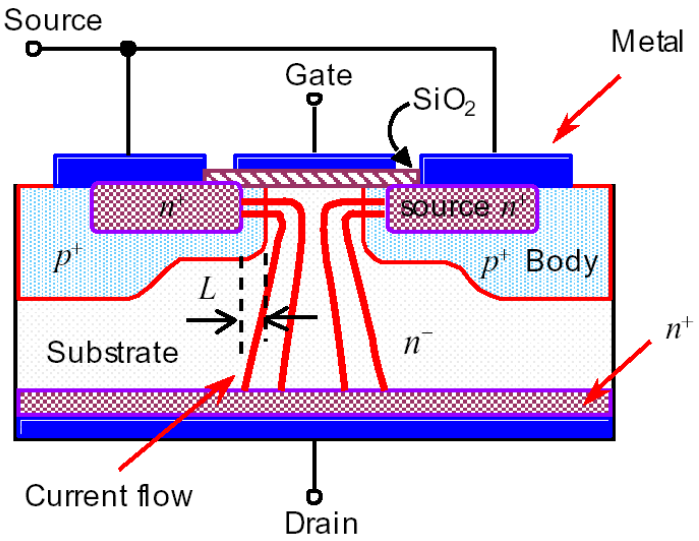


- 双极型晶体管、三极管、晶体管
- NPN/PNP，基极B、发射极E、集电极C
- 电流控制型：I_C电流正比于I_B的电流；通过I_B控制来控制I_C（负载）电流；导通压降小（0.6~0.7V）；单向工作

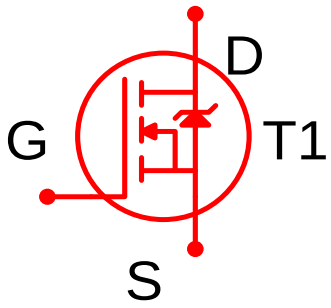
- 功率开关2：晶体管/Power switch 2: Transistor

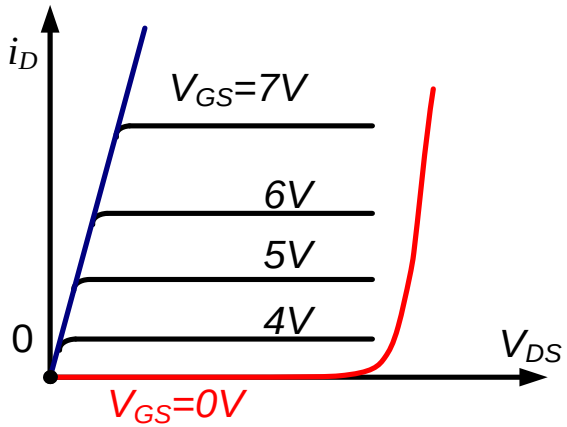


功率开关3：金属氧化物场效应晶体管 (MOSFET)



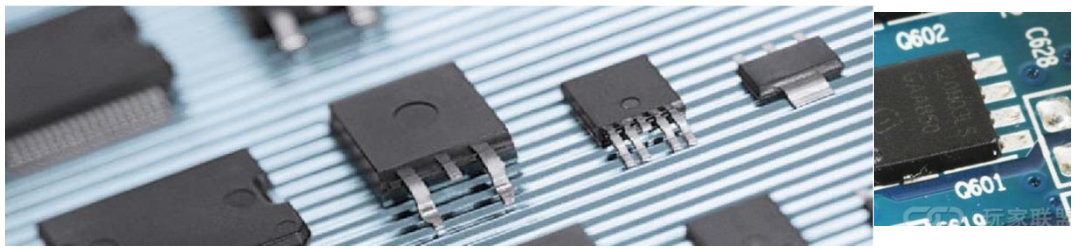
N沟道增强型场效应管



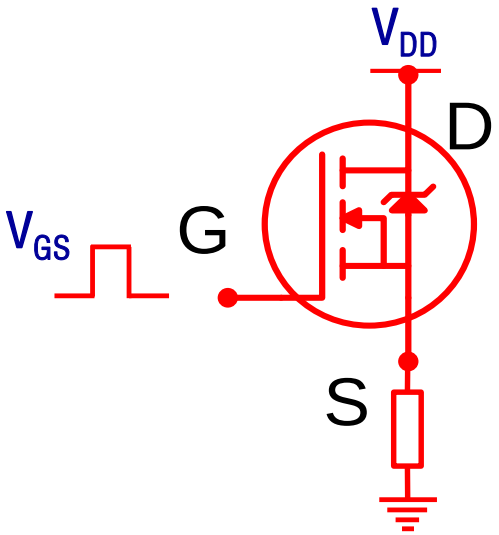


源极S、漏极D、栅极G

电压控制型；输入电阻高、不需要信号源提供电流；导通内阻低，开关速度快；内阻具有正温度系数，可以多个并联，应用最广。

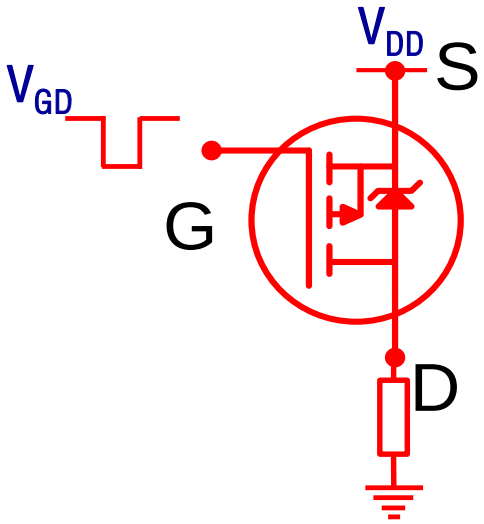


功率开关3：金属氧化物场效应晶体管 (MOSFET)



N沟道(耗尽型/增强型，制作工艺不同，工作参数不同)绝缘栅场效晶体管

- V_{GS}>0（如12V），则导通
- 电流由D流向S

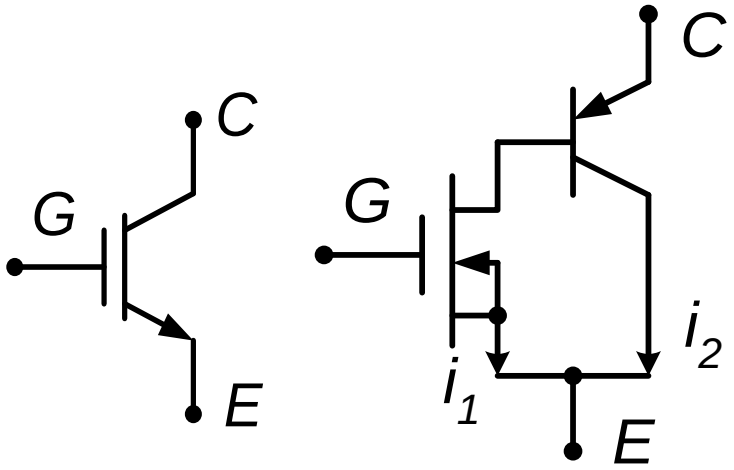


P沟道(耗尽型/增强型，制作工艺不同，工作参数不同)绝缘栅场效晶体管

- V_{GS}<0 （如-12V），则导通
- 电流由S流向D

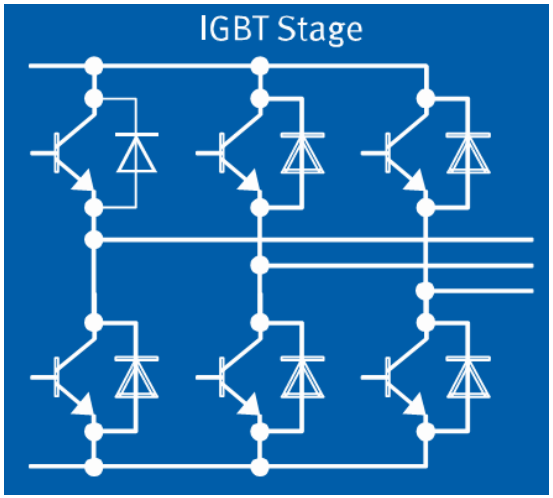
功率开关4：绝缘栅型晶体管(IGBT)

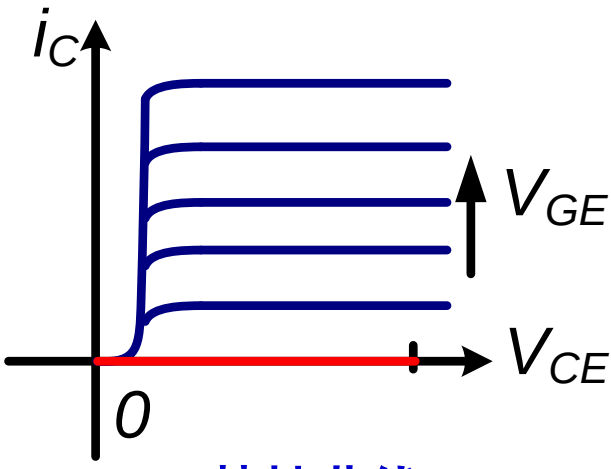
- 电压控制型；
- 综合了三极管和MOS管的优点；
- 大功率控制领域应用广泛；
- 电动汽车（电机控制器、逆变器）
- 集电极C、发射极E、栅极G



(a) 符号　　(b) 等效电路







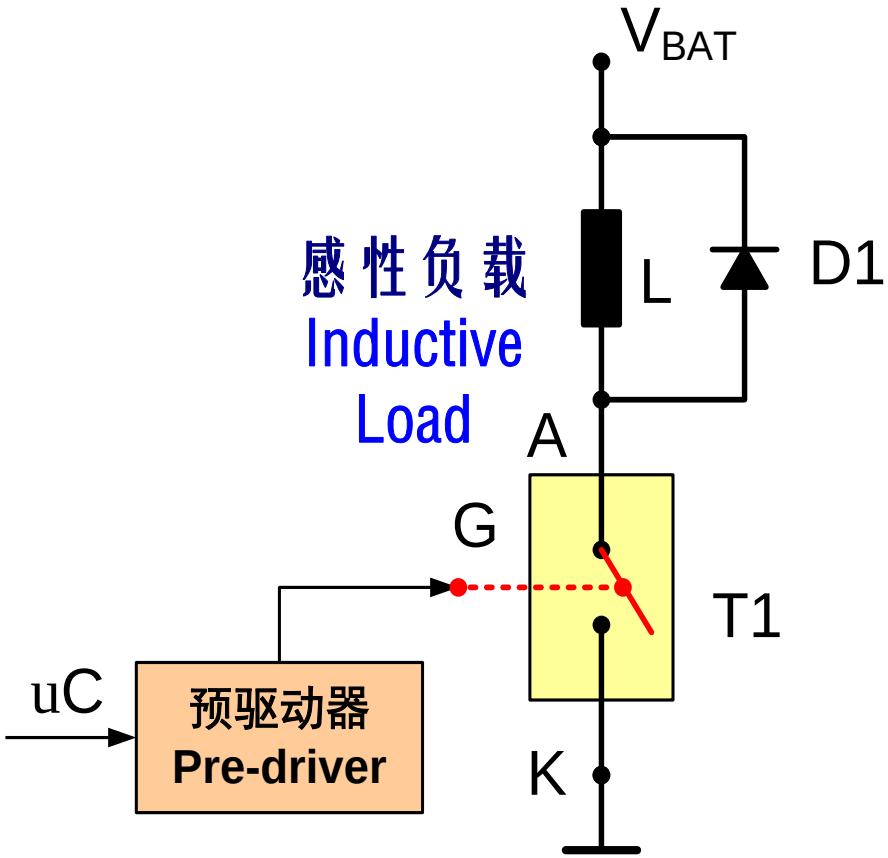
(c) 特性曲线

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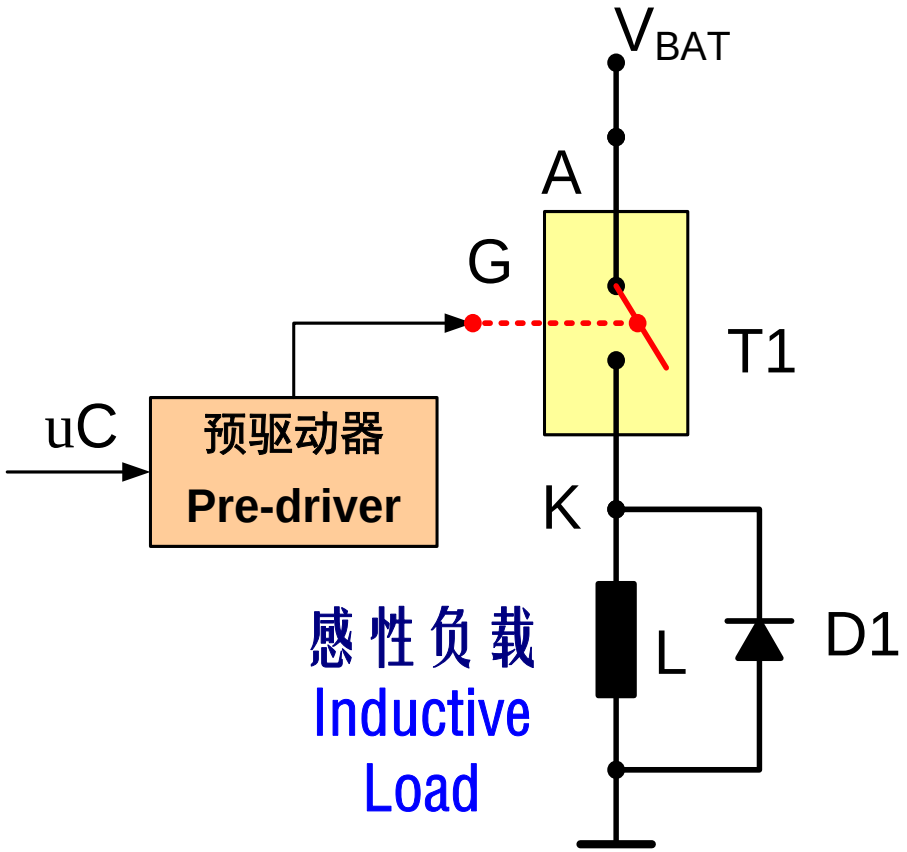
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

(1)低边驱动典型电路/Typical low side drive circuit



继电器作为功率开关的低边驱动
Low side drive circuit: relay as power switch

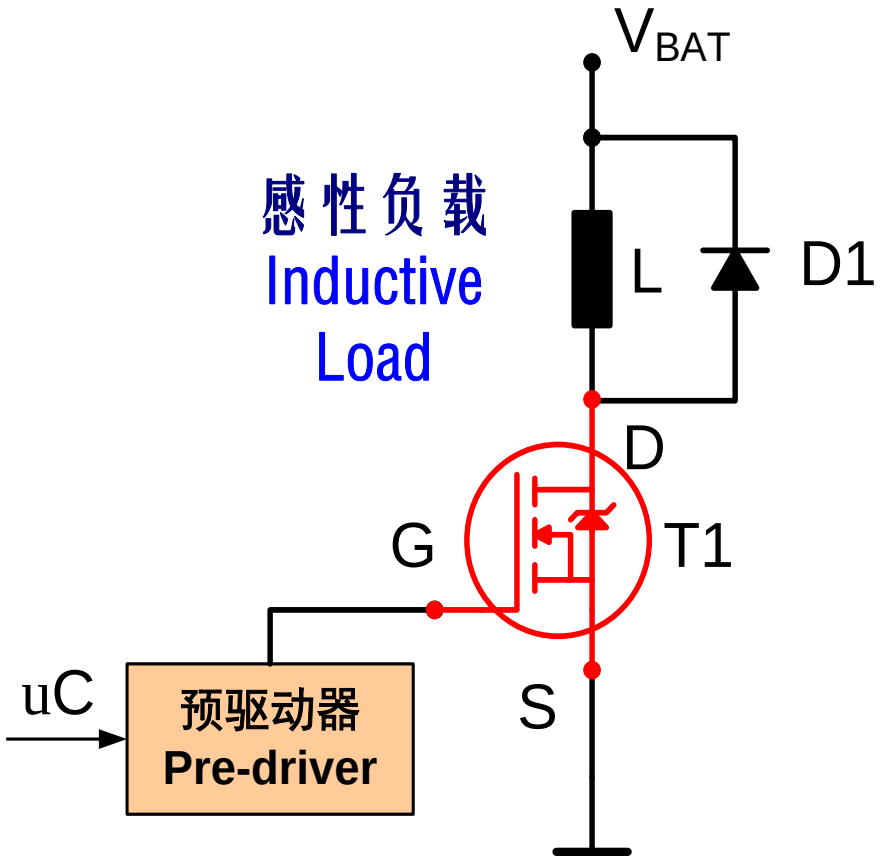
(2)高边驱动典型电路/Typical high side drive circuit



继电器作为功率开关的高边驱动
High side drive circuit: relay as power switch

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

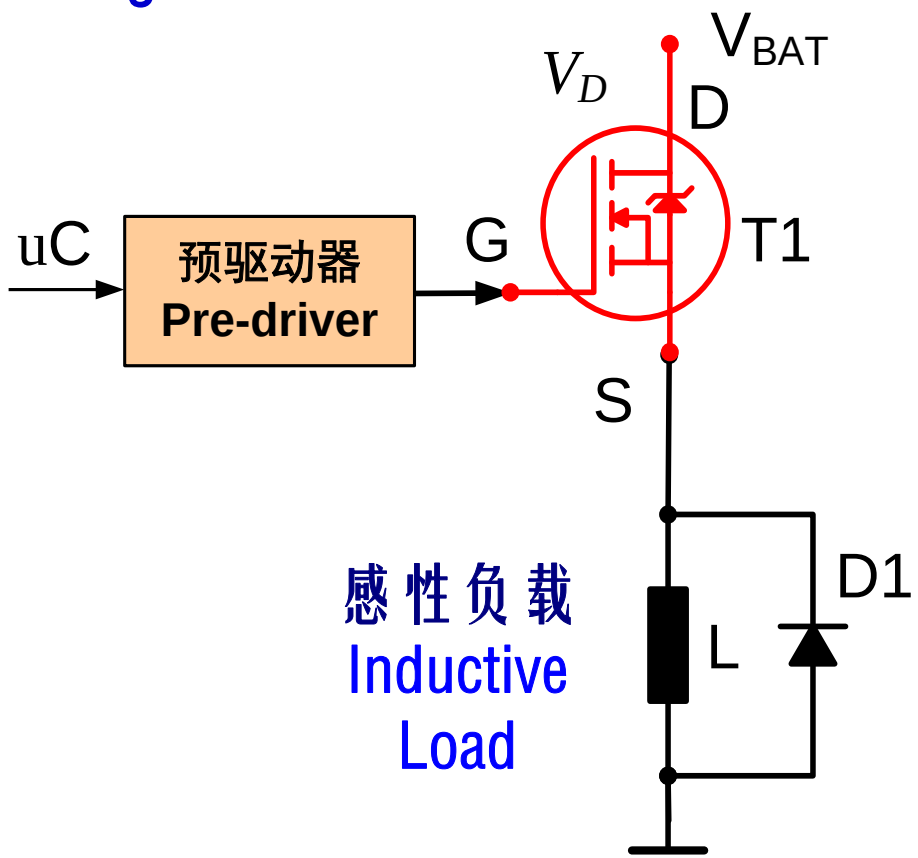
(1)低边驱动典型电路/Typical low side drive circuit



MOSFET作为功率开关的低边驱动

Low side drive circuit: MOSFET as power switch

(2)高边驱动典型电路/Typical high side drive circuit



MOSFET作为功率开关的高边驱动

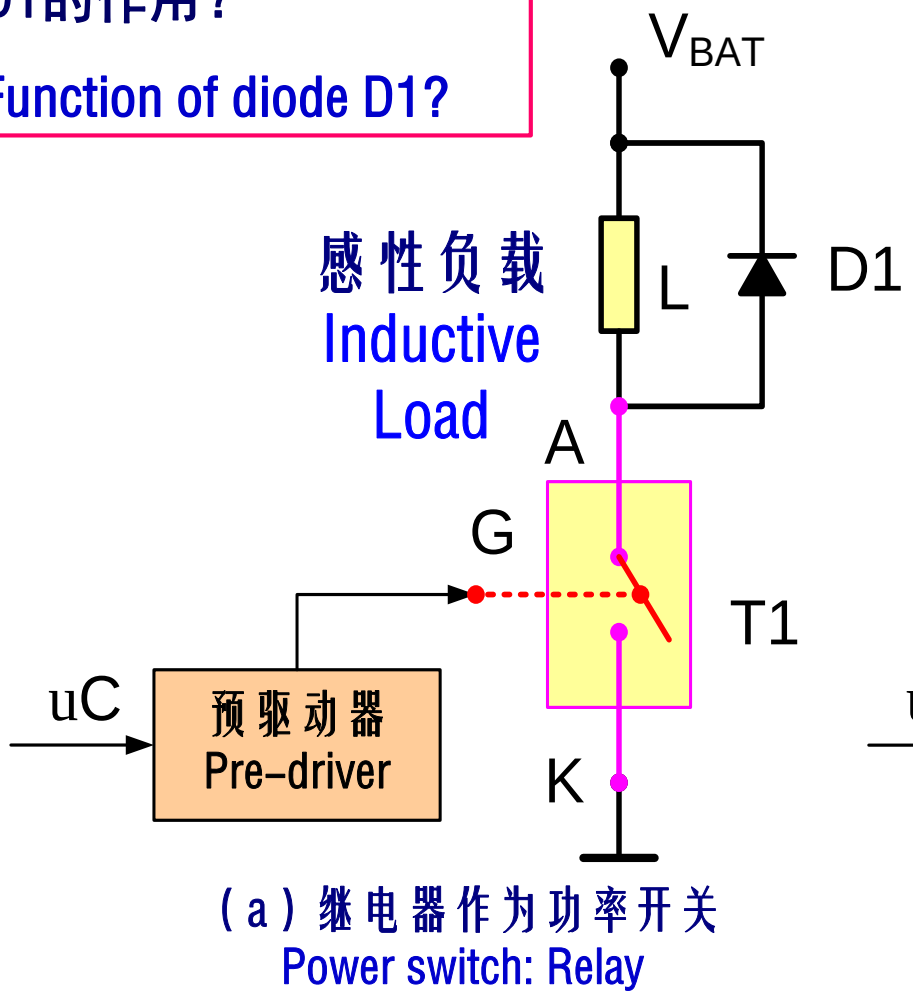
High side drive circuit: MOSFET as power switch

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

(1)低边驱动典型电路/ Typical Low Side Driver Circuit

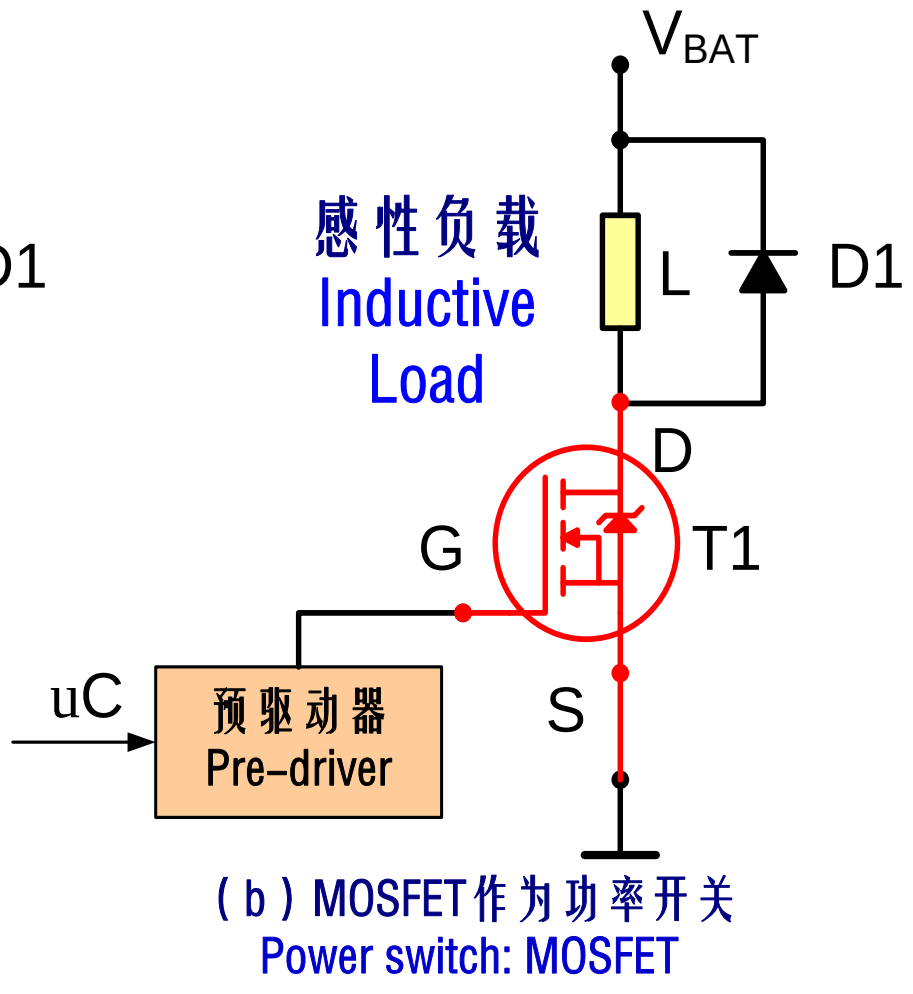
D1的作用？

Function of diode D1?



（a）继电器作为功率开关

Power switch: Relay



（b）MOSFET作为功率开关

Power switch: MOSFET

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

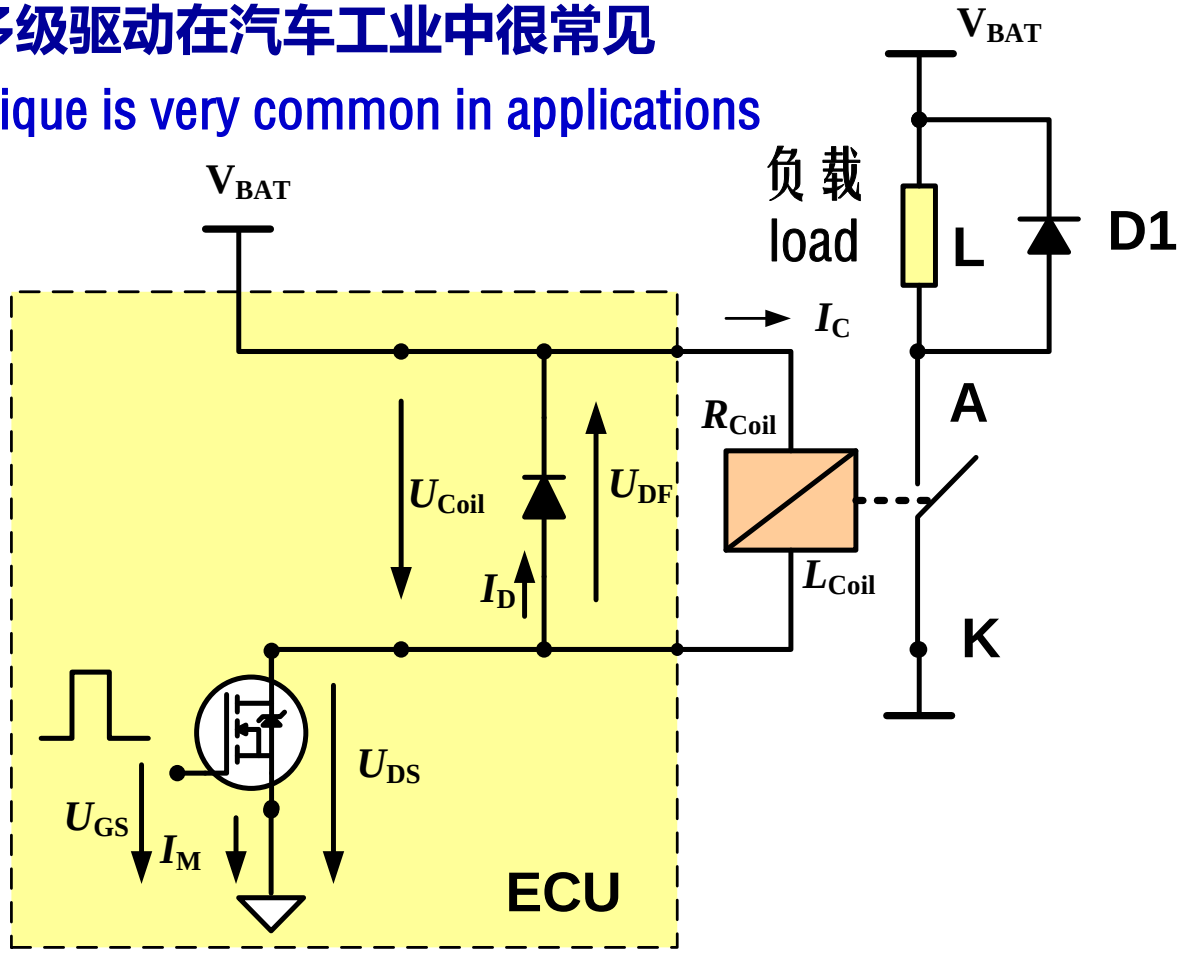
●低边驱动典型电路/ Typical low side drive circuit

- ◆ ECU通过继电器控制负载，继电器由小MOSFET驱动

/The ECU drive the relay via a small MOSFET

- ◆ 这种两级甚至多级驱动在汽车工业中很常见

/The pilot drive technique is very common in applications



§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

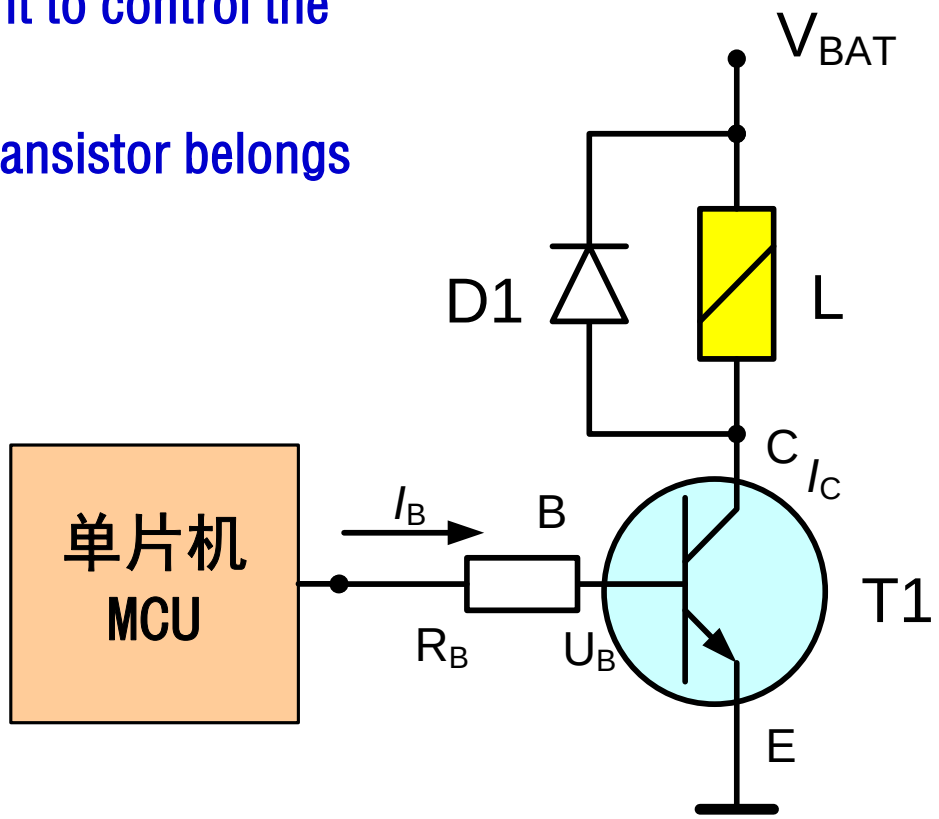
●低边驱动典型电路/ Typical low side drive circuit

- ◆ 采用三极管控制负载电流/ Control load current by using bipolar transistor
- ◆ 通过控制基极电流来控制集电极（负载）电流

Generally, MCU control the Base current to control the Collector (load) current.

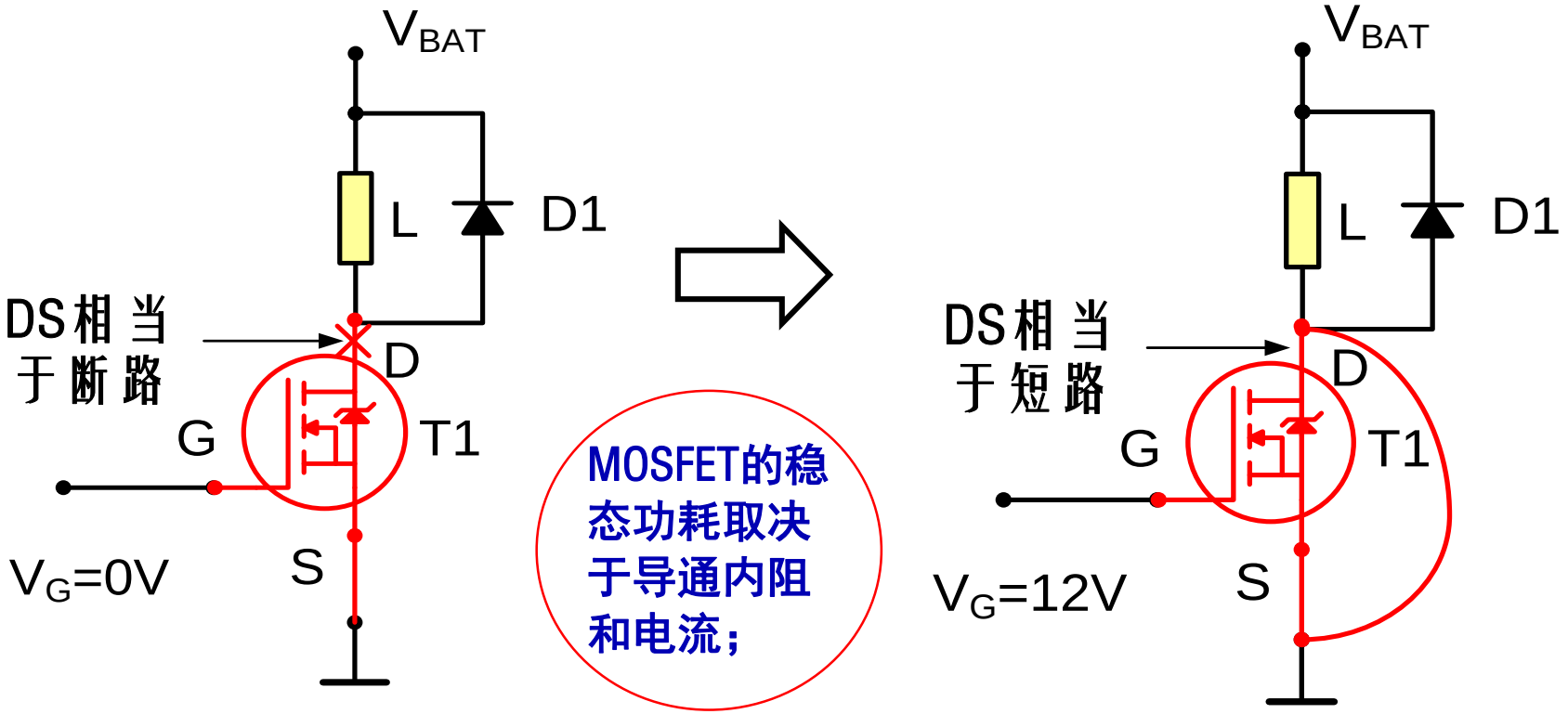
→三极管为电流控制型/ Hence the transistor belongs to current control type power switch.

$$I_C = I_B \times \beta$$



§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- **低边驱动典型电路/ Typical low side drive circuit**
  - **稳态分析：导通和关断状态电路分析/ Circuit analysis in On/Off state**



$V_{GS} = 0, V_{DS} = V_{BAT}$;
$V_S = 0, V_L = 0$; $R_{DS} = \infty$;
$I_D \cong 0$, $P_{DS} = I_D \times V_{DS} \cong 0$;

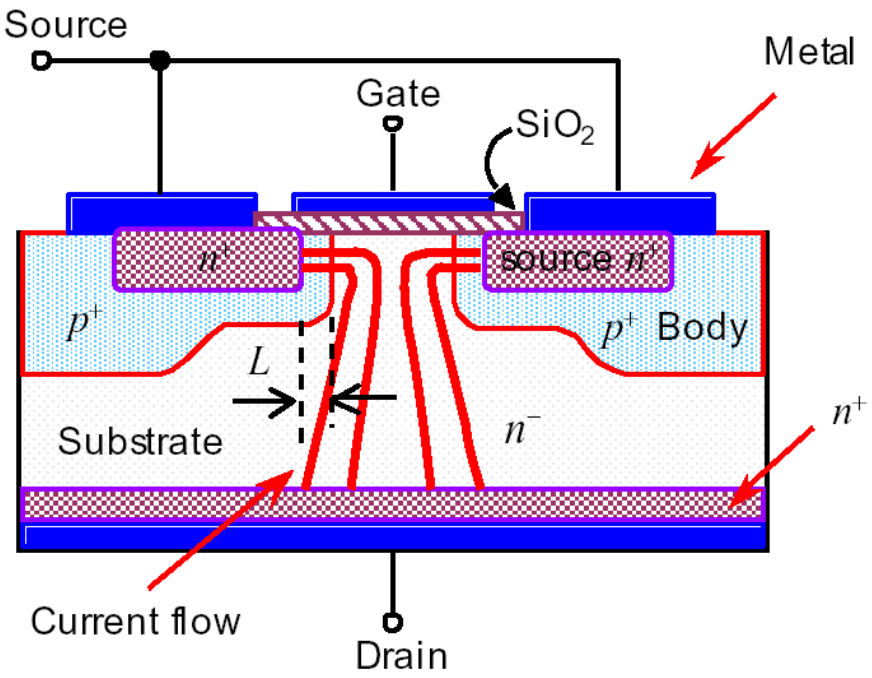
$V_{GS} = 12V, V_{DS} \cong 0V$;
$V_S = 0, V_L = V_{BAT}$;
$R_{DS} > 0$; $P_{DS} = I^2_D \times R_{DS} > 0$;

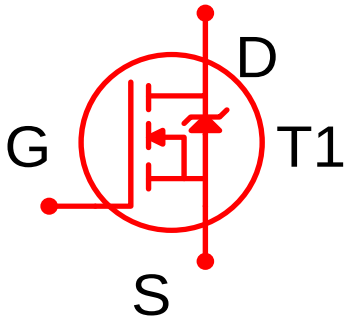
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

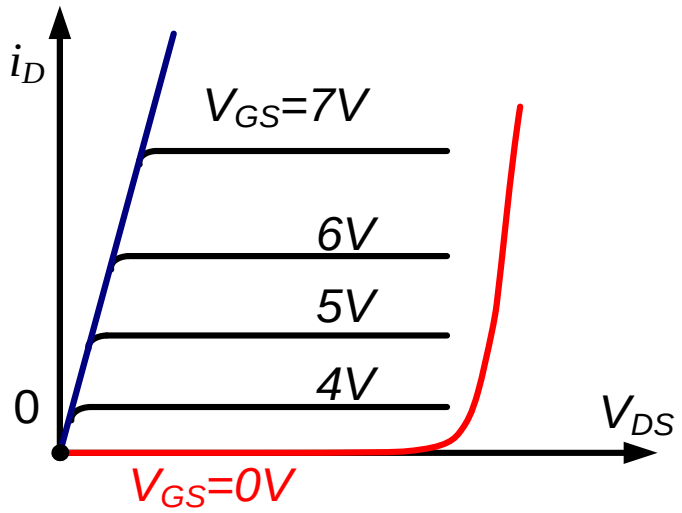
●低边驱动典型电路/Typical low side drive circuit

◆ 瞬态分析：MOS管的导通过程分析/ Transient

Analysis of turn on process of MOSFET



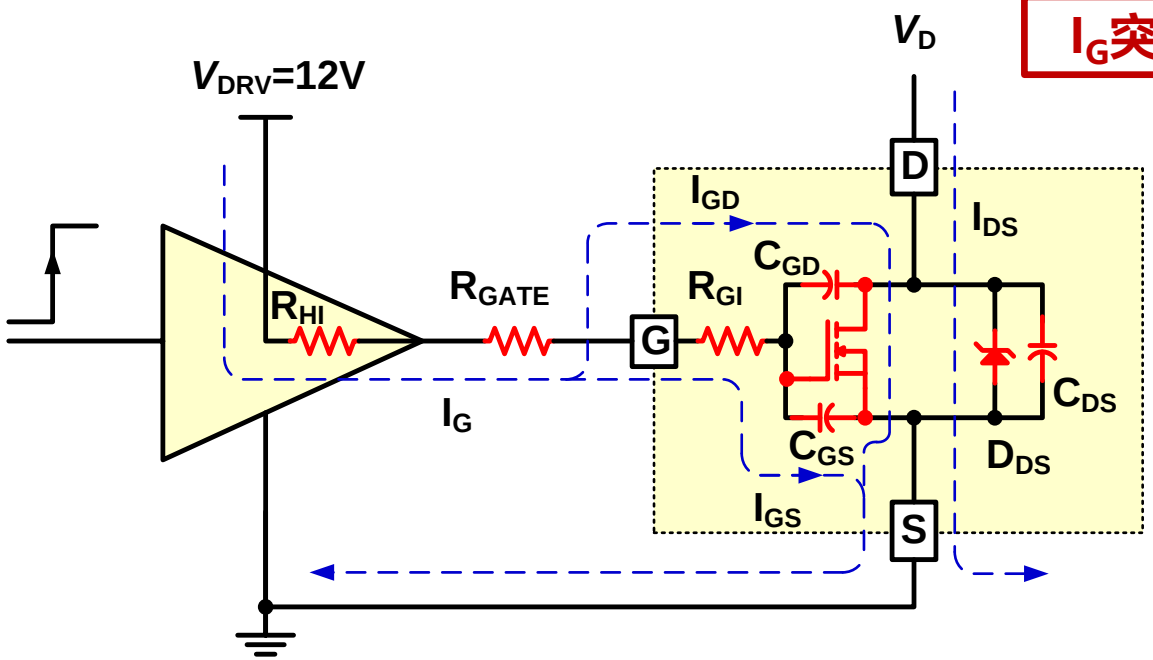


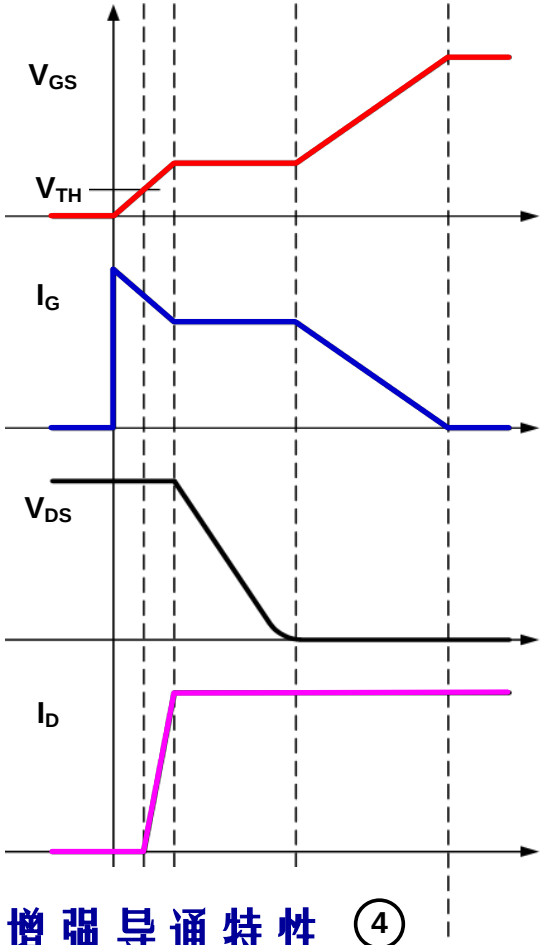


§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

●低边驱动典型电路/Typical low side drive circuit

◆ 瞬态分析：MOS管的导通过程分析/ Transient Analysis of turn on process of MOSFET





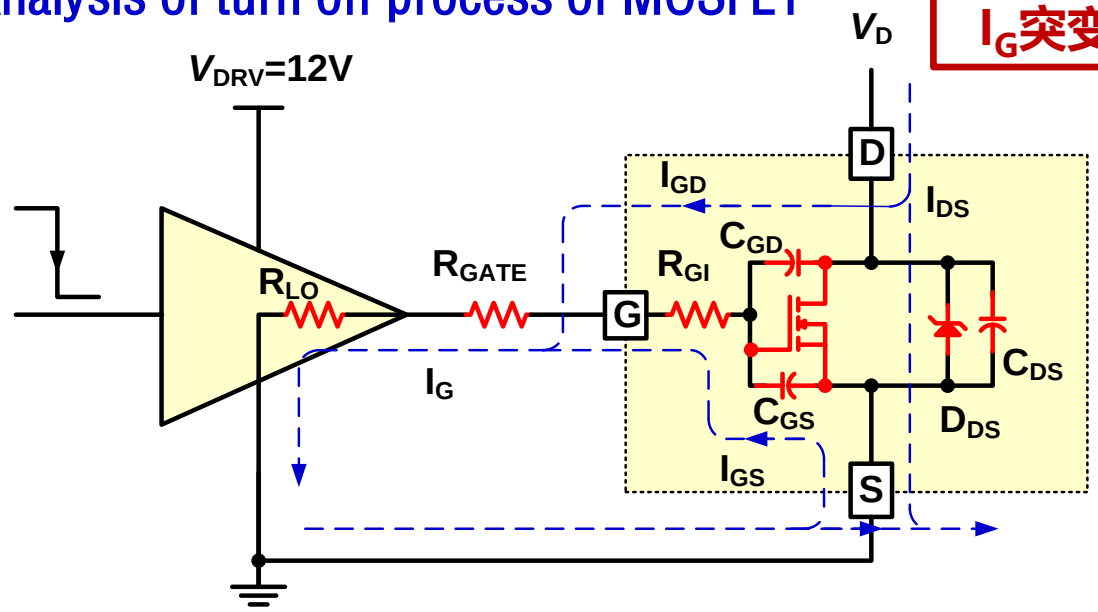
①导通延迟	②建立导电通道	③米勒效应	④增强导通特性
$V_{GS}=0\uparrow$, $V_{DS}=V_{BAT}$, $I_G\downarrow$, $I_D=0$	$V_{GS}\geq V_{TH}\uparrow$, $V_{DS}=V_{BAT}$, $I_G\downarrow$, $I_D\geq 0\uparrow$	$V_{GS}=V_{Miller}$, $V_{DS}\leq V_{BAT}\downarrow$, I_G, I_D=Max	$V_{GS}\geq V_{Miller}$, $V_{DS}=0$, $I_G\downarrow, I_D$=Max

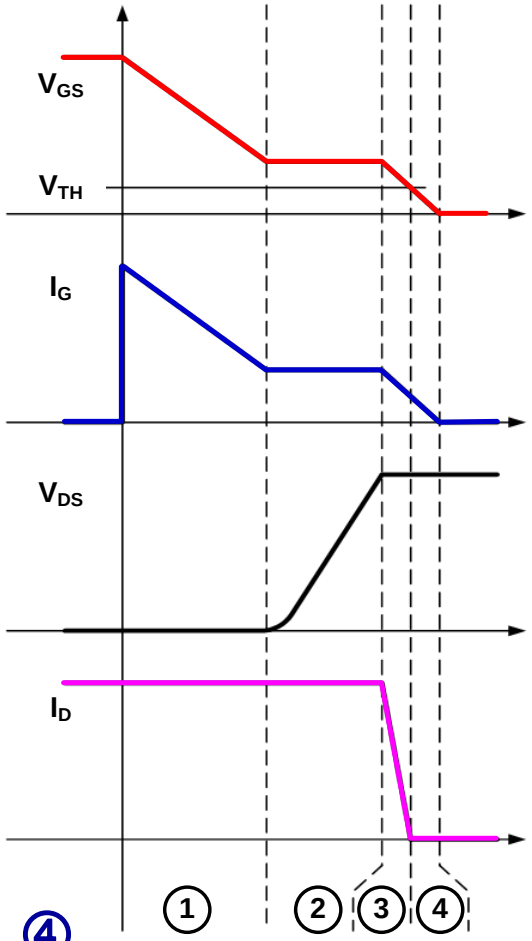
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

●低边驱动典型电路/Typical low side drive circuit

◆ 瞬态分析： MOS管关断过程分析/ Transient

Analysis of turn off process of MOSFET





① $V_{GS} \geq V_{Miller}\downarrow$, $V_{DS}=0$, $I_G\downarrow$, I_D=Max

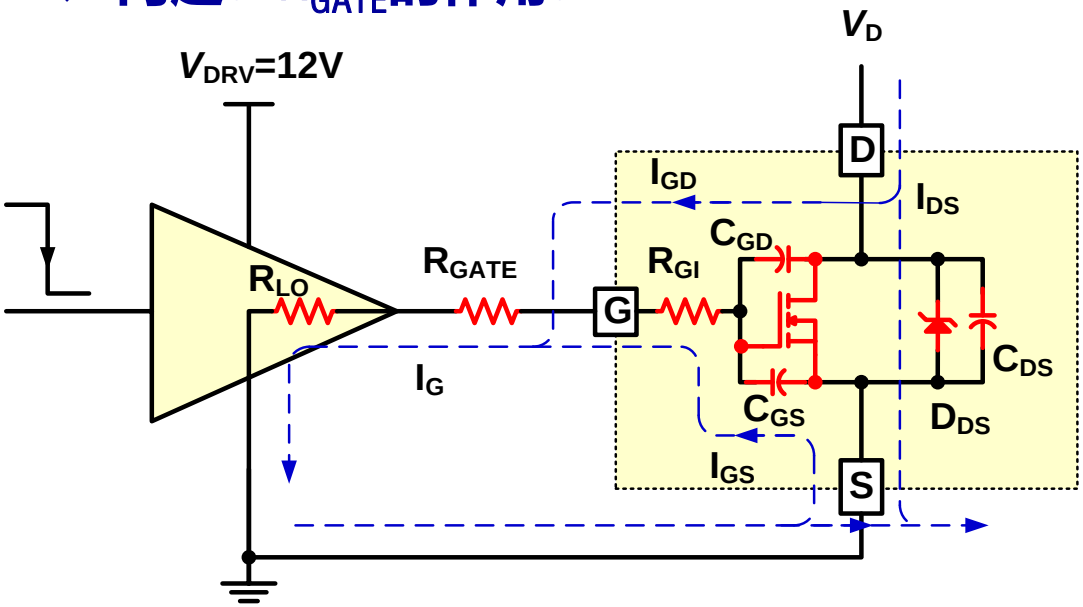
② $V_{GS}=V_{Miller}$, $V_{DS}\geq 0\uparrow$, I_G, I_D=Max

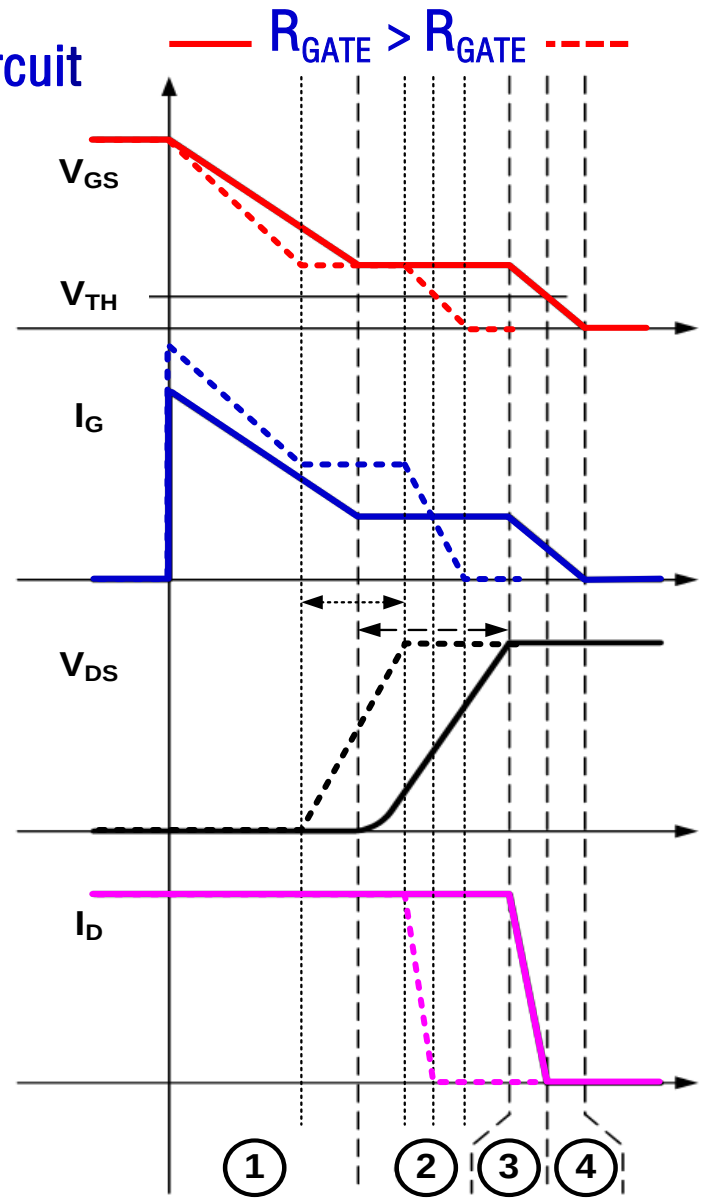
③ $V_{GS} \geq V_{TH}\downarrow$, $V_{DS}=V_{BAT}$, $I_G\downarrow$, $I_D\rightarrow 0\downarrow$

④ $V_{GS}\rightarrow 0\downarrow$, $V_{DS}=V_{BAT}$, $I_G\rightarrow 0\downarrow$, I_D=0

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- **低边驱动典型电路**/Typical low side drive circuit
  - MOSFET的动态功耗分析/Transient power Dissipation analysis of MOSFET
  - 动态功耗与电压、电流和工作频率相关/ Transient power dissipation depends on voltage, Current and frequency.
  - 高频(>10KHz)将以动态的开关损耗为主
  - 问题：R_{GATE}的作用?

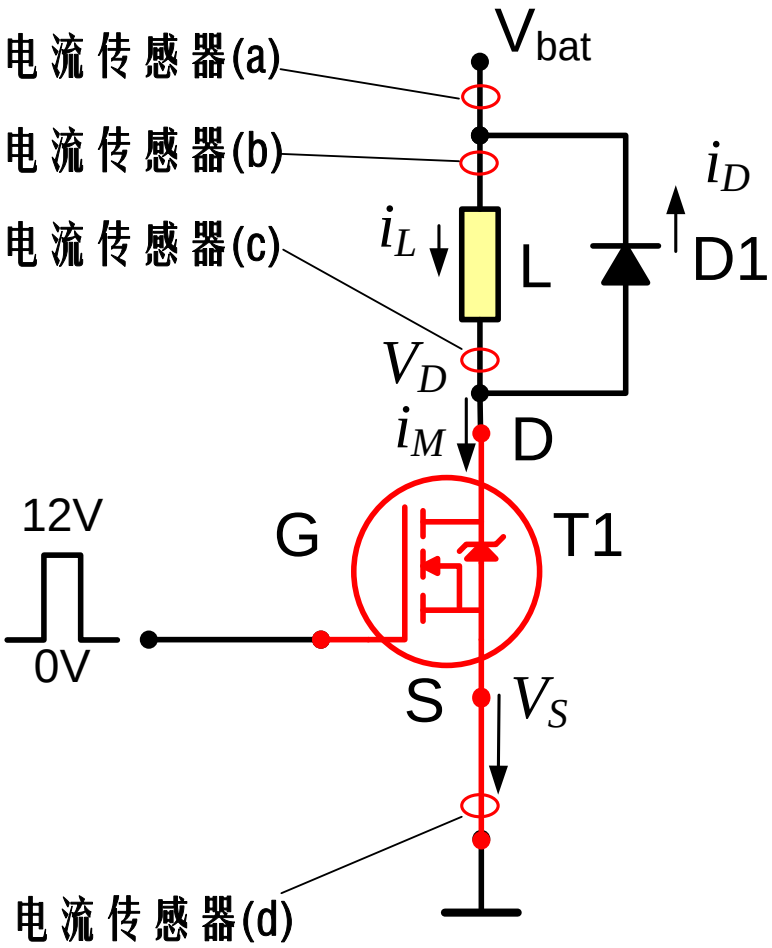


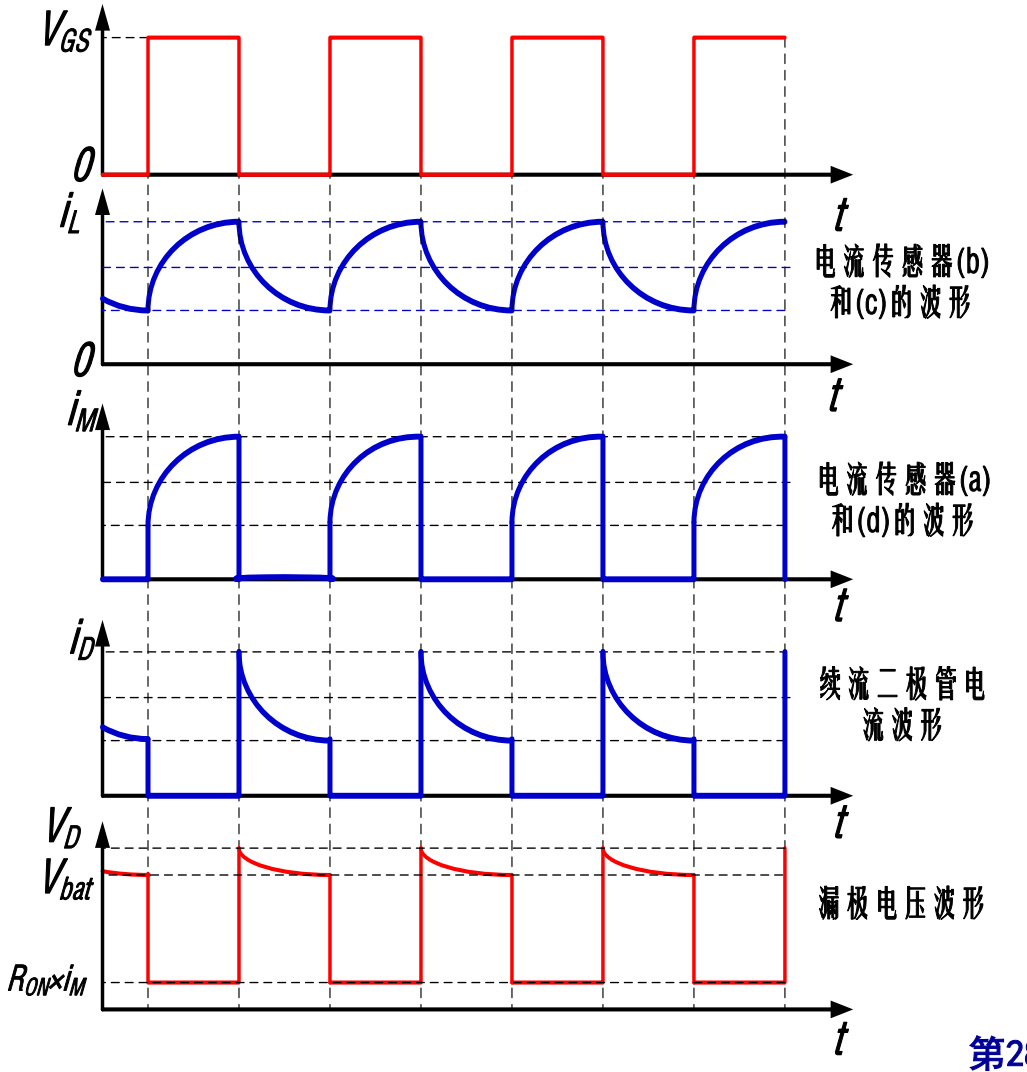


§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

● 低边驱动典型电路/ Typical Low Side Driver Circuit

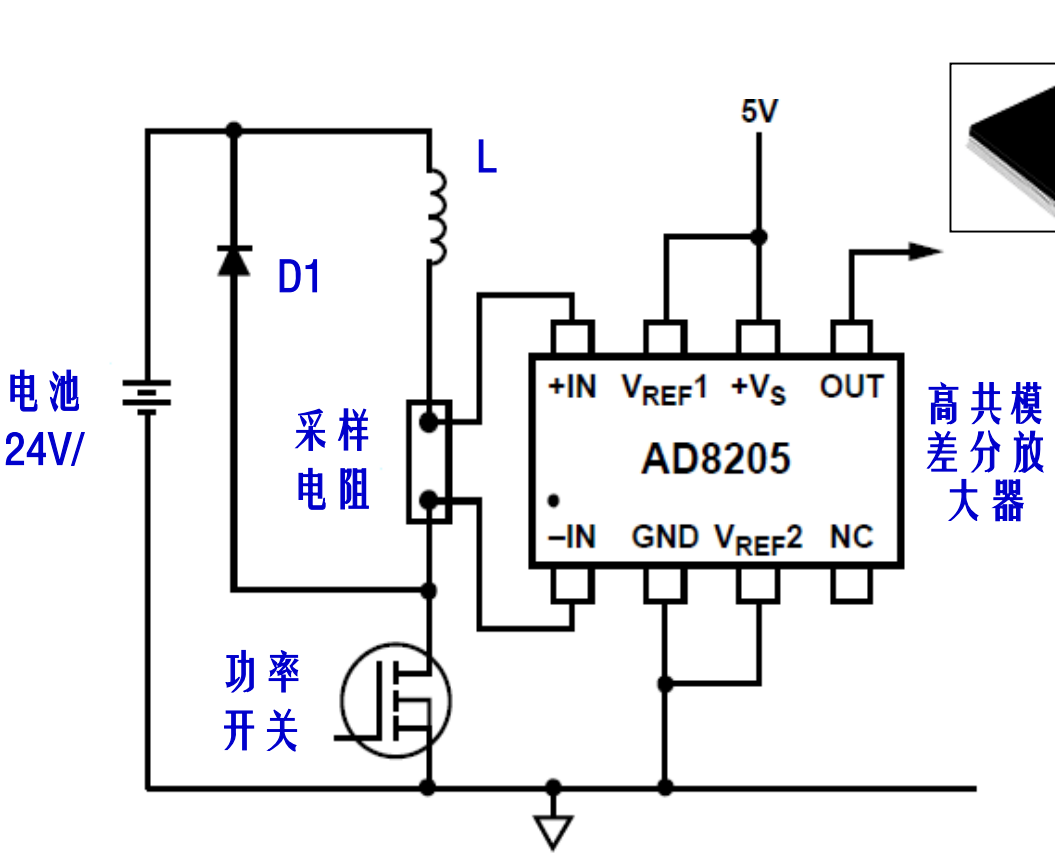
◆ 电流检测和故障诊断方法/ Methods of current measurement and diagnosis

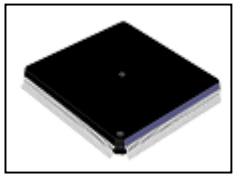


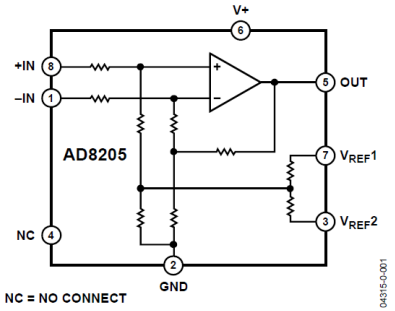


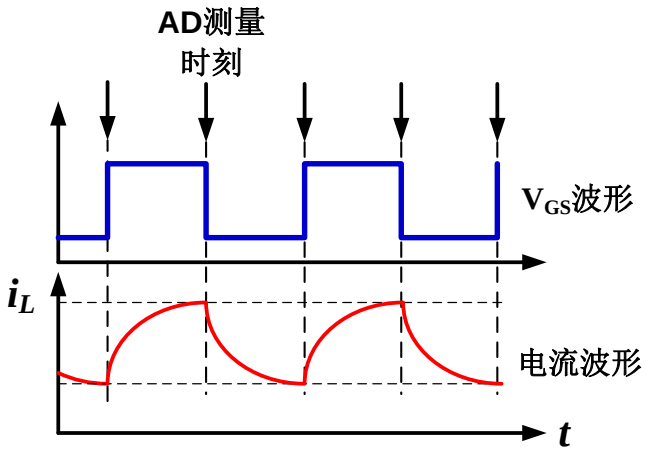
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- **低边驱动典型电路/ Typical Low Side Driver Circuit**
  - **电流检测和故障诊断方法/ Methods of current measurement and diagnosis**



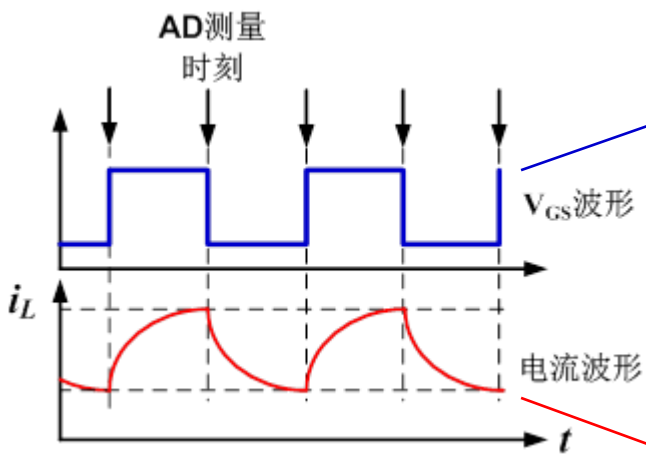




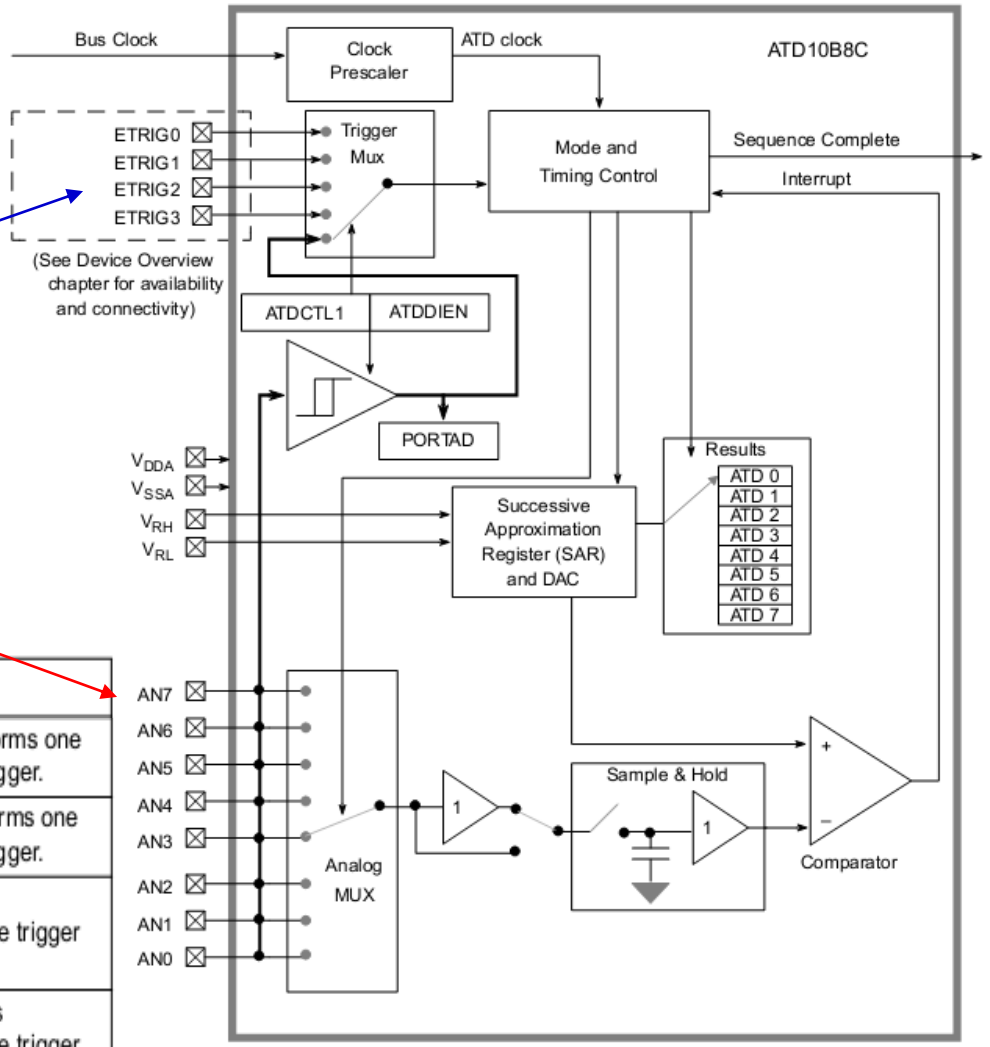


§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- 低边驱动典型电路
- 电流检测和故障诊断方法

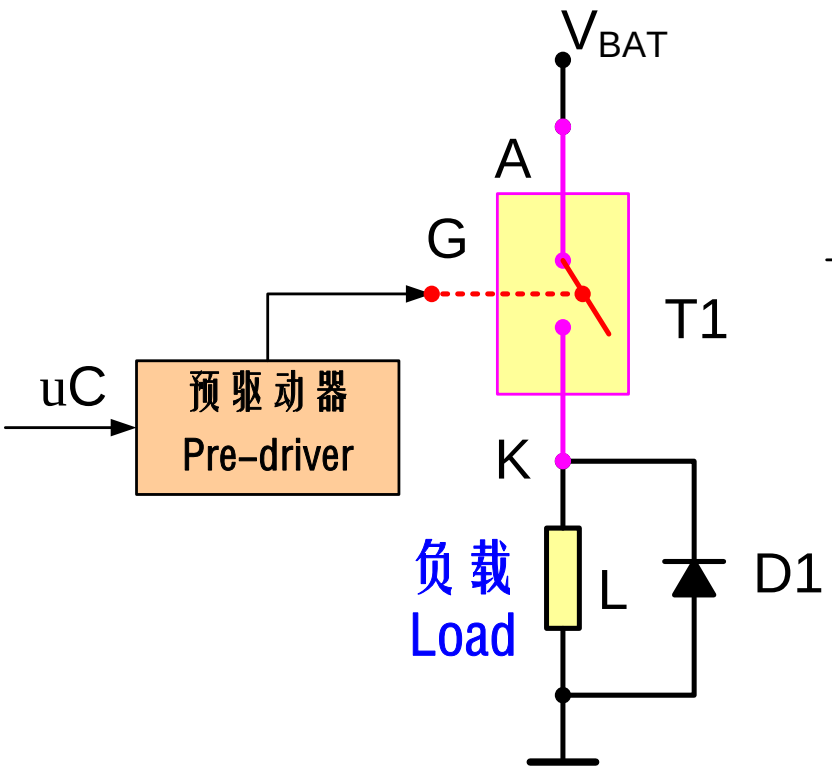


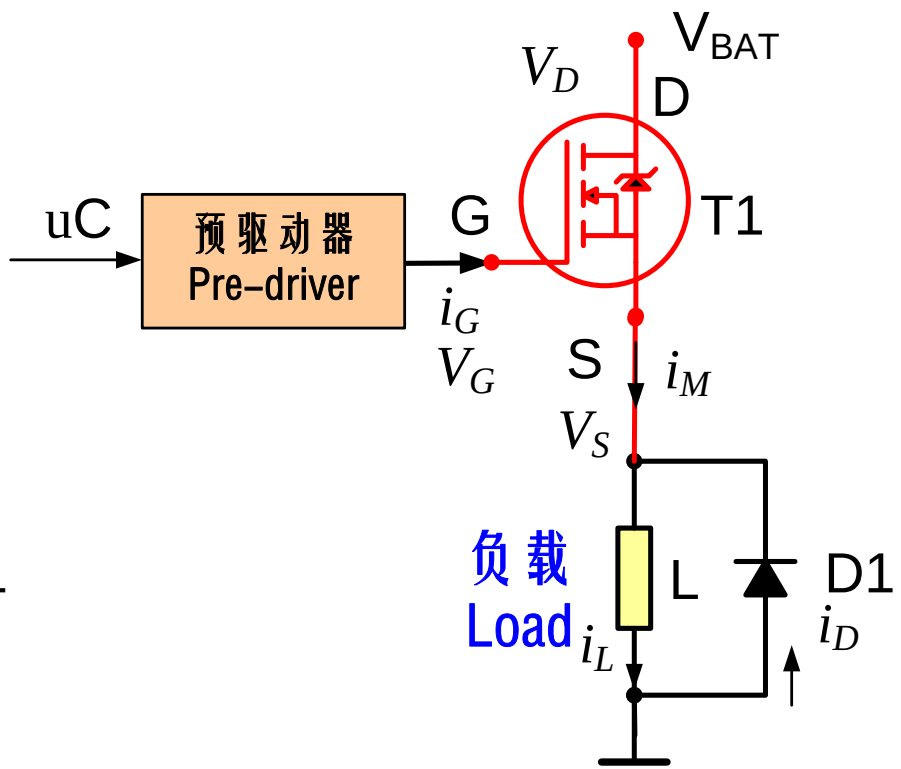
ETRIGLE	ETRIGP	ETRIGE	SCAN	Description
0	0	1	X	Falling edge triggered. Performs one conversion sequence per trigger.
0	1	1	X	Rising edge triggered. Performs one conversion sequence per trigger.
1	0	1	X	Trigger active low. Performs continuous conversions while trigger is active.
1	1	1	X	Trigger active high. Performs continuous conversions while trigger is active.



§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

(2)高边驱动典型电路/ Typical high side driver circuit

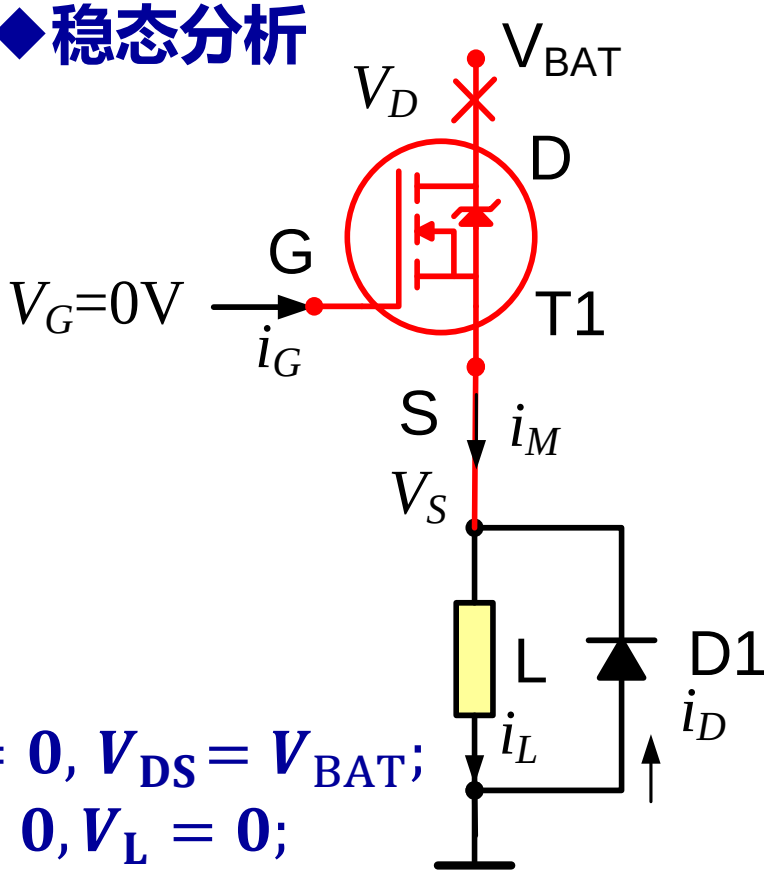




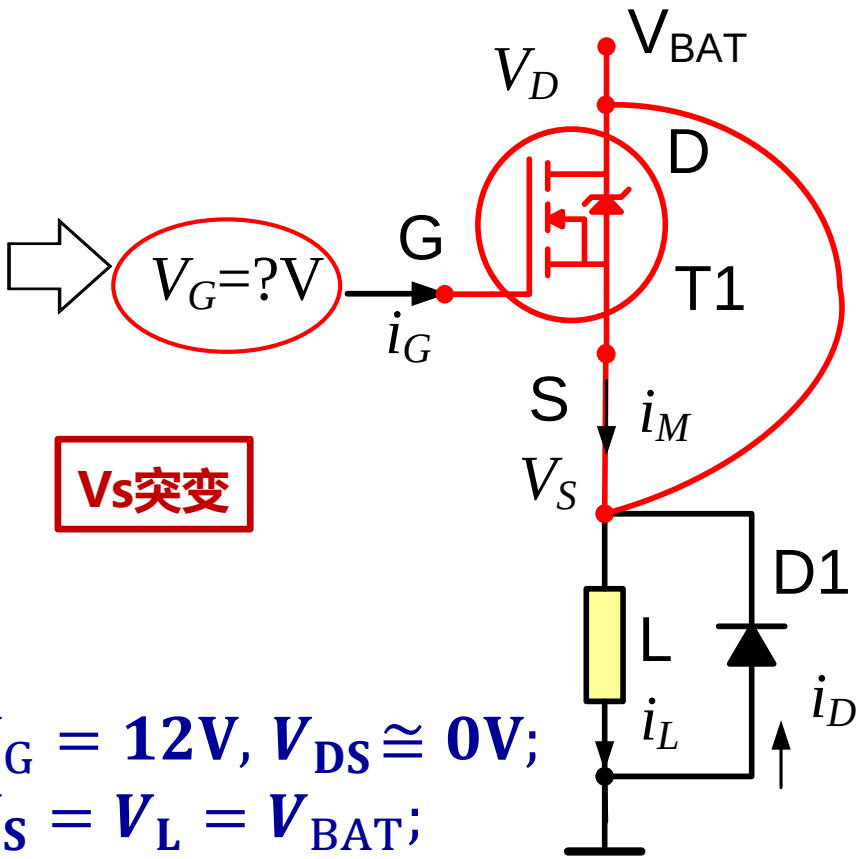
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- 高边驱动典型电路/ Typical high side driver circuit

◆稳态分析



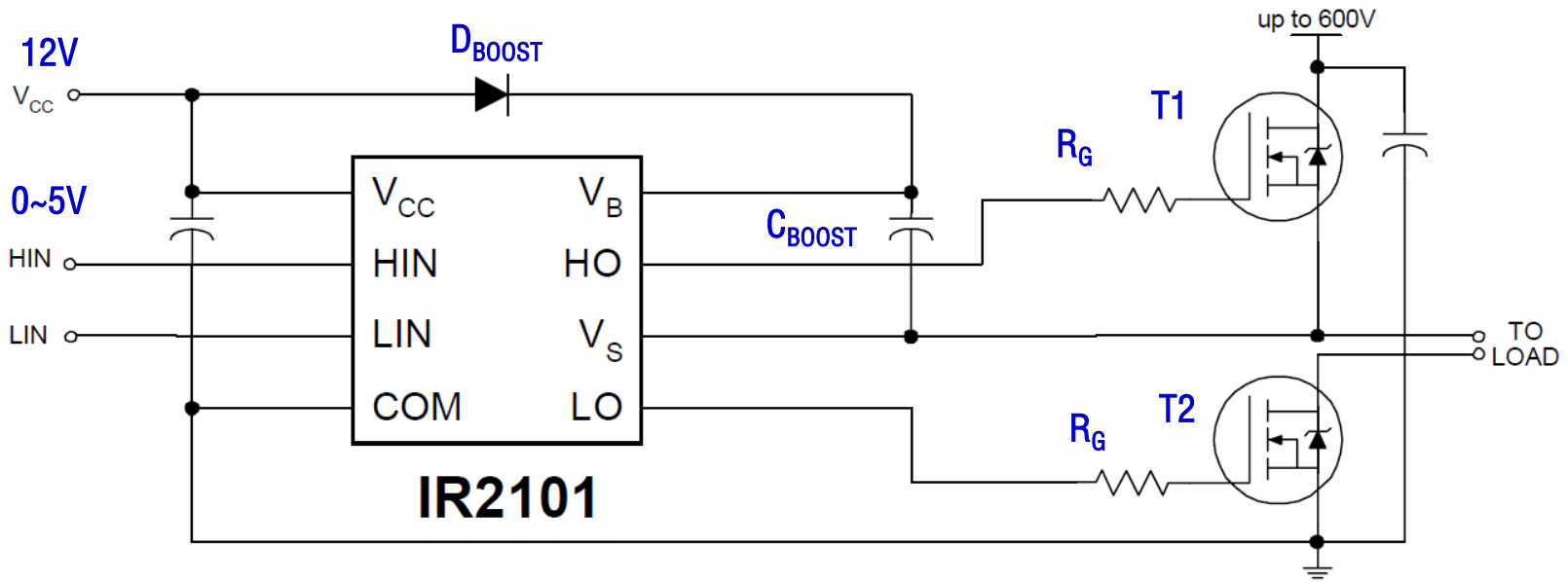
$V_G = 0, V_{DS} = V_{BAT};$
$V_S = 0, V_L = 0;$
$R_{DS} = \infty;$
$I_D \cong 0,$
$P_{DS} = I_D \times V_{DS} \cong 0;$



$V_G = 12V, V_{DS} \cong 0V;$
$V_S = V_L = V_{BAT};$
$V_G = 12V + V_{BAT};$
$R_{DS} > 0;$
$P_{DS} = I^2_D \times R_{DS} > 0;$

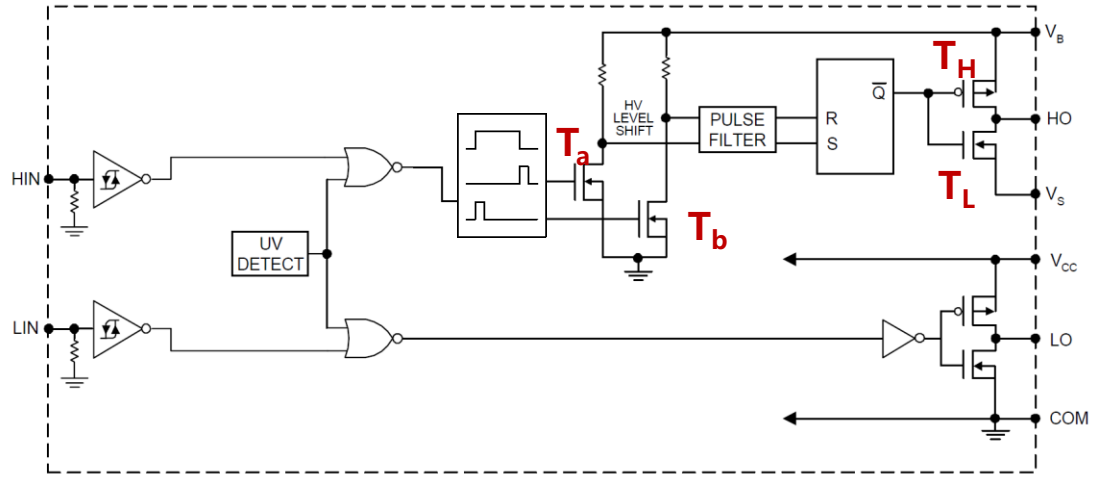
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- 高边驱动在MOSFET导通的瞬间，栅极电压要从12V迅速上升到V_{bat} + 12V，需要有专门的自举电路提升该电压/ The high side driver need a special circuit to boost the gate voltage from 12V up to Vbat+12V when turn on the MOSFET.
- 这种栅极浮动驱动的技术在很多的执行器控制中被用到，也已经形成了专门的高边驱动接口电路/ This float gate drive technology has been widely applied in actuate control, it has become a special high side pre-driver chip.



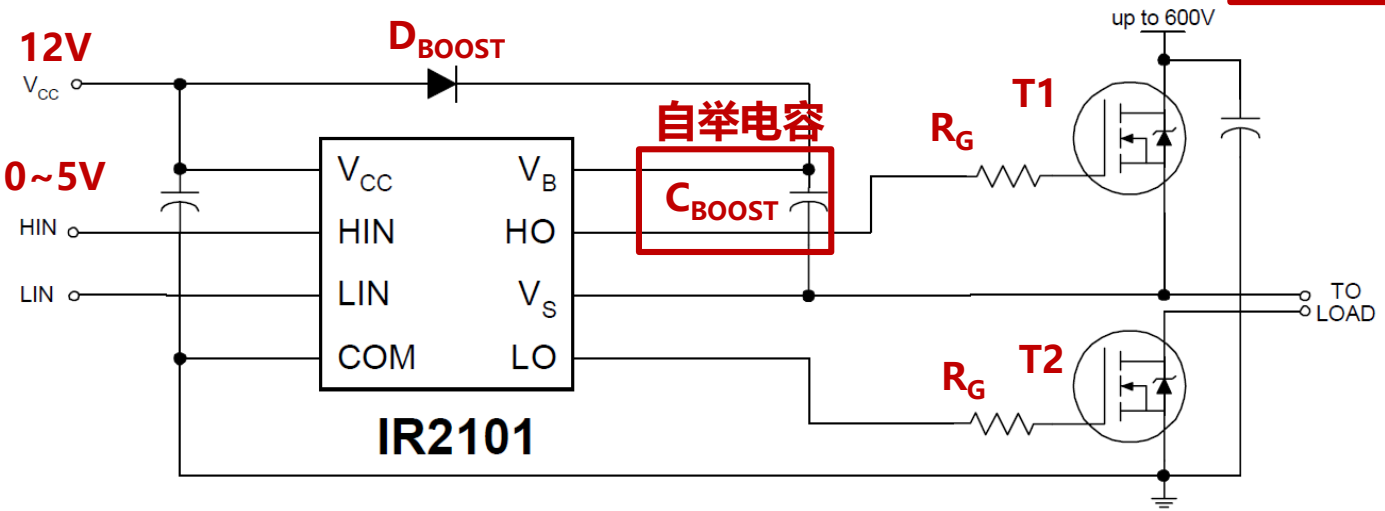
§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- 高边驱动器内部结构/ internal structure high side driver



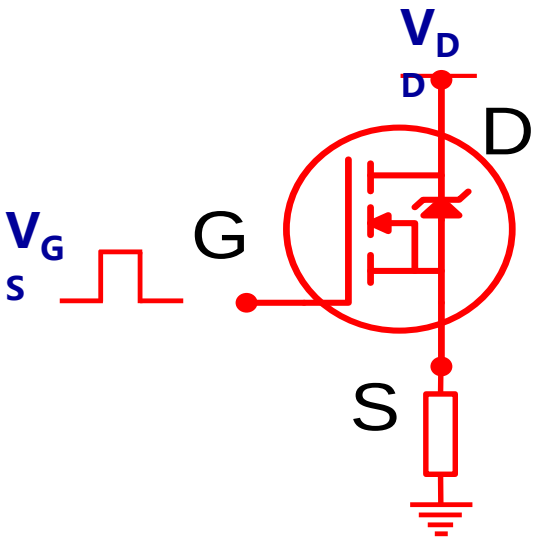
$V_{HO}=V_B$

$V_B=V_{Cboost}+V_s$

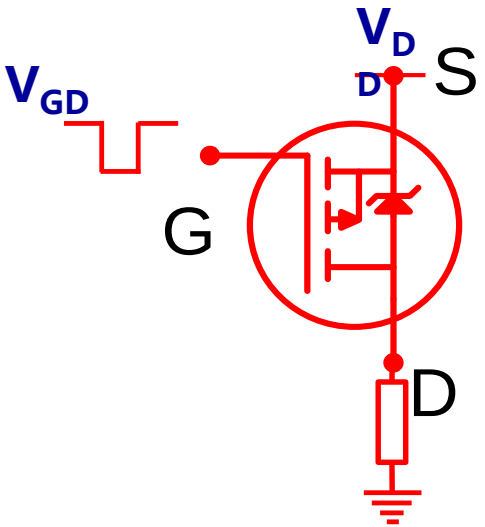


§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

不同类型金属氧化物场效晶体管 (MOSFET) 高边驱动对比



N沟道(耗尽型/增强型)绝缘栅场效晶体管



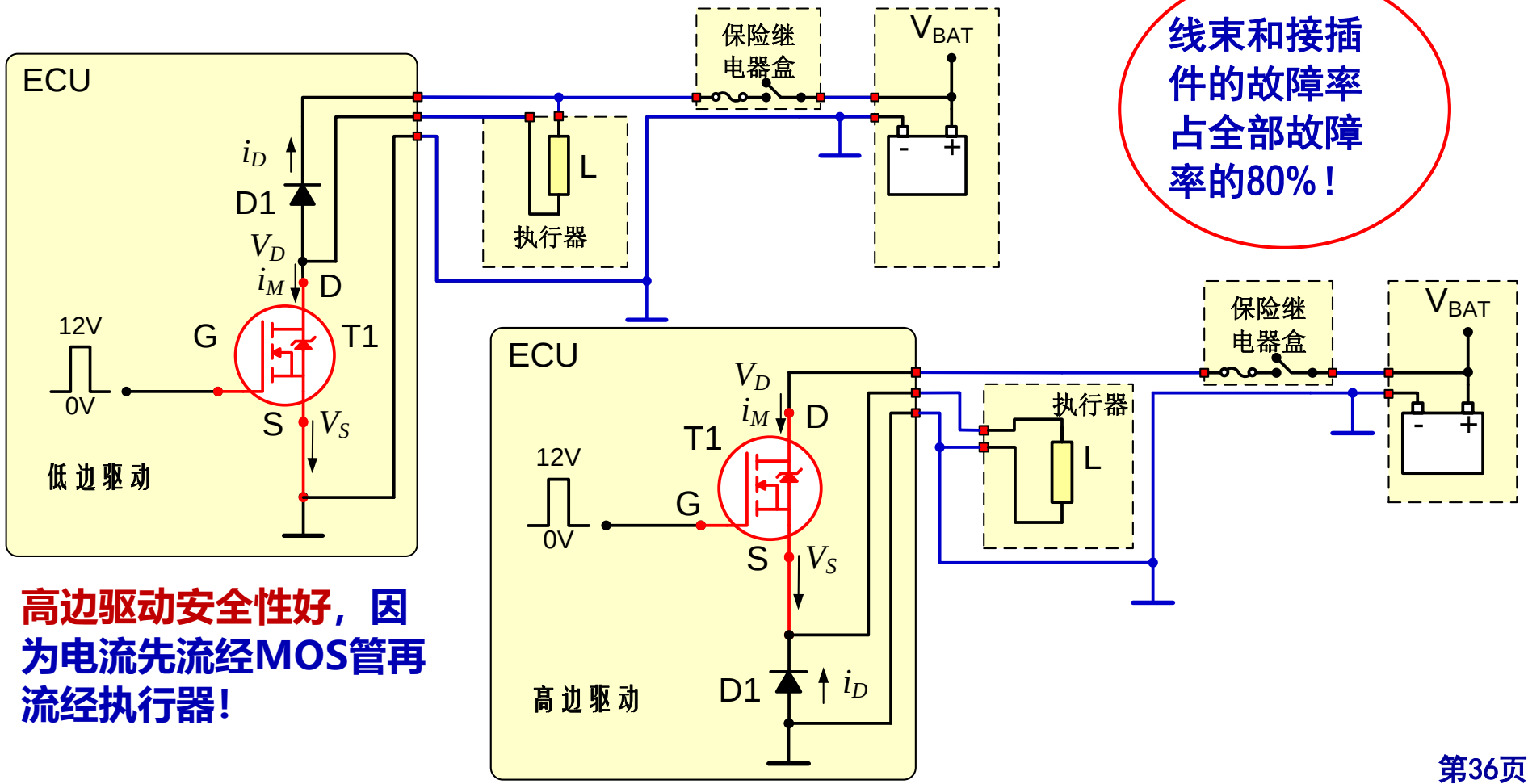
P沟道(耗尽型/增强型)绝缘栅场效晶体管

- V_{GS}>0（如12V），则导通
- 电流由D流向S
- **浮动栅极**：V_G需在导通瞬间突升

- V_{GS}<0 （如-12V），则导通
- 电流由S流向D
- **固定栅极**：V_G在导通瞬间电位不变

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

- 高边和低边的比较/ Compare between low side and high side
  - 在相同的功率和频率要求条件下/ Under same power and frequency
  - 成本：高边驱动成本高，浮动栅极驱动器

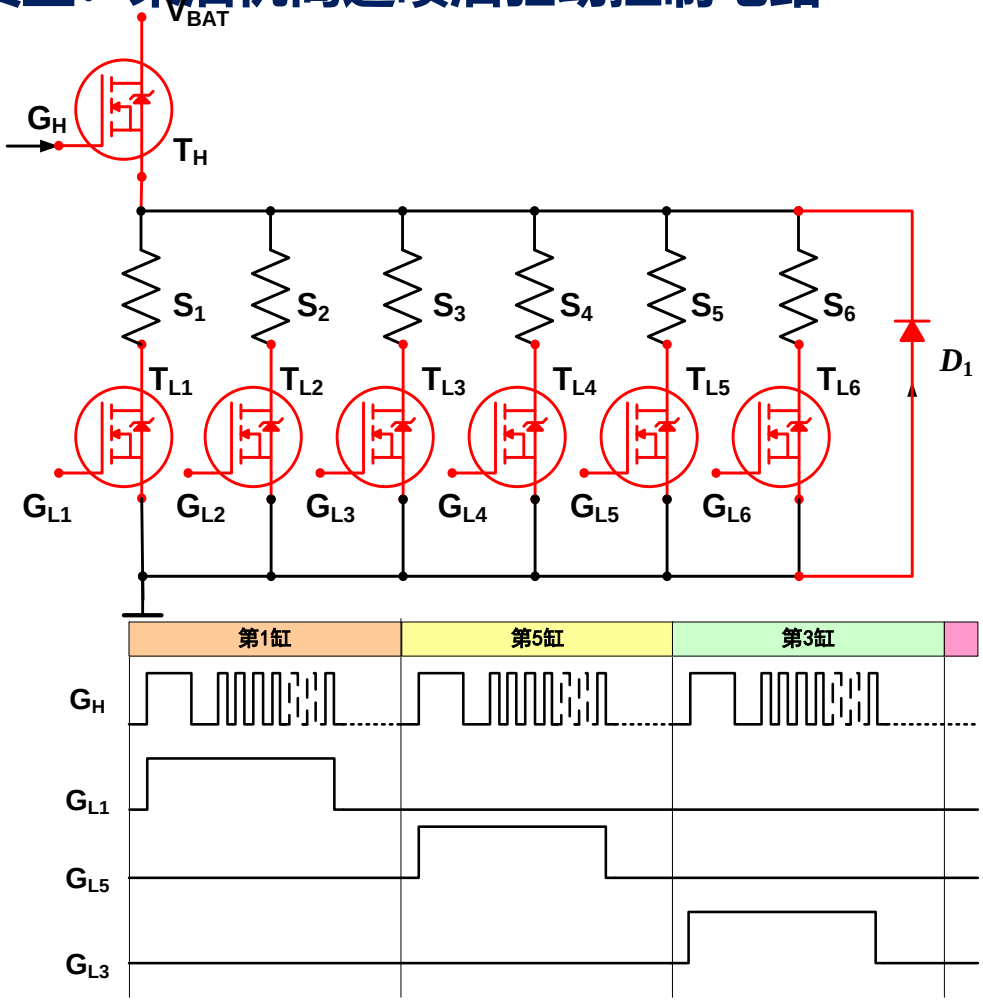


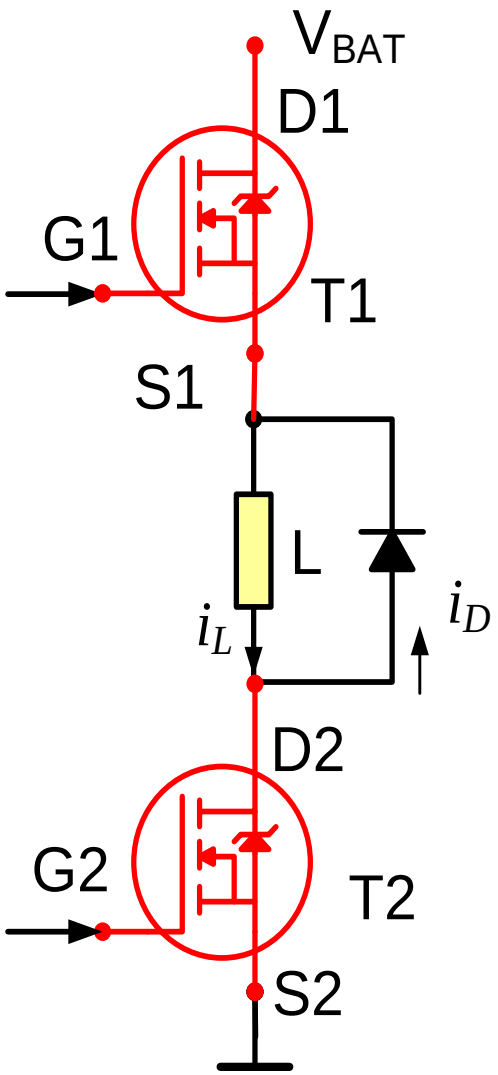
高边驱动安全性好，因为电流先流经MOS管再流经执行器！

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

(3) 半桥驱动/Half bridge driver

- 高边+低边，电流方向唯一
- 典型：柴油机高速喷油驱动控制电路

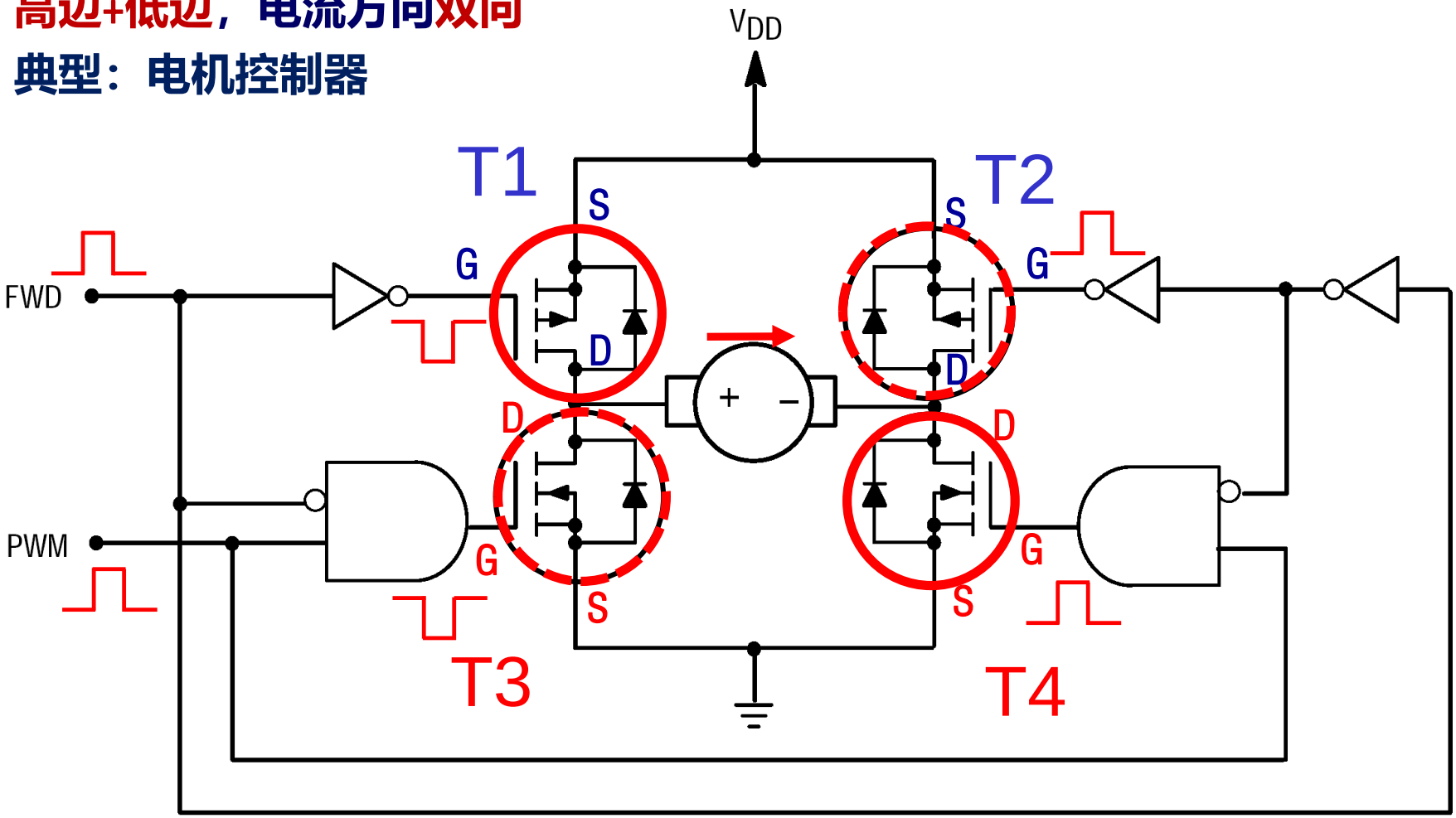




§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

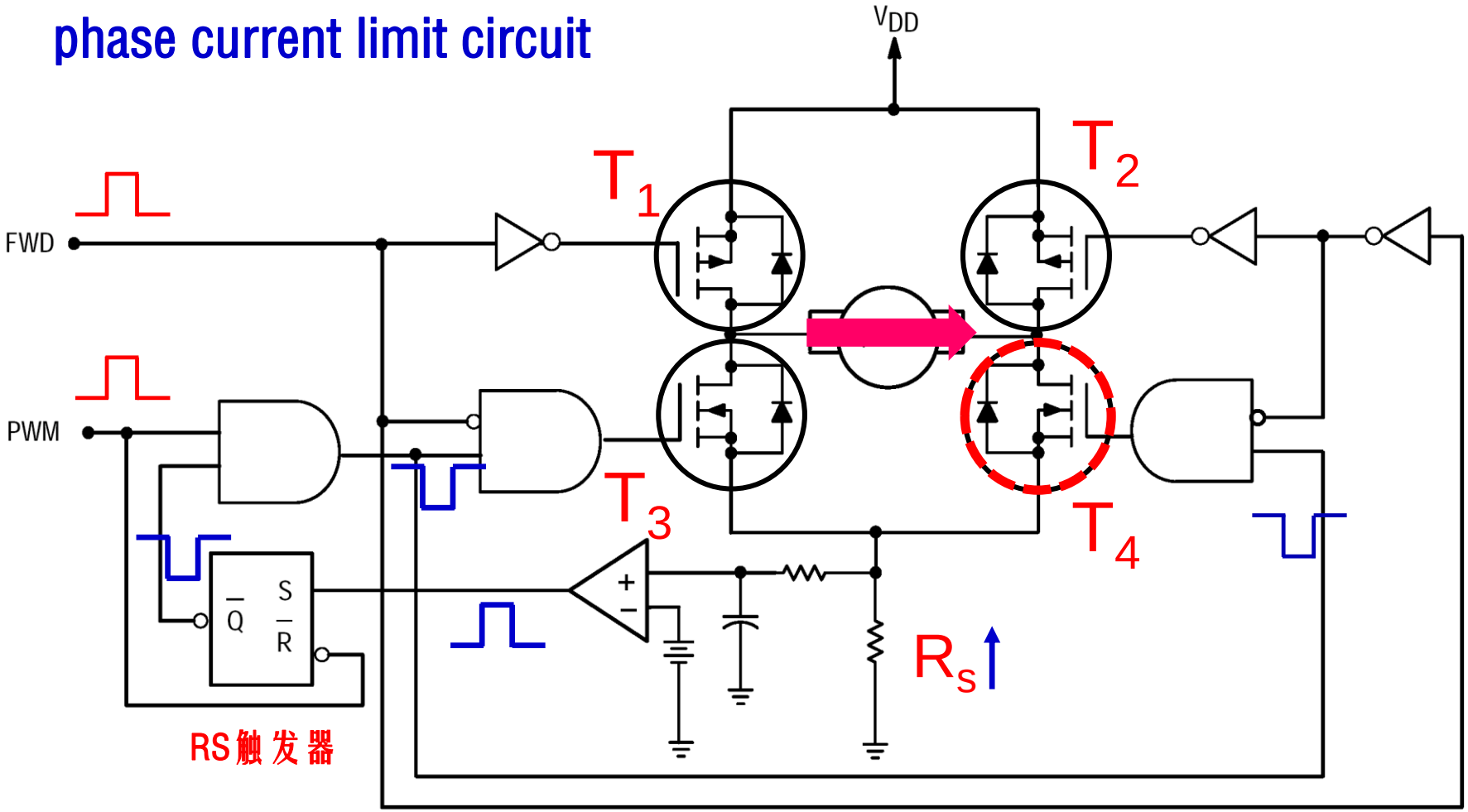
(4) 全桥驱动/Full bridge driver

- 高边+低边，电流方向双向
- 典型：电机控制器



§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

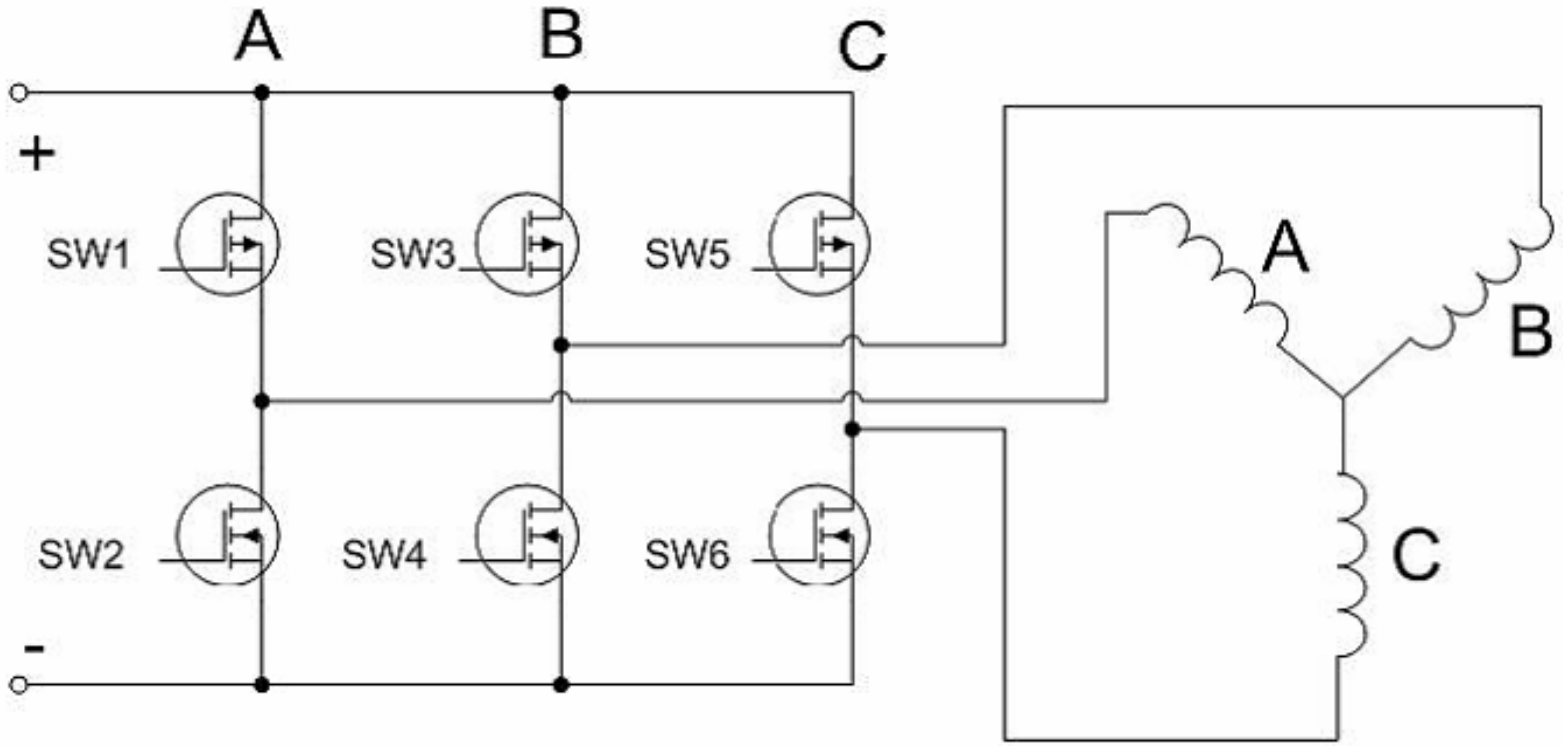
带有相电流反馈和保护的全桥电机驱动电路/Full–bridge drive with phase current limit circuit



线路故障->驱动电流突然增大->Rs两端电压升高->T4关断保护电机

§2四拓扑结构驱动信号接口电路
Four topologies of drive interface circuits

三相电机的全桥电机驱动电路/Full-bridge drive of 3 phase motor

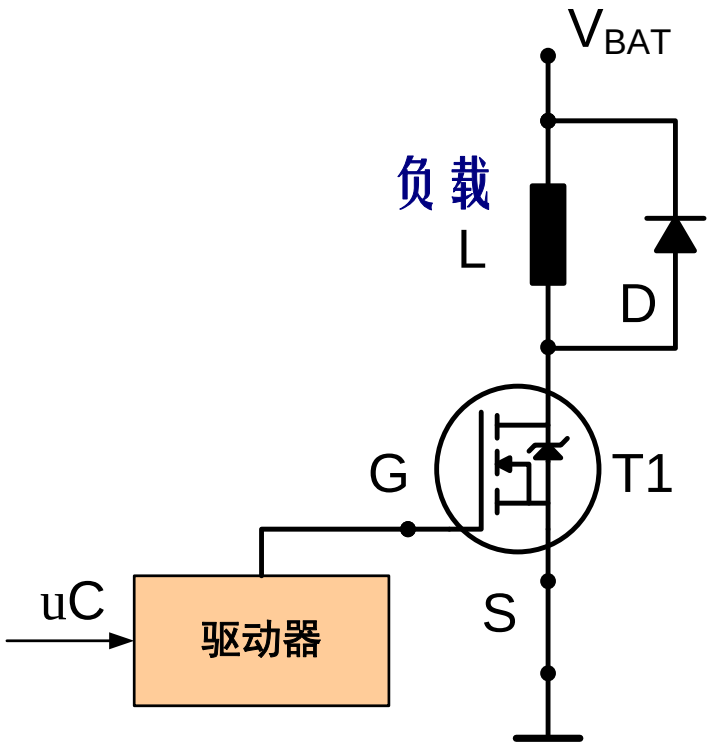


§目录 / Index

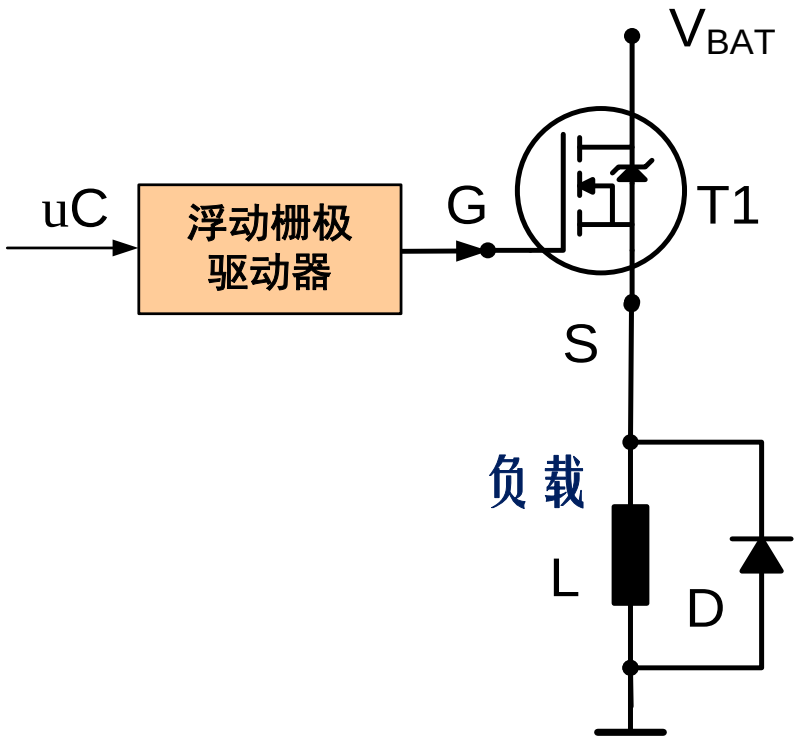
1. 传感器、信号接口电路及运算放大器
   ① 传感器工作原理
   ② 传感信号接口电路
   ③ 典型运算放大器
2. 执行器、开关器件及驱动接口电路
   ① 执行器及四种功率开关器件（Actuators and power switches）
   ② 驱动信号接口电路：四拓扑结构（Four topologies of drive interface circuits）
   ③ 驱动电路的续流回路（Freewheel circuit design）
   ④ 驱动电路的故障诊断和容错控制（Diagnosis and fault tolerant control for drive circuits）
   ⑤ 汽车专用功率驱动集成芯片（Integrated Power Chip）

§3驱动电路的续流回路
Freewheel circuit design

(1) 低边驱动电路/Low side drive circuit

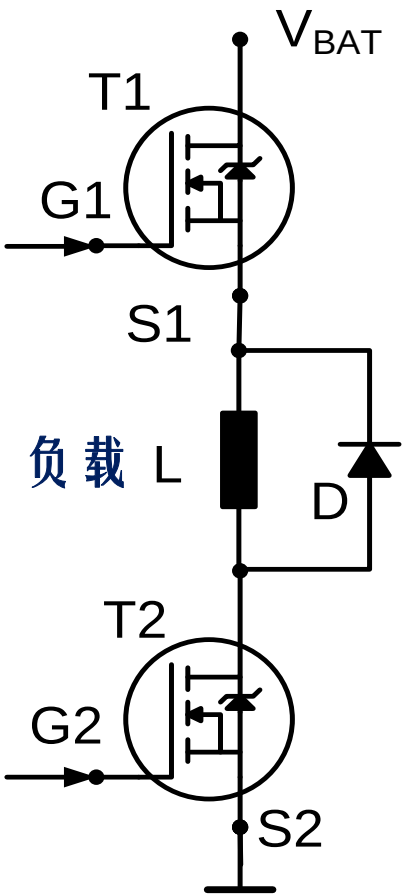


(2) 高边驱动电路/High side drive circuit

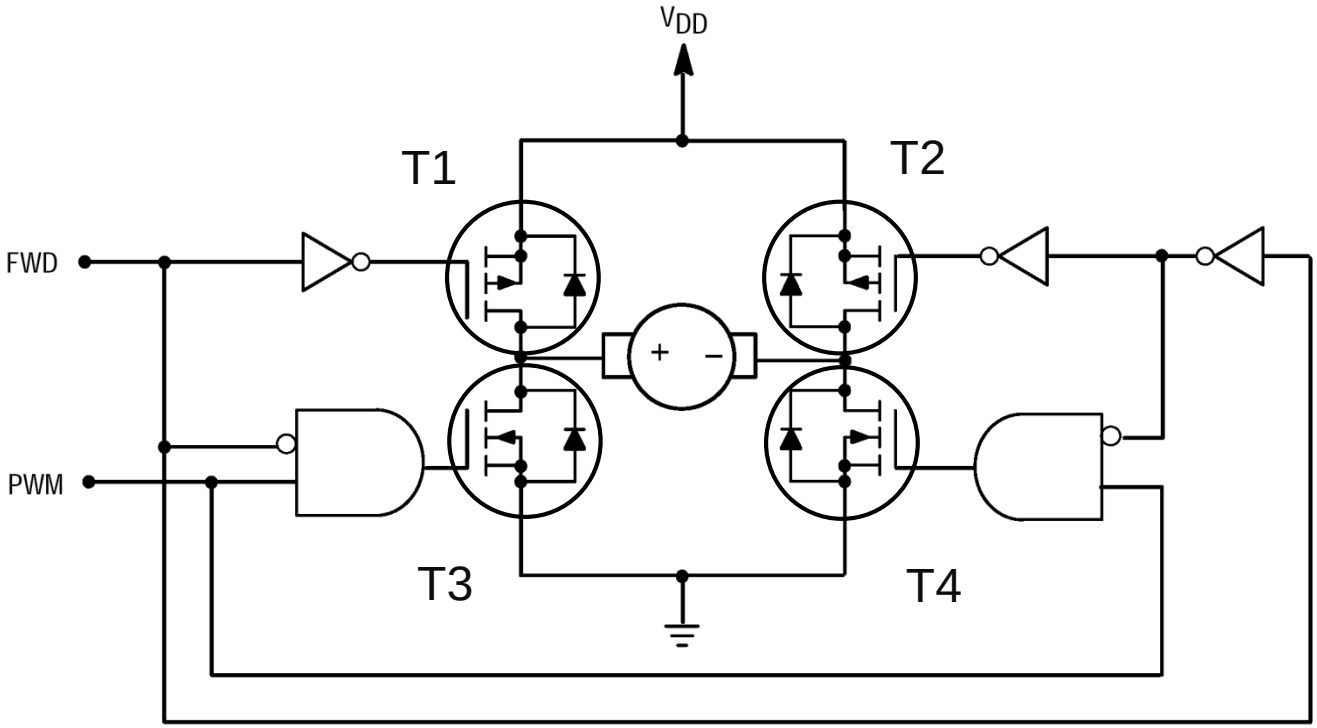


§3驱动电路的续流回路
Freewheel circuit design

(3) 半桥驱动/Half bridge drive circuit



(4) 全桥驱动/Full (H) bridge drive circuit



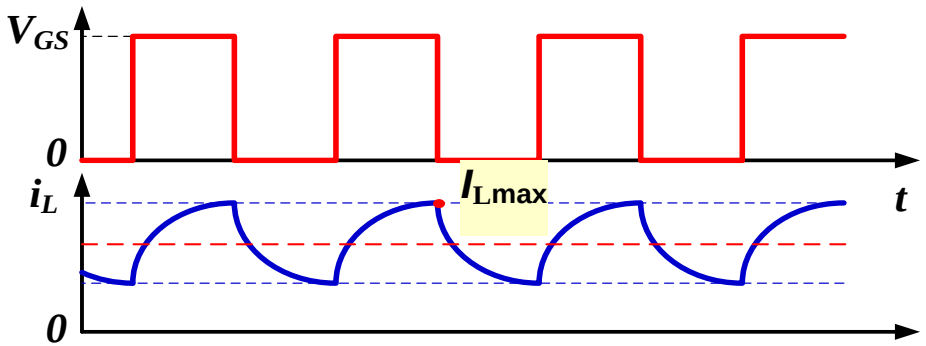
§3驱动电路的续流回路
Freewheel circuit design

续流回路设计1：缓慢续流型/Freewheel circuit type I: Slow continually drop down

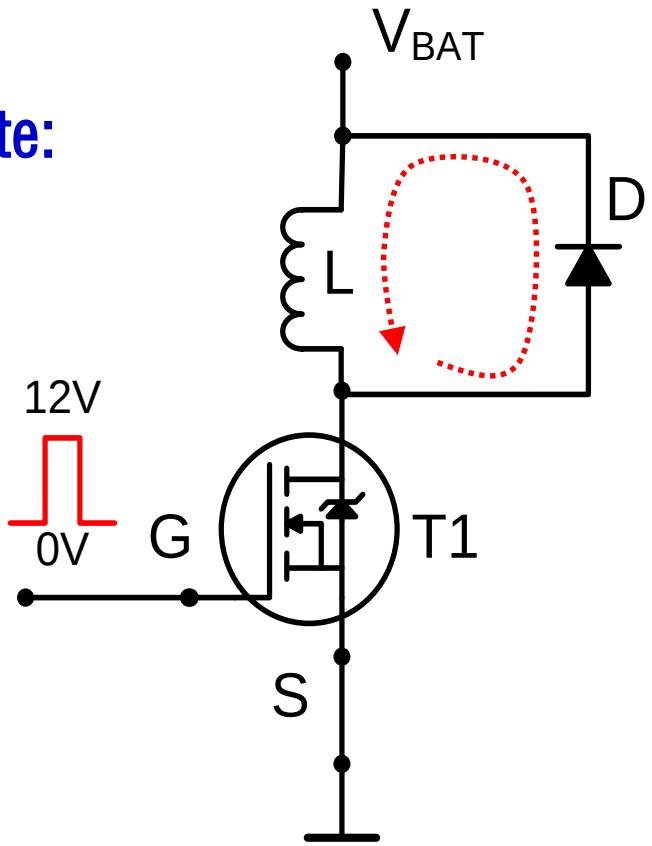
◆ 执行器能量释放速度慢/Slow energy release for actuator

初始释能功率/Initial Energy release rate:

$$P_L = I_{Lmax} \times 0.7V;$$



●二极管D的作用
/Function of Diode D?



典型应用：ABS液压制动电磁阀控制。

Typical Application: ABS valve control

续流回路设计2：快速切断型/ Freewheel circuit type II: Fast Cut off

◆阶段1：A→B

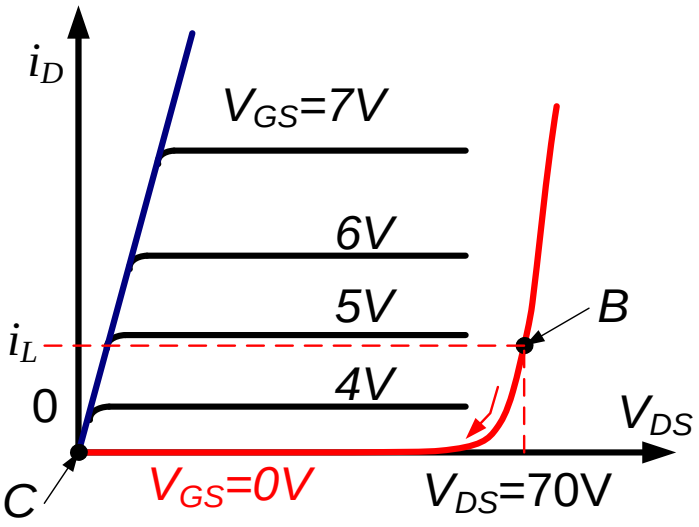
V_{GS1}=0, V_{GS2} = 12V

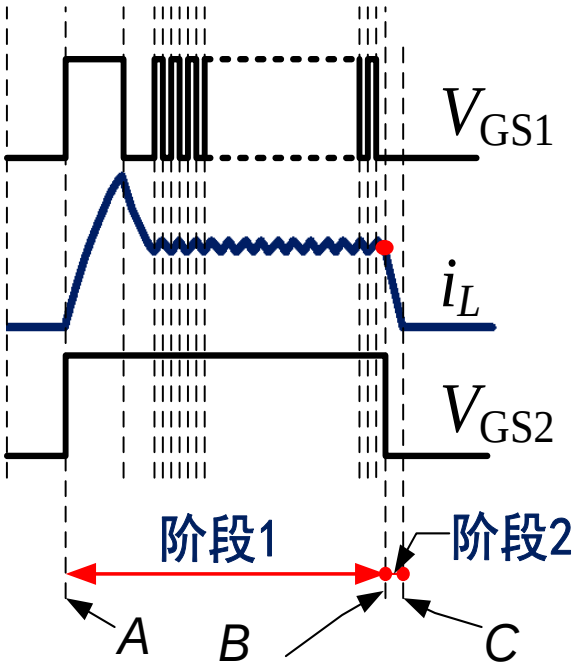
$P_L \approx I_L \times 0.7V$;

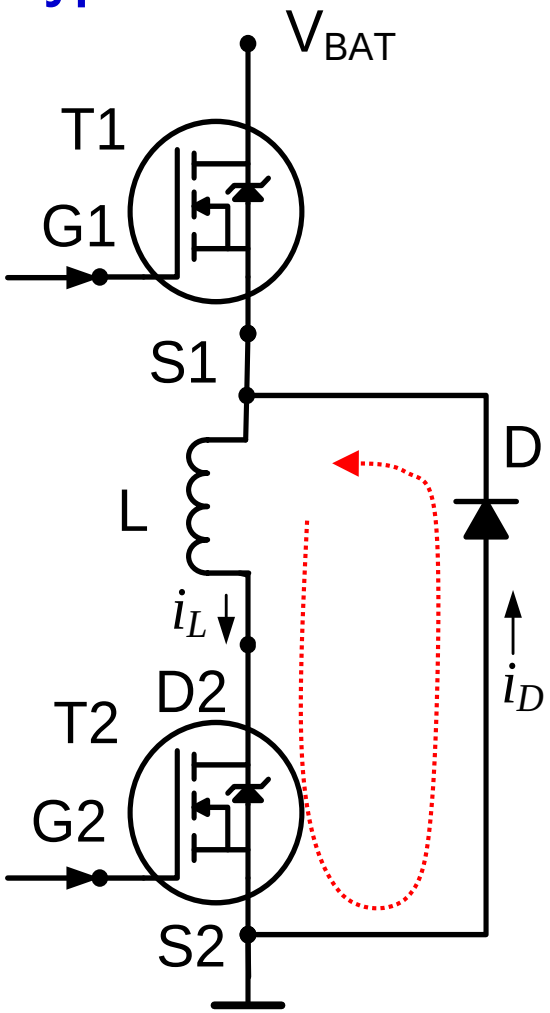
◆阶段2：B→C

V_{GS1}=0, V_{GS2} =0V

$P_L \approx I_L \times 70.7V$;

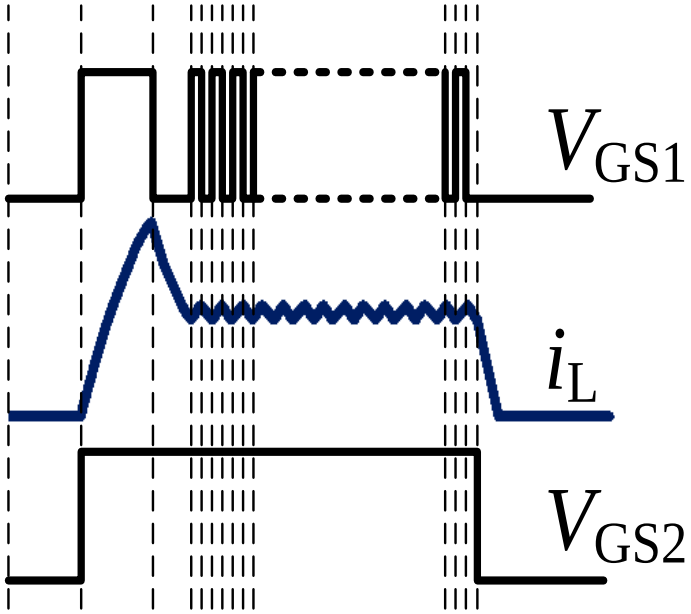


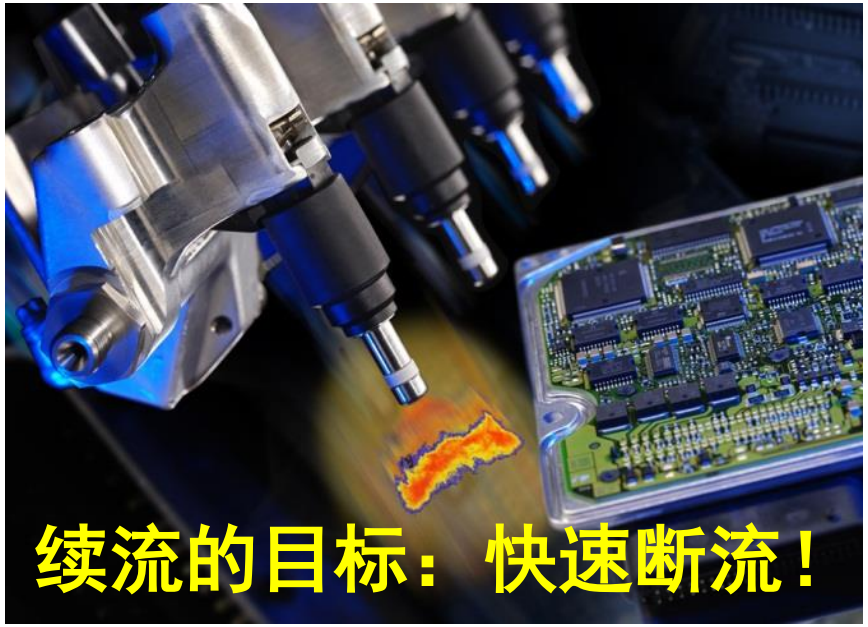




§3驱动电路的续流回路
Freewheel circuit design

- **典型应用：发动机喷油器电磁阀，在喷油结束时迅速释放电磁阀线圈中的能量，使得电磁力迅速减到零，干净利落结束喷油**/ Typical application: Injection valve control of engine. To achieve fast stopping injection, it is necessary to fast release the energy stored in the coil of the valve and make sure the magnetic force could drop down to zero rapidly.



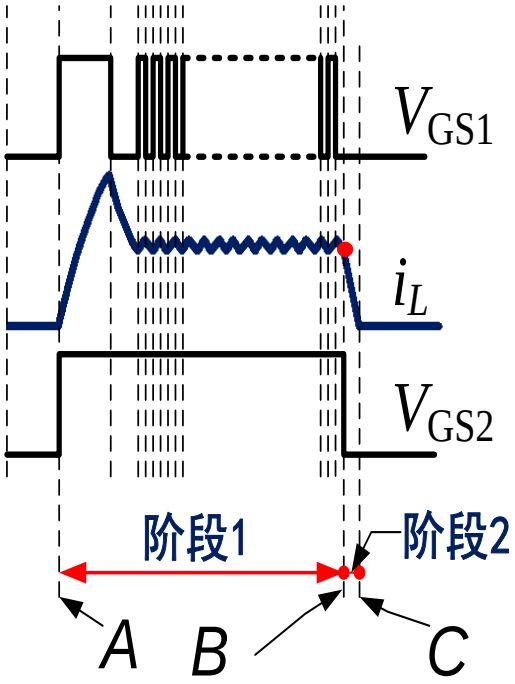


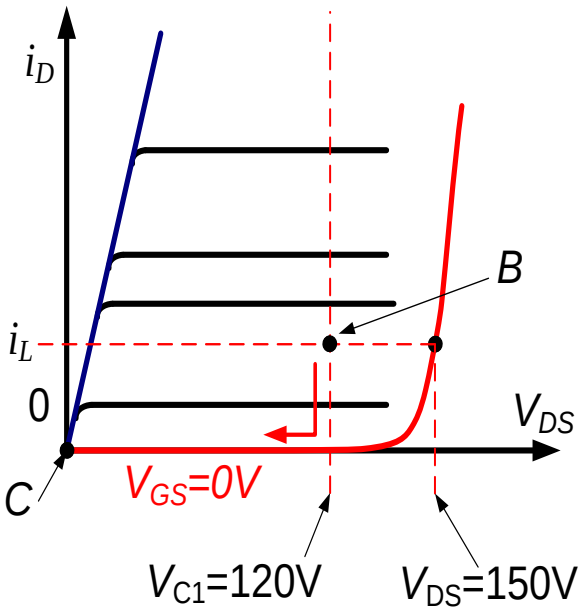
- 能量：执行器→ MOS管，转变为MOS管的发热/ Energy was transfer from actuator and converted into heat by MOSFET.

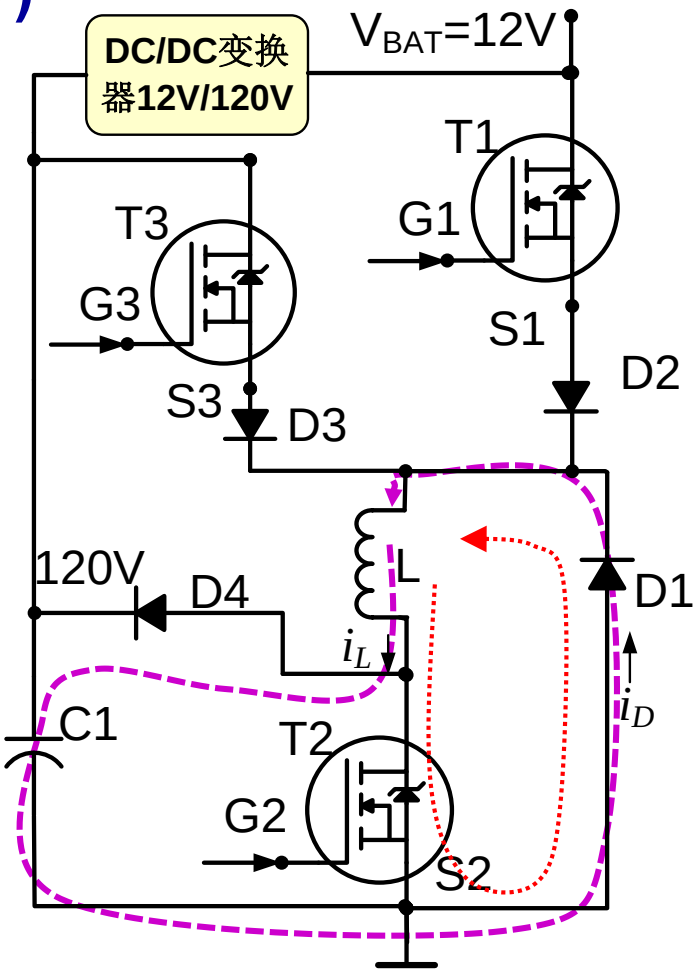
§3驱动电路的续流回路
Freewheel circuit design

续流回路设计3：能量回馈型/ Freewheel circuit type III: Fast energy recovery (前提条件：T2管的V_{DSS} = 150V)

- V_{GS1}=0, V_{GS2} =0



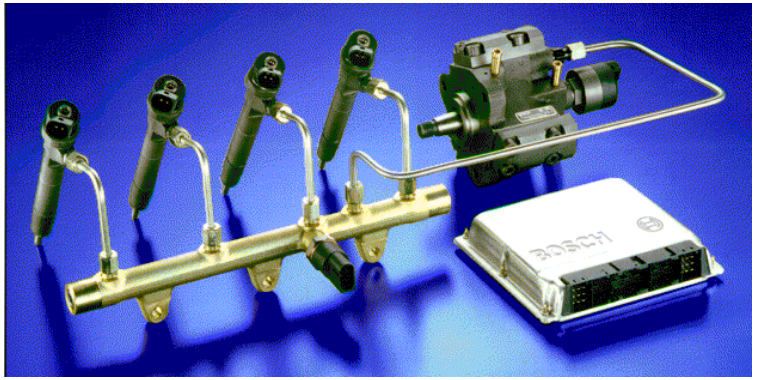


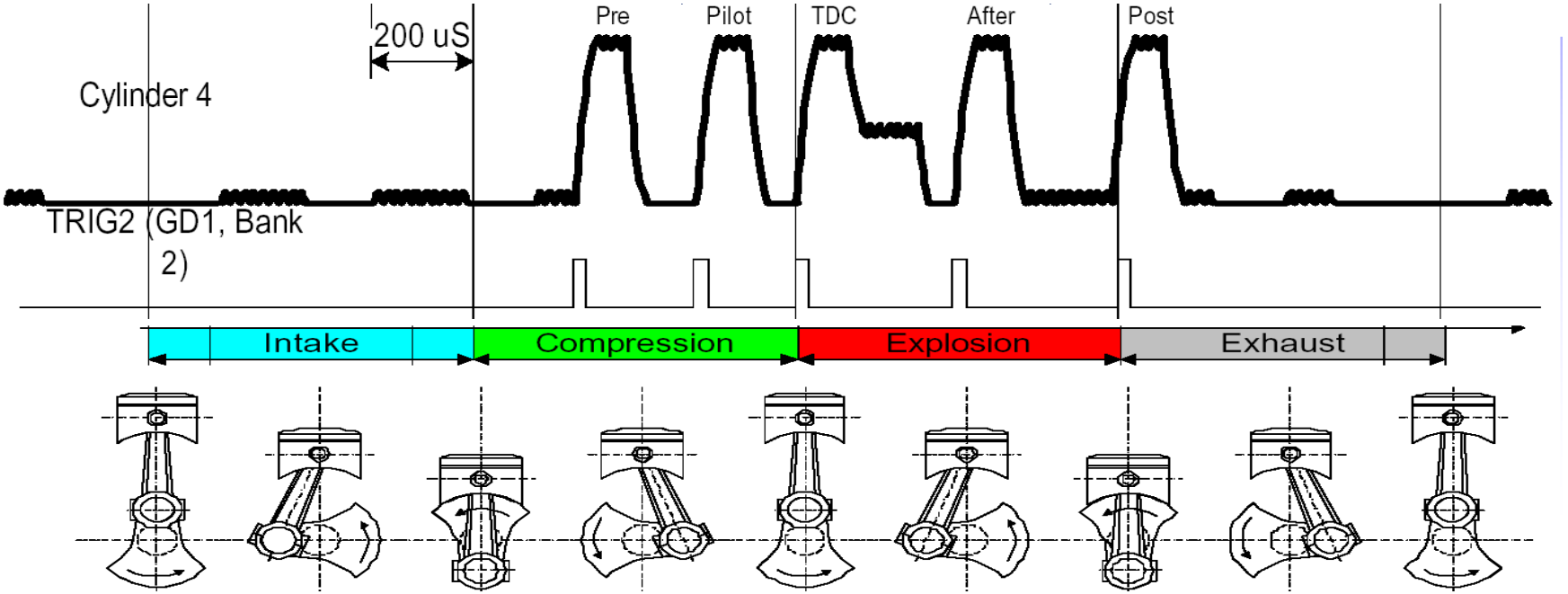


执行器释能功率： $P_L \approx I_L \times 121.4V$;

§3驱动电路的续流回路
Freewheel circuit design

●典型应用：缸内直喷柴油机的高速强力电磁阀驱动！1500Bar的喷射压力，300微秒的响应时间，完成一个缸每个循环5次喷射/Typical application: High speed injection solenoid valve for diesel engines, which response was limited in 300us under 1500 bar injection pressure, 5 times of injections for each cylinder per cycle.





§3驱动电路的续流回路
Freewheel circuit design

设计举例：柴油机高速强力电磁阀的驱动电路设计

是成果“车用柴油发动机新型电控系统及其应用”的核心内容，该成果获得国家技术发明二等奖；

填补了我国柴油机电控的空白，并开始在国民经济和国防领域获得产业化应用。



国家技术发明奖

证 书

为表彰国家技术发明奖获得者，特颁发此证书。

项目名称：车用柴油发动机新型电控系统及其应用

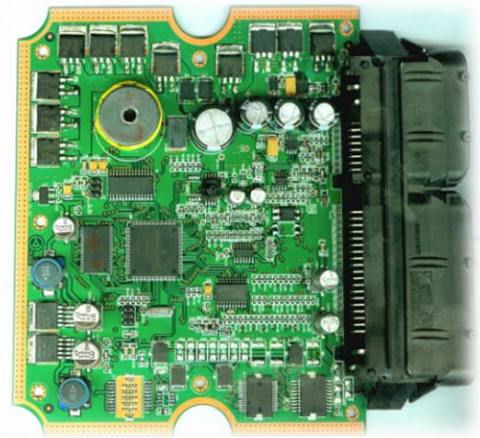
奖励等级：二等

获 奖 者：李建秋(清华大学)

中华人民共和国国务院

2007年12月11日

证书号：2007-F-223-2-01-R02





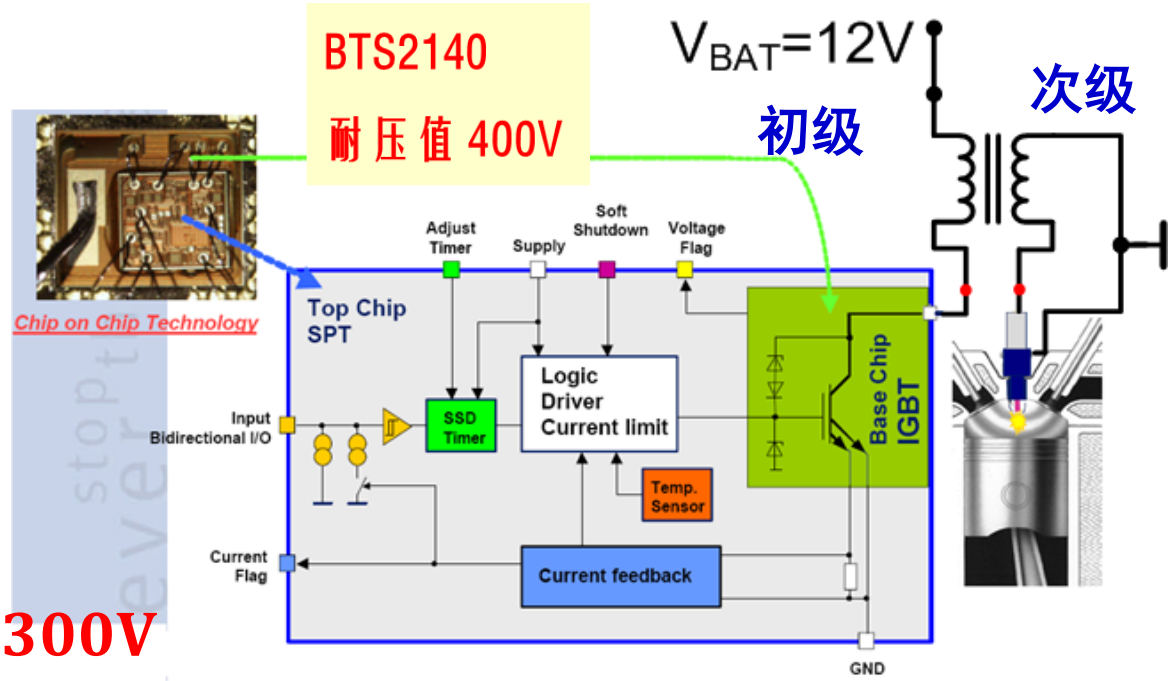
§3驱动电路的续流回路
Freewheel circuit design

续流回路设计4：点火线圈-自感变压器/ Freewheel type IV: the spark plug coil self–inductive transformer.

匝数比(初级:次级)/winding ratio (Pri: Sec) =1:40~1:50

- 初级先导通→线圈储能，然后关断IGBT/The primary was conductive→ coil store energy, then switch off IGBT
- 次级产生超过1万伏的高压，击穿空气，产生火花；The secondary coil generates voltage higher than 10000 V, ionizes the air and generates spark.







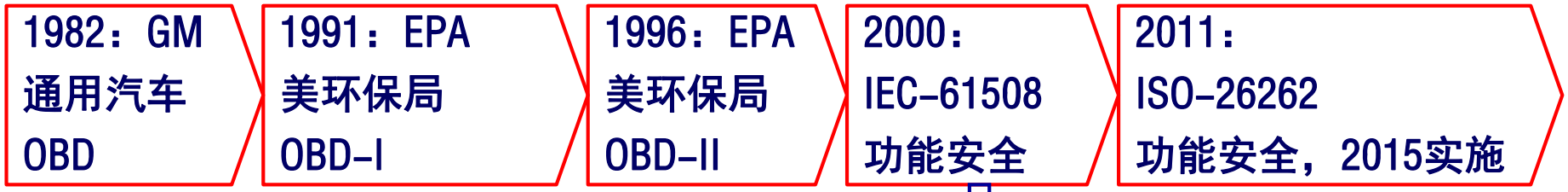
$P_L > 10A \times 300V$

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   ① 执行器及四种功率开关器件（Actuators and power switches）
   ② 驱动信号接口电路：四拓扑结构（Four topologies of drive interface circuits）
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   ⑤ 汽车专用功率驱动集成芯片（Integrated Power Chip）

§4 驱动电路的故障诊断和容错控制
Diagnosis and fault tolerant control for drive circuits

- 为什么要故障诊断/ Why Diagnosis?



- 故障和容错的层次
  - ① 诊断Diagnosis
  - ② 容错Fault Tolerant
  - ③ 安全Fail Safe

对整个系统的故障检测的覆盖率>99%

- 如何诊断故障/ How to diagnosis?
  - ■温度检测，温度过高/ Temperature detection： Over temperature
  - ■电流检测，过流，过载/ Current detection: Over current, Over load
  - ■电压检测，过压，对电源或对地短路，开路/Voltage detection: Over voltage, short circuit, open circuit.

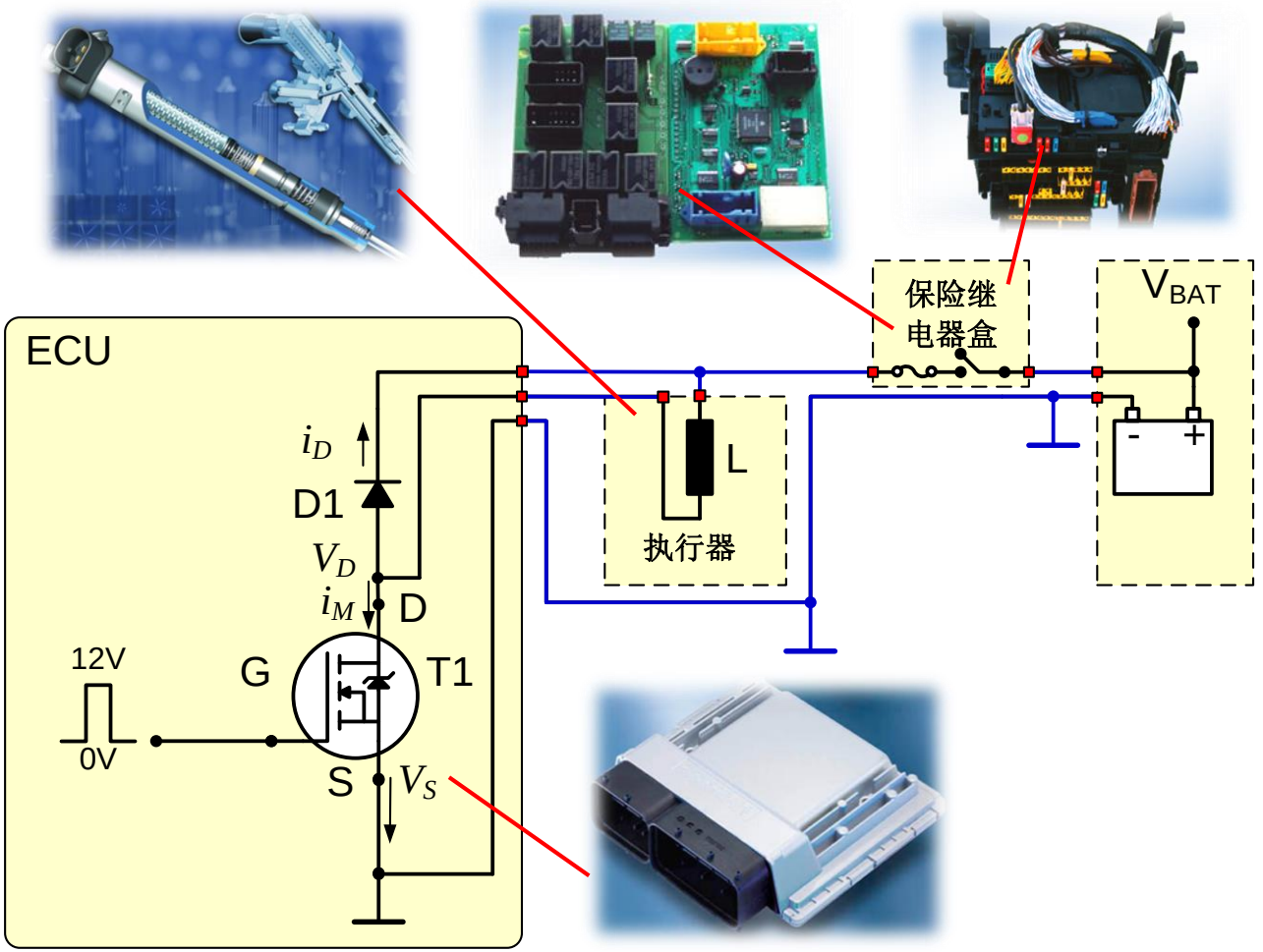
§4 驱动电路的故障诊断和容错控制
Diagnosis and fault tolerant control for drive circuits

- 低边驱动的故障诊断方法/ Diagnosis method for low side drivers
  - 利用电压电流信号进行故障诊断/ Diagnosis by using current and voltage signals

①if $V_{GS} = 12V$, $V_D = V_{BAT}$,
对电源短路！

②if $V_{GS} = 0V$, $V_D = 0V$,
对地短路！

③if $V_{GS} = 12V$, $V_D \equiv 0V$, $I_M = 0A$
开路！



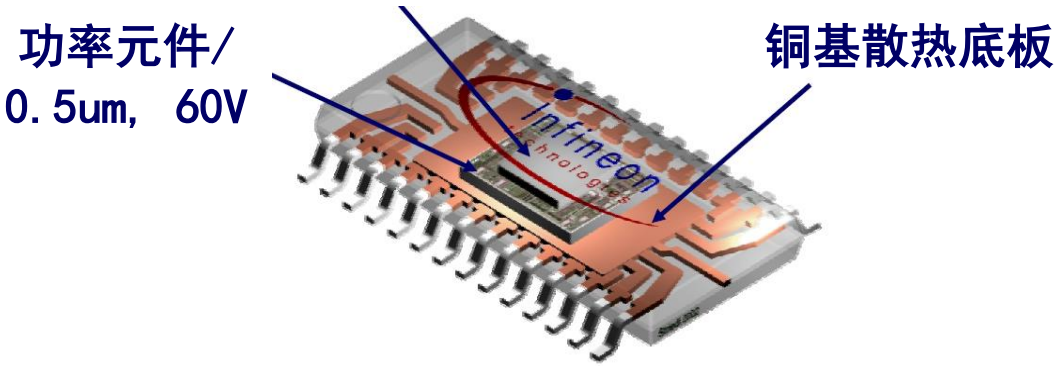
- 低边驱动的故障诊断方法/ Diagnosis method for low side drivers
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对电源短路！

②if $V_{GS} = 0V$, $V_D = 0V$,
对地短路！

③if $V_{GS} = 12V$, $V_D \equiv 0V$, $I_D = 0A$
开路！

正常控制、故障诊断和容错控制逻辑，先进CMOS逻辑电路0.22um/0.13um，包括Flash/EEPROM

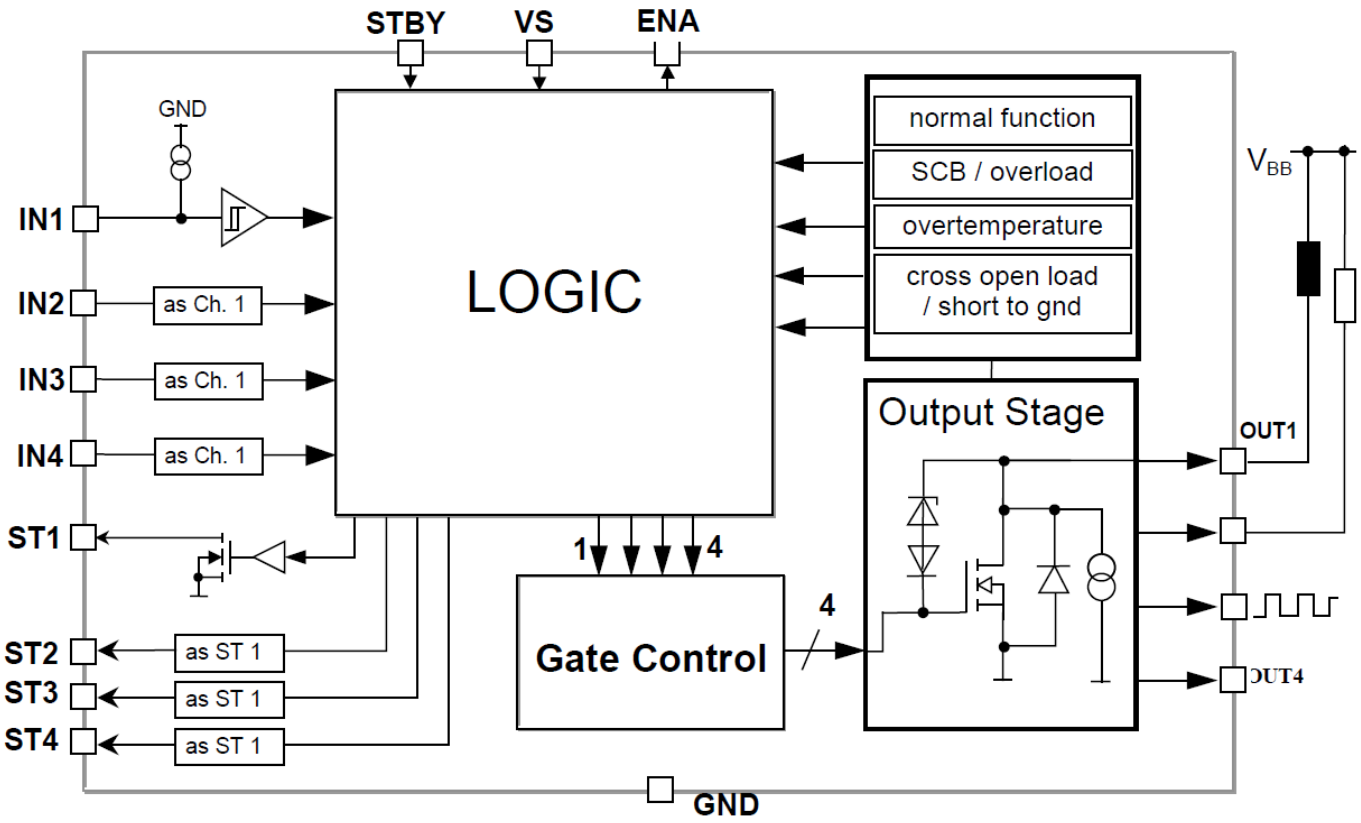


多路集成驱动芯片/Multi-channel integrate drive circuit

§4 驱动电路的故障诊断和容错控制
Diagnosis and fault tolerant control for drive circuits

- 集成驱动芯片TLE-6216的故障诊断
  - ◆特点：并行输入，并行反馈，4 路低边



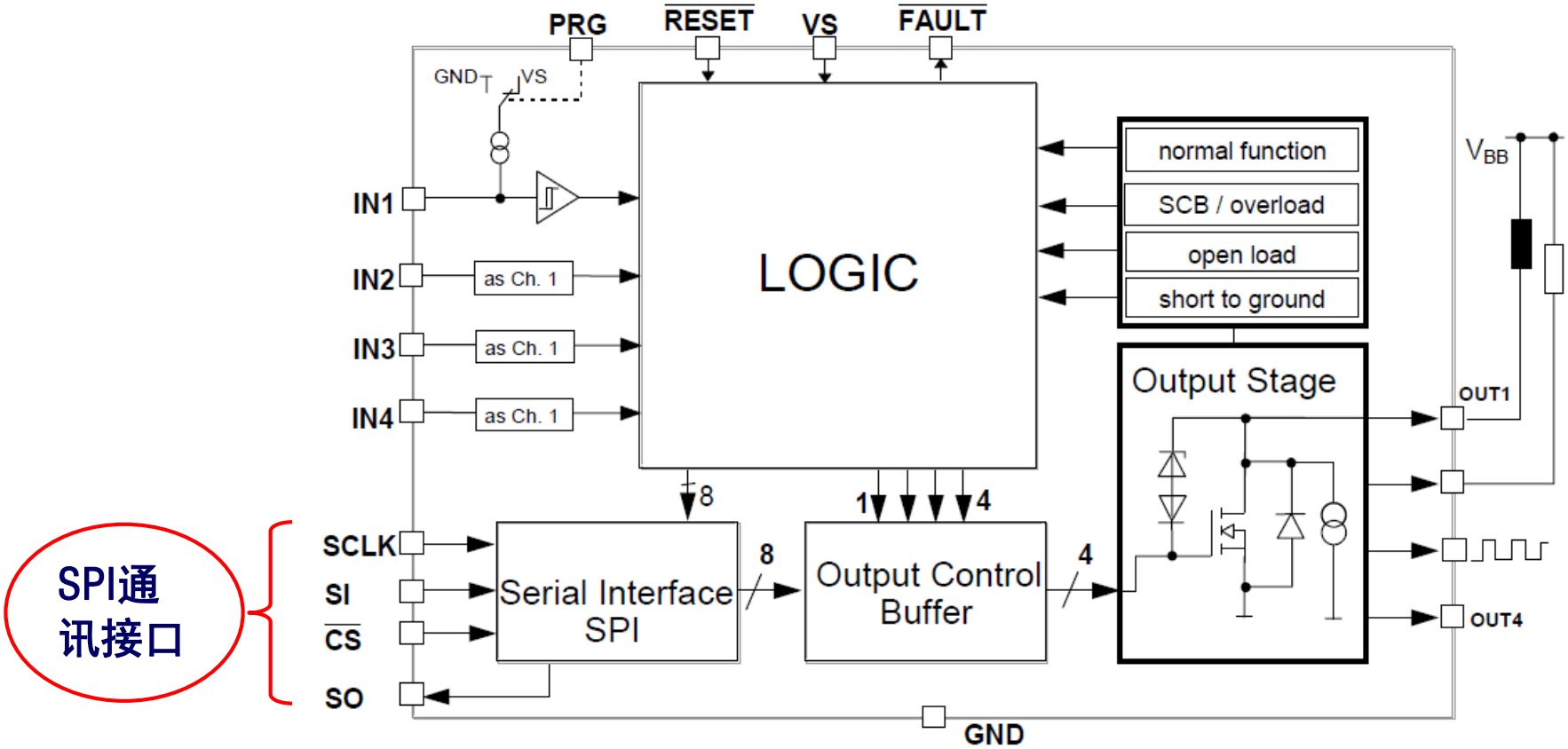


- ●诊断和容错功能
  - ◆ 过载关断
  - ◆ 过热关断
  - ◆ 过压保护
  - ◆ 短路保护
  - ◆ 休眠模式

§4 驱动电路的故障诊断和容错控制
Diagnosis and fault tolerant control for drive circuits

- 集成驱动芯片-TLE6220（4路低边驱动芯片）
  - 并行输入，SPI串行输入和SPI串行反馈，
  - 通过SPI接口返回故障码；





- 集成驱动芯片-TLE6220（4路低边驱动芯片）
  - 并行输入，SPI串行输入和SPI串行反馈，
  - 通过SPI接口返回故障码；



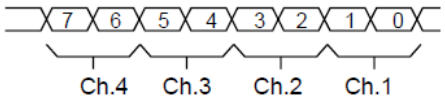
MSB LSB
CCCC DDDD : Serial input byte
Control Bits Data Bits

Five specific control words are recognised, having the following functions:

No.	Serial Input Byte	Function
1	LLLL XXXX	Only 'Full Diagnosis' performed. No change to output states.
2	HHLL XXXX	State of four parallel inputs and '1-bit Diagnosis' outputted.
3	HLHL XXXX	Echo-function of SPI; SI direct connected to SO
4	LLHH DDDD	IN1...4 and serial data bits 'OR'ed. 'Full Diagnosis' performed.
5	HHHH DDDD	IN1...4 and serial data bits 'AND'ed. 'Full Diagnosis' performed.

Note: 'X' means 'don't care', because this bit will be ignored
'D' represents the data bit, either being H (=ON) or L (=OFF)

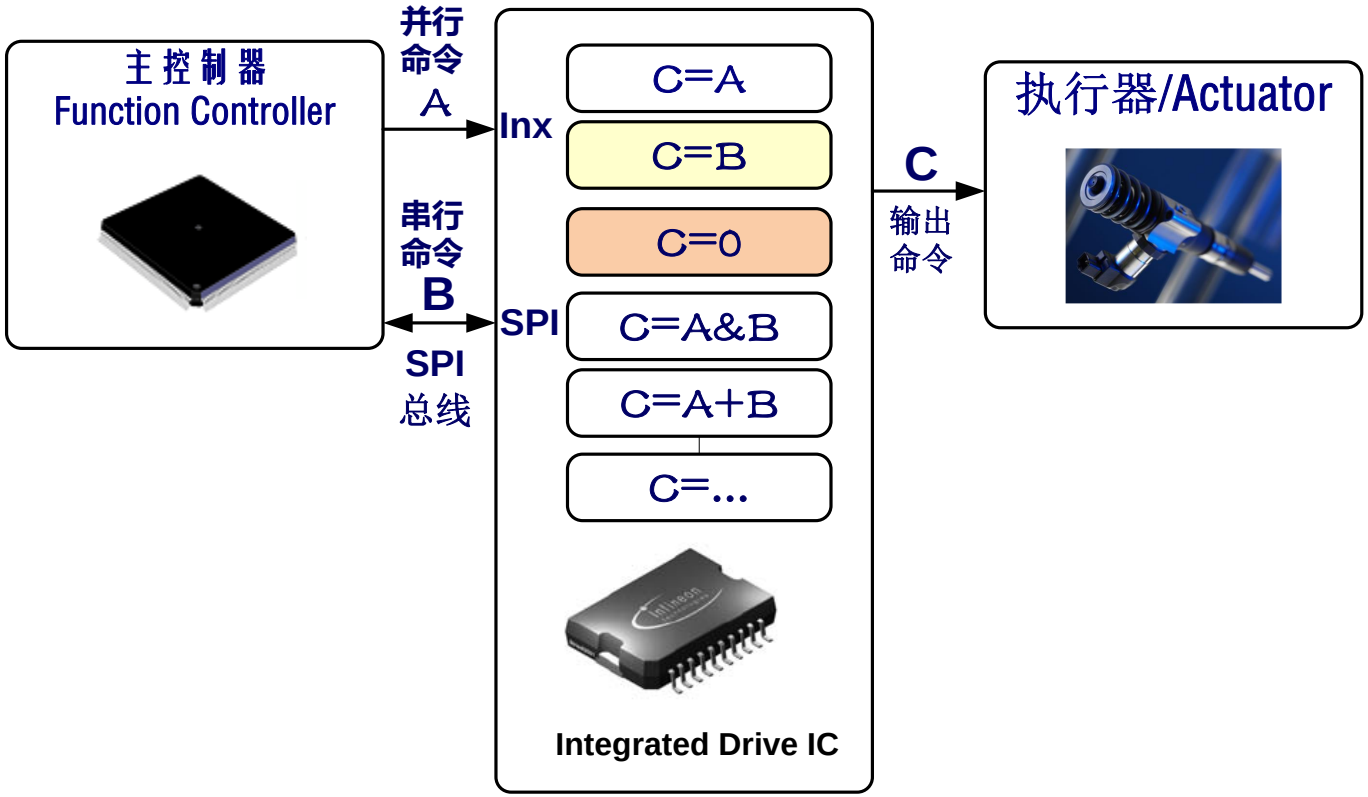
Diagnostic Serial OUT (SO)



HH	Normal function
HL	Overload, Shorted Load or Overtemperature
LH	Open Load
LL	Shorted to Ground

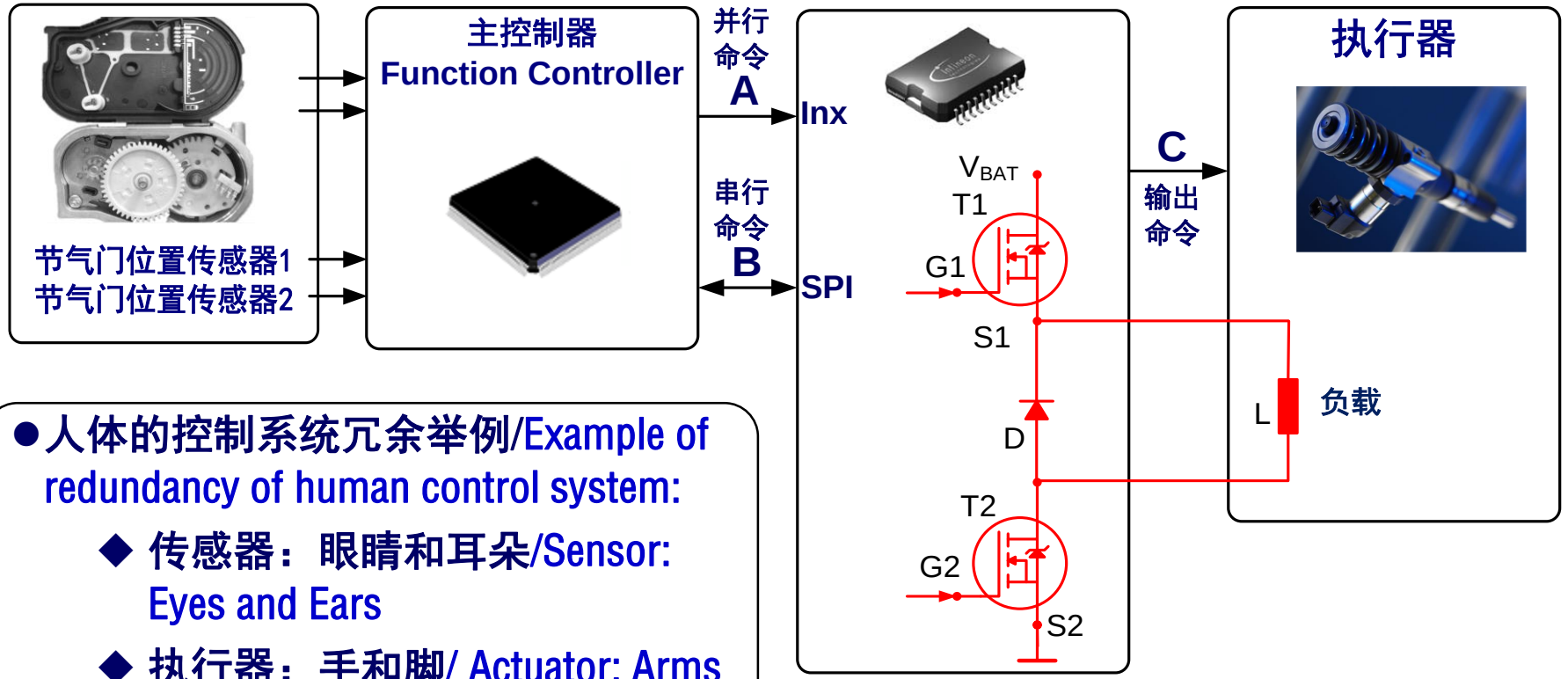
●容错控制逻辑/Logic of fault tolerant control

- ◆输出可配置/ Configurable output
- ◆输出通道冗余/ Redundancy output channel



§4 驱动电路的故障诊断和容错控制
Diagnosis and fault tolerant control for drive circuits

- 容错控制的主要措施：冗余通道配置/ Channel Redundancy
  - 输入信息冗余/Input channel redundancy→ Throttle Position Sensor 1 &2
  - 输出信息通道冗余/Output channel redundancy → Parallel and Serial
  - 输出功率器件冗余/ Power switch redundancy→ Half bridge Drive circuit

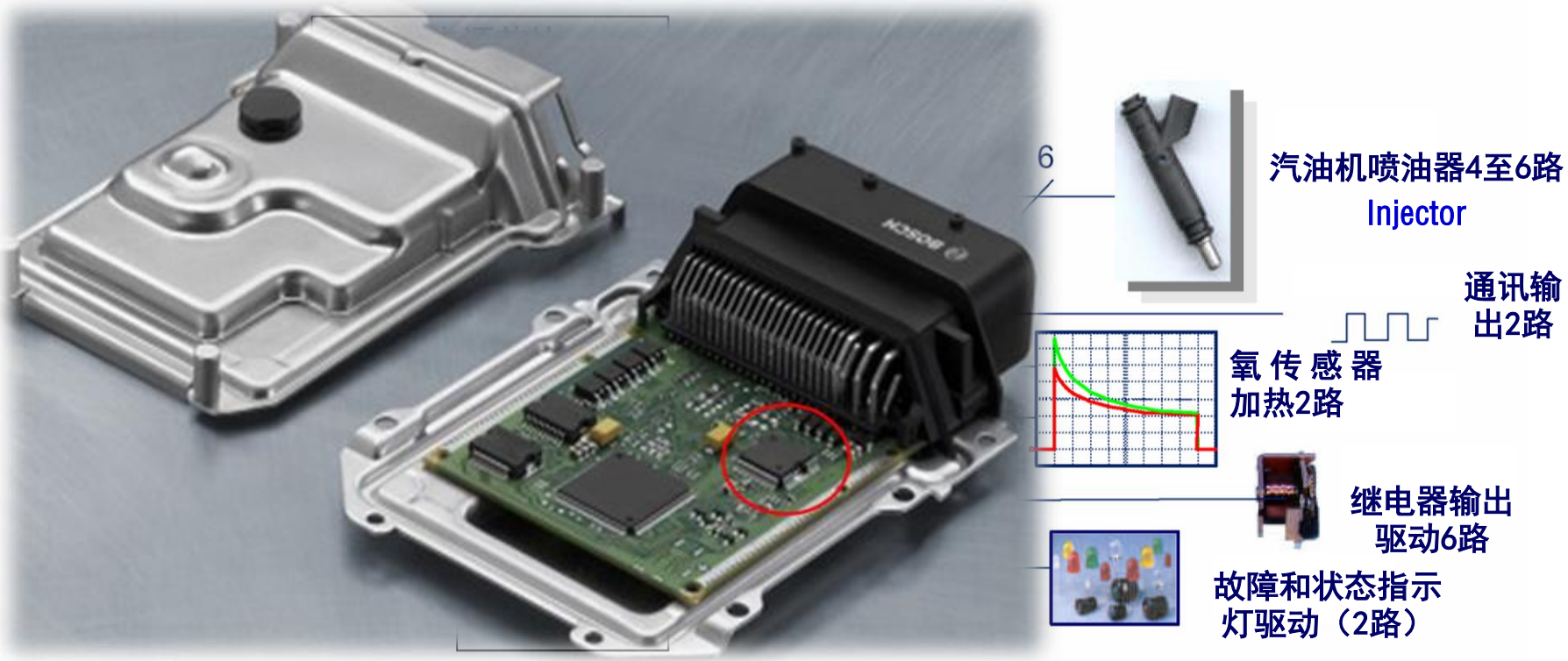


- 人体的控制系统冗余举例/Example of redundancy of human control system:
  - 传感器：眼睛和耳朵/Sensor: Eyes and Ears
  - 执行器：手和脚/ Actuator: Arms and legs

●应用举例：汽油机电子控制系统/Example of application: Gasoline control system

◆特点：并行输入，串行输入和反馈， 18路低边—TLE6244

Features: Parallel input, Serial inputs and feedback, 18 channels



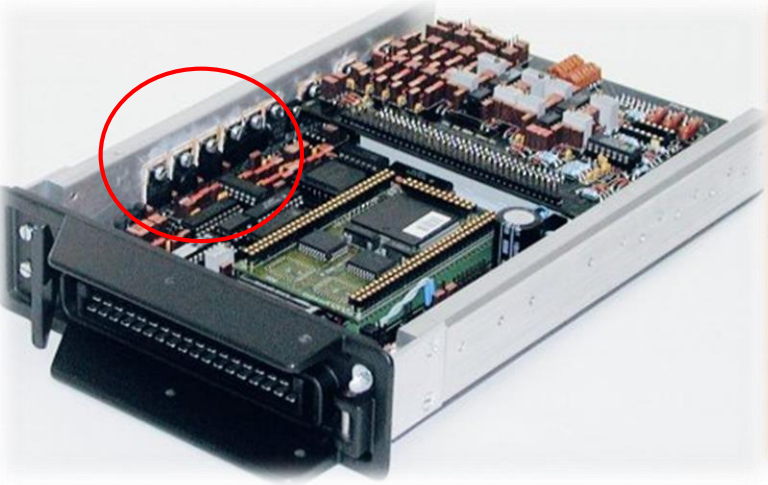
§4 驱动电路的故障诊断和容错控制
Diagnosis and fault tolerant control for drive circuits

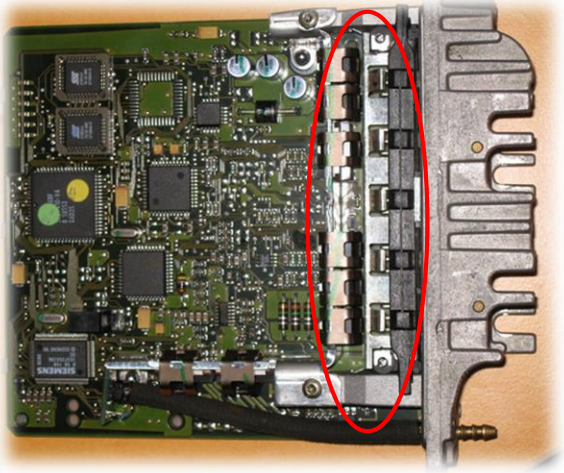
ECU已经历了3个发展阶段/ The design of ECU had been passed 3 phases:

① 在集成驱动芯片产生以前，利用分离元器件焊接在PCB上集成功率驱动和诊断电路/Directly Weld the discrete power components onto the PCB before generation of Integrate Drive ICs.

----1990s以前，ECU体积大，功能少，散热效果差，可靠性低；

Before 1990s, ECU has large volume, less functions, lower heat exchange effect and lower reliability.





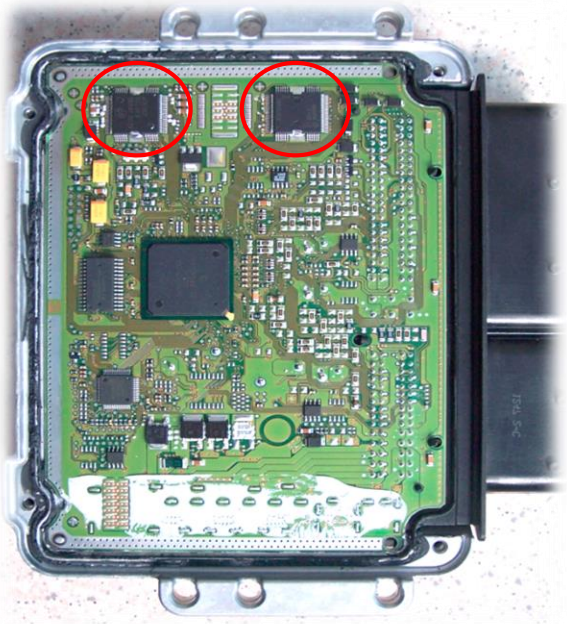


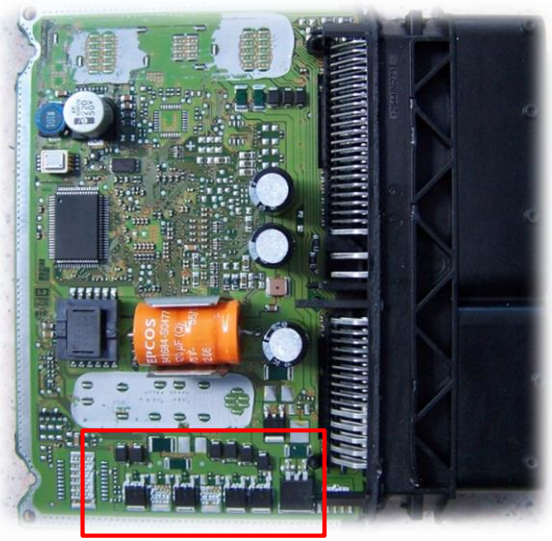
② 在专用芯片上集成（SOC），封装好后焊接到PCB上/ Function was integrated and packaged onto Chips (System on Chip, SOC), and then was wold onto the PCB board.

----1995年左右，专用的集成功率芯片，大大降低了器件的数量，缩小了ECU的体积，降低了成本，提高了性能和可靠性。

About 1995, this Integrate drive chip domestically decreased the number of components thus achieved less volume, lower cost and higher performance and reliability.



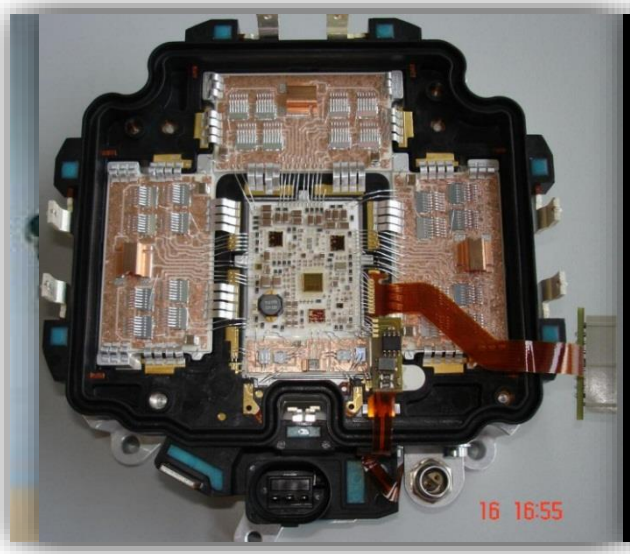


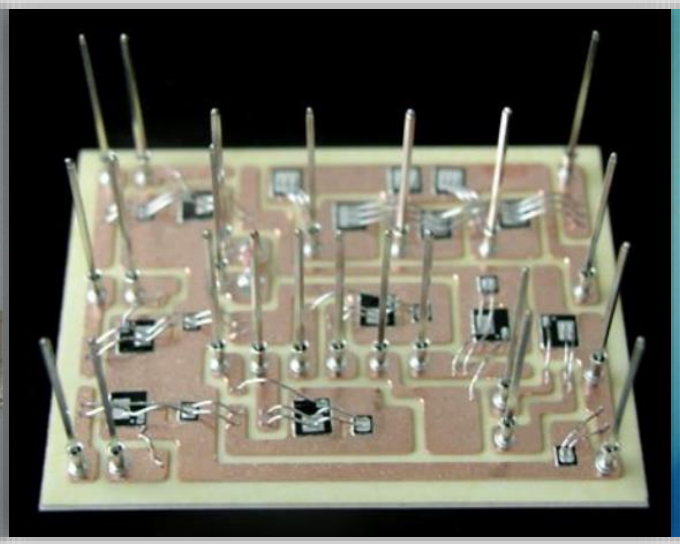


③ 无封装的集成芯片在PCB上形成混合电路/Integrate the drive circuit (die) directly weld onto the PCB board without packages.

-----2005前后开始，功率驱动和逻辑芯片均不带封装，进一步缩小了体积，降低成本，开始在高端汽车电子控制系统中应用。

About 2005, Both the power circuit and logic chips are direct welding onto the PCB board. It further decreased the volume and cost, and started apply in high level automotive control systems.

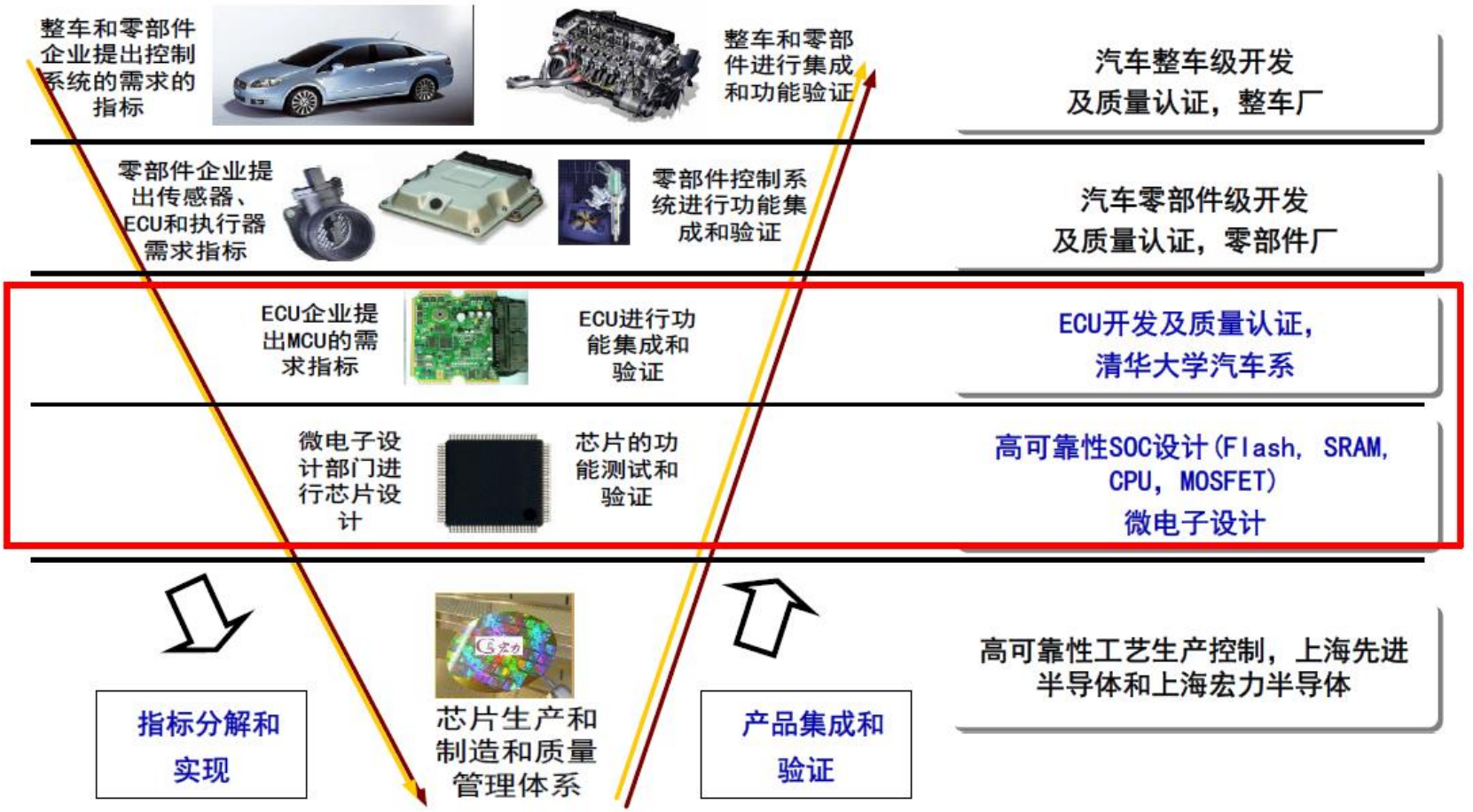






●汽车电子芯片的设计

科技部国家重大科技专项（02）专项“汽车电子芯片工艺研发及产业化”，2012-2015，推动国产专用芯片的研发和产业化！



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   - ④ 驱动电路的故障诊断和容错控制（Diagnosis and fault tolerant control for drive circuits）
   - ⑤ 汽车专用功率驱动集成芯片（Integrated Power Chip）

●功率驱动芯片的散热设计/ Power dissipation of drive circuit

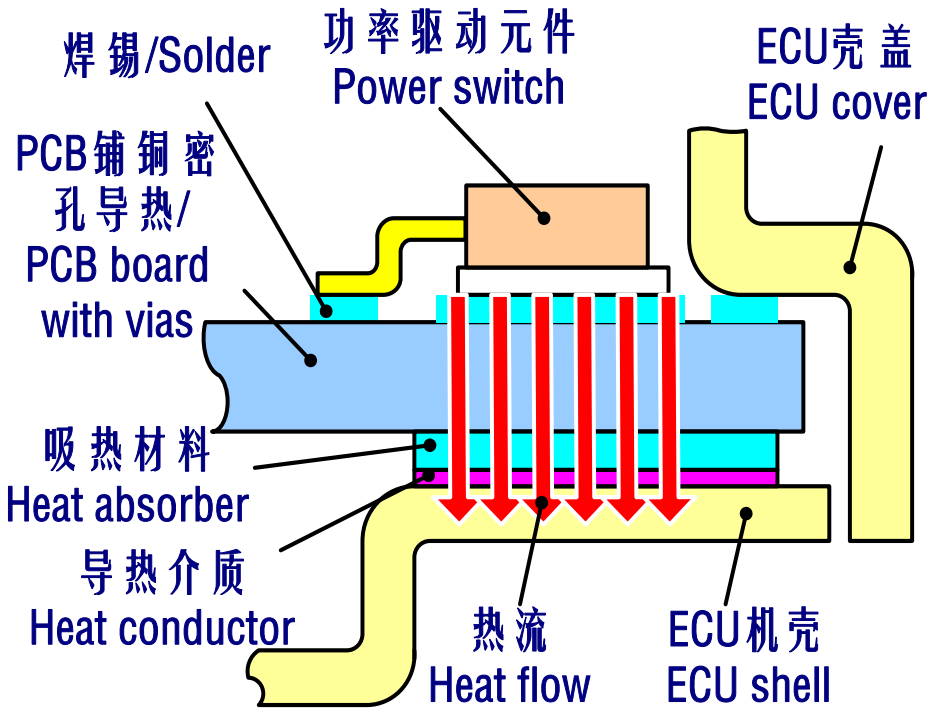
◆ 器件温升和热阻的关系/ Relationship of temperature rise and thermal resistance

$$\Delta T = P_{loss} \times R_{heat}$$

◆ 器件温度和寿命的关系 /Relationship of temperature and lifetime

$$\ln\theta = a + b/T$$

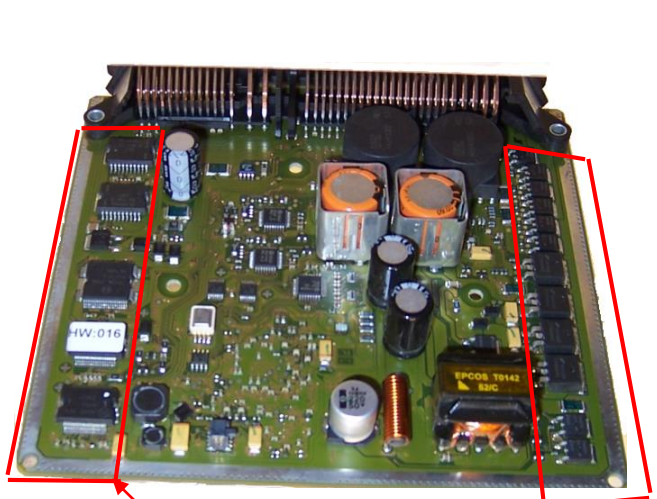
$$\theta = a' e^{b/T}$$

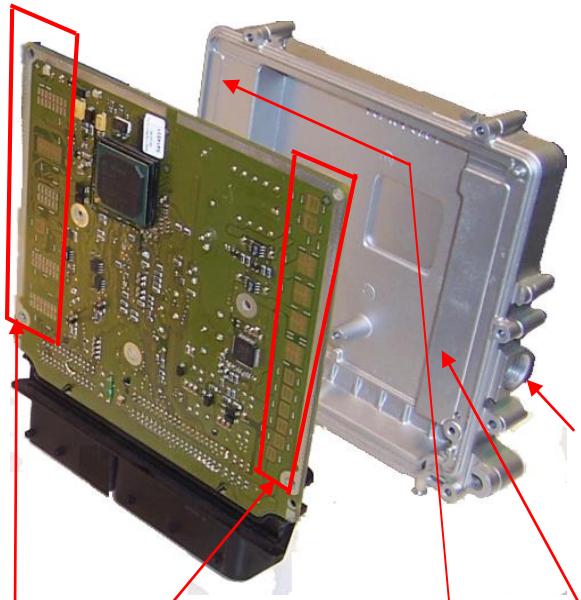


§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

●功率驱动芯片的散热设计/ Power dissipation of drive circuit

◆ECU的散热设计/ Heat release deign of ECU





PCB正面的功率驱动器件，布置在PCB两侧/Toplayer of PCB, the power switches

PCB板上密集的过孔，可将热量从PCB正面传递到背面/The heat is transmitted from toplayer to bottom layer through the high density vias

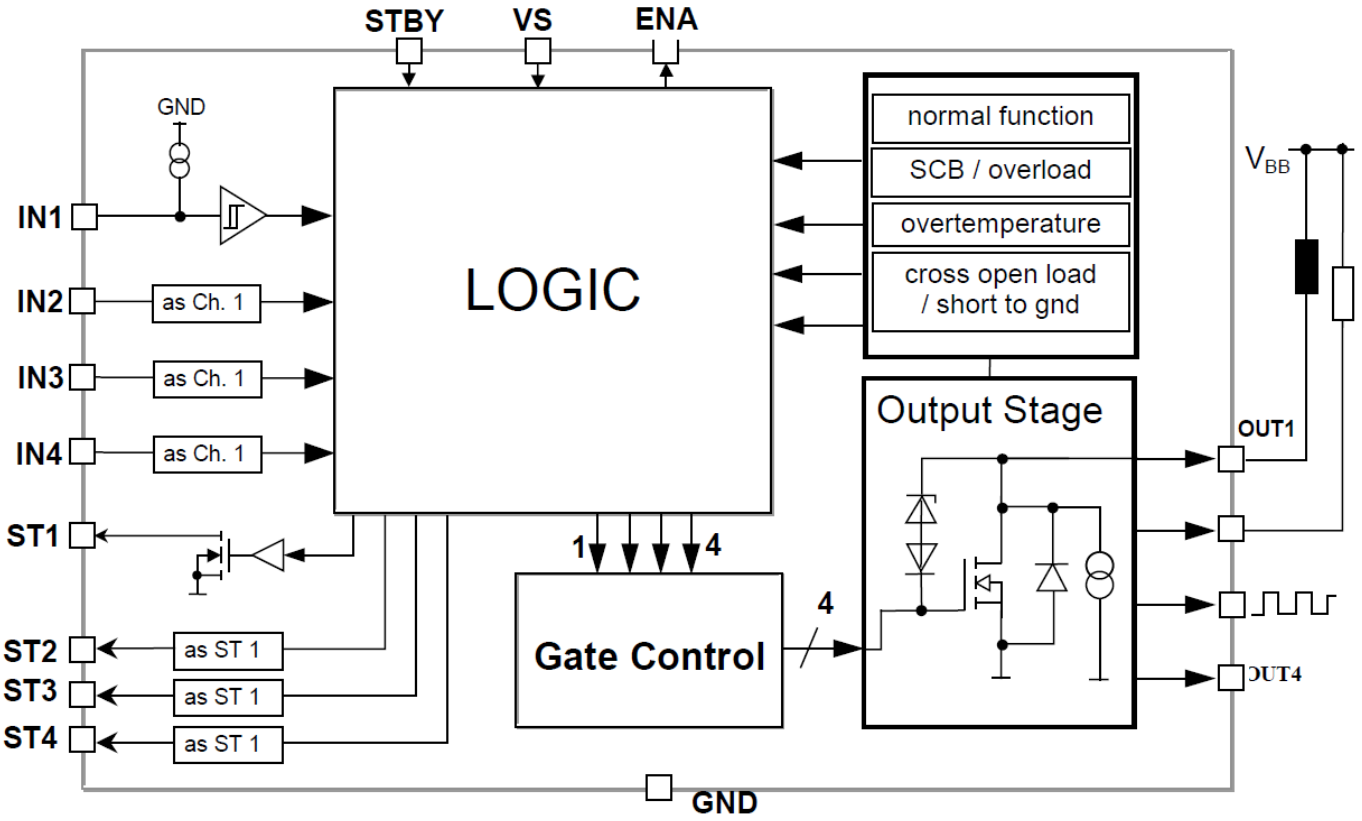
机壳上对应的散热区域，机壳紧贴PCB背面/The area of heat exchanger on the shell, which contacts with bottom layer of PCB

§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

- 集成驱动芯片1/ Integrated Smart MOS–TLE6216
  - ◆特点：并行输入，并行反馈，4 路低边
  - ◆Features：Parallel input and feedback，4XLS



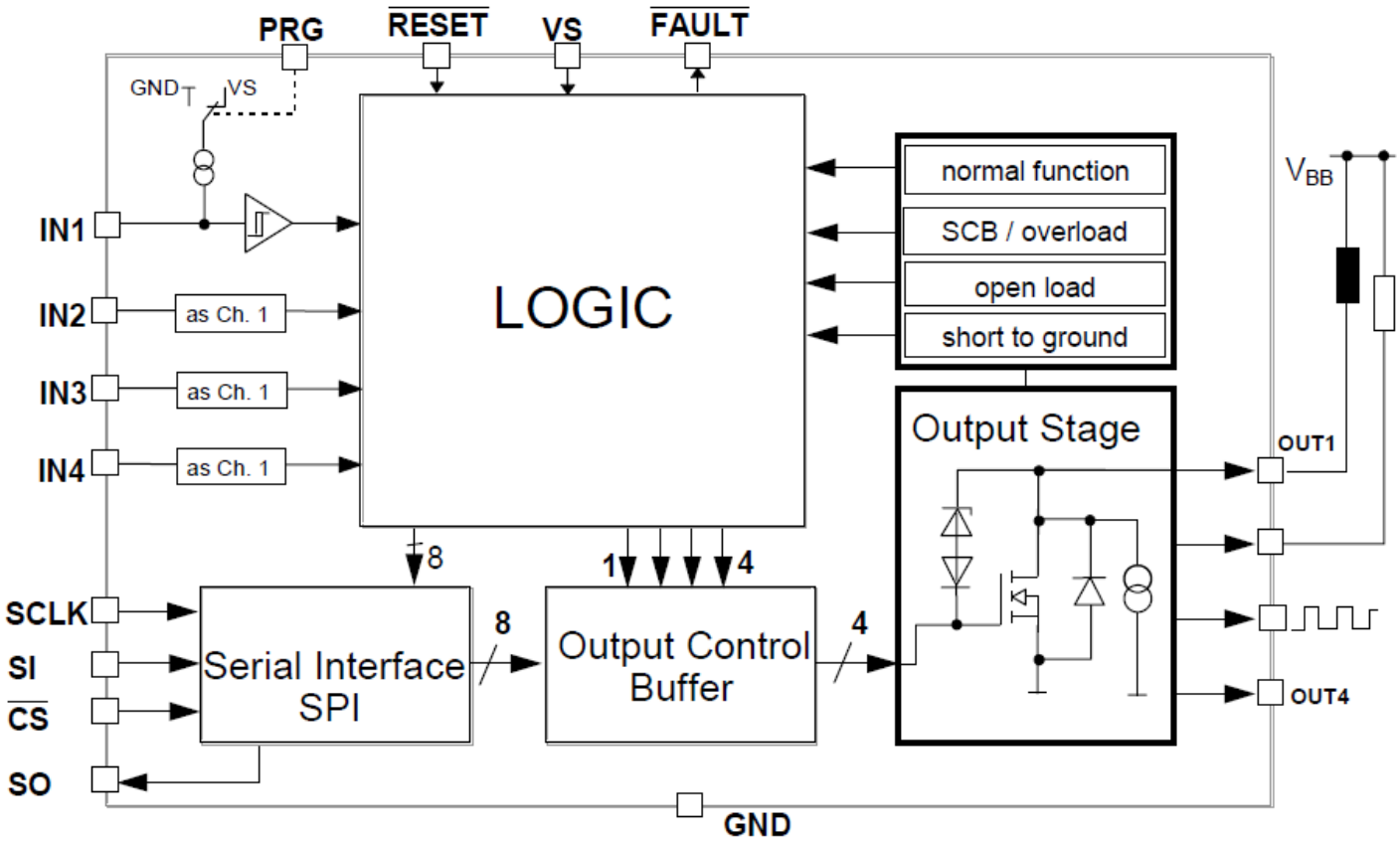
- 过载关断/Overload shutdown
- 过热关断/ Thermal shutdown
- 过压保护/ Over–voltage protection
- 短路保护/ Shorted circuit protection
- 休眠模式/Standby mode



§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

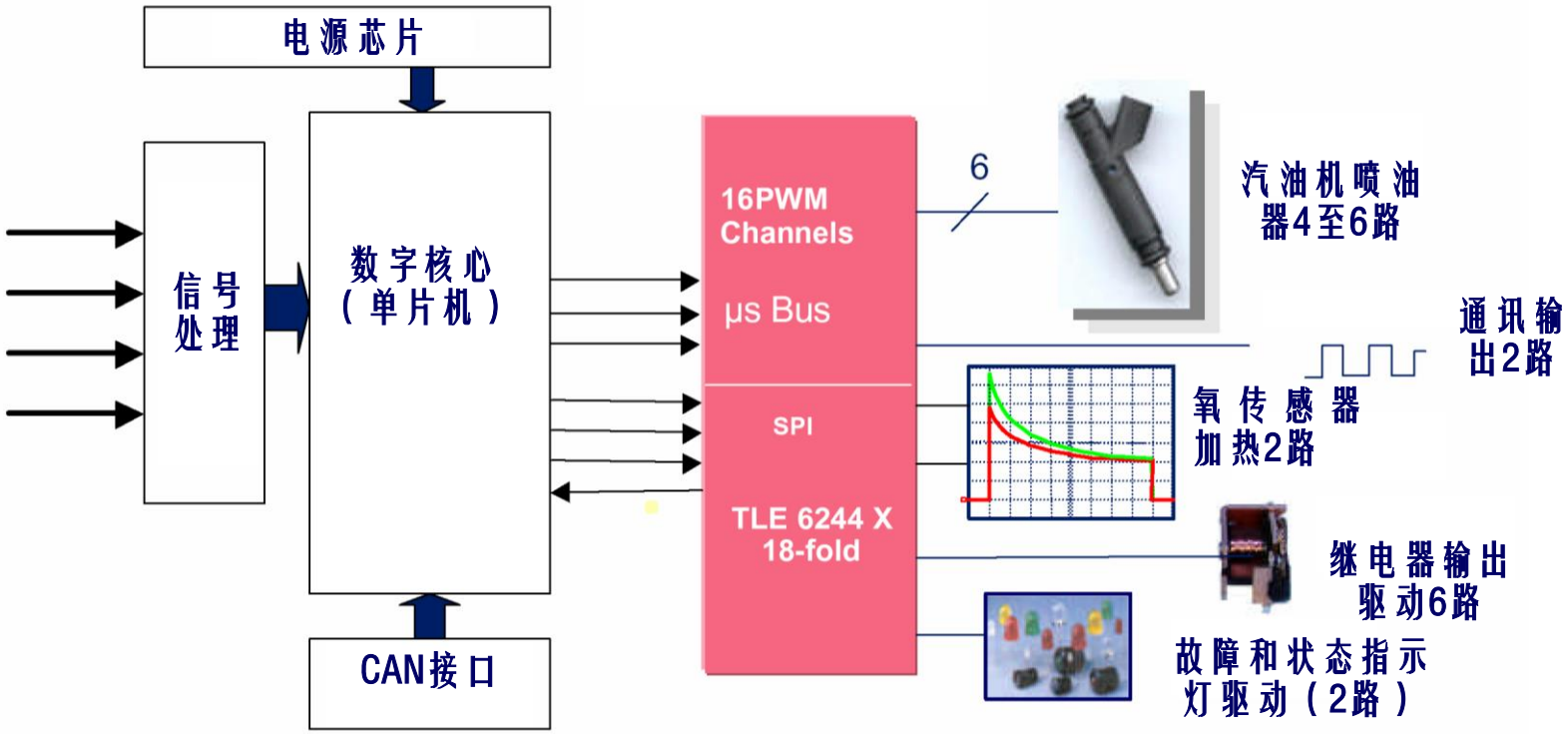
- 集成驱动芯片2/ Integrated Smart MOS–TLE6220
  - 特点：并行输入，串行输入和反馈， 4 路低边
  - 每路可以单独返回故障码；



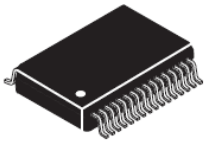


§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

- 集成驱动芯片3/ Integrated Smart MOS–TLE6244x
  - ◆特点：并行输入，串行输入和反馈， 18路低边
  - ◆Features： Parallel input, serial input and feedback



●集成驱动芯片4/ Integrated Smart MOS–MC33996



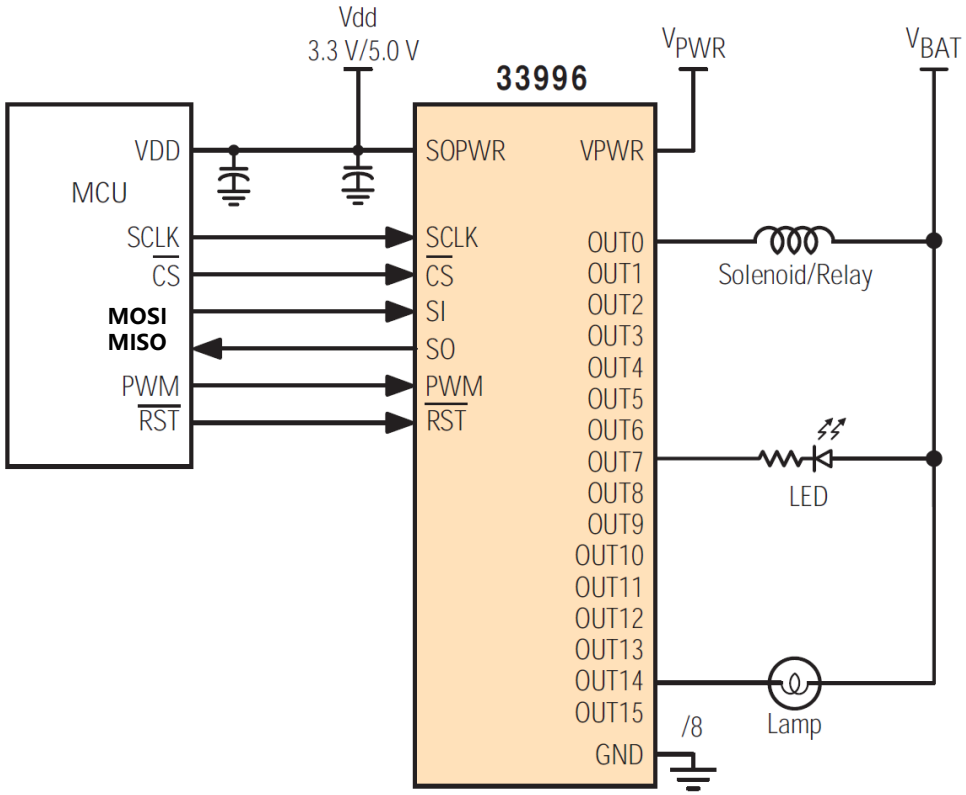
◆特点：串行输入和反馈， 16路低边

◆Features：Serial input and feedback

DESCRIPTION

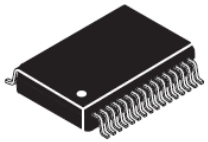
The 33996 is a 16-output low-side switch with a 24-bit serial input control. It is designed for a variety of applications including inductive, incandescent, and LED loads. The Serial Peripheral Interface (SPI) provides both input control and diagnostic readout. A Pulse Width Modulation (PWM) control input is provided for pulse width modulation of multiple outputs at the same duty cycle. A dedicated reset input provides the ability to clear all internal registers and turn all outputs off.

The 33996 directly interfaces with micro controllers and is compatible with both 3.3 V and 5.0 V CMOS logic levels. The 33996, in effect, serves as a bus expander and buffer with fault management features that reduce the MCU's fault management burden.



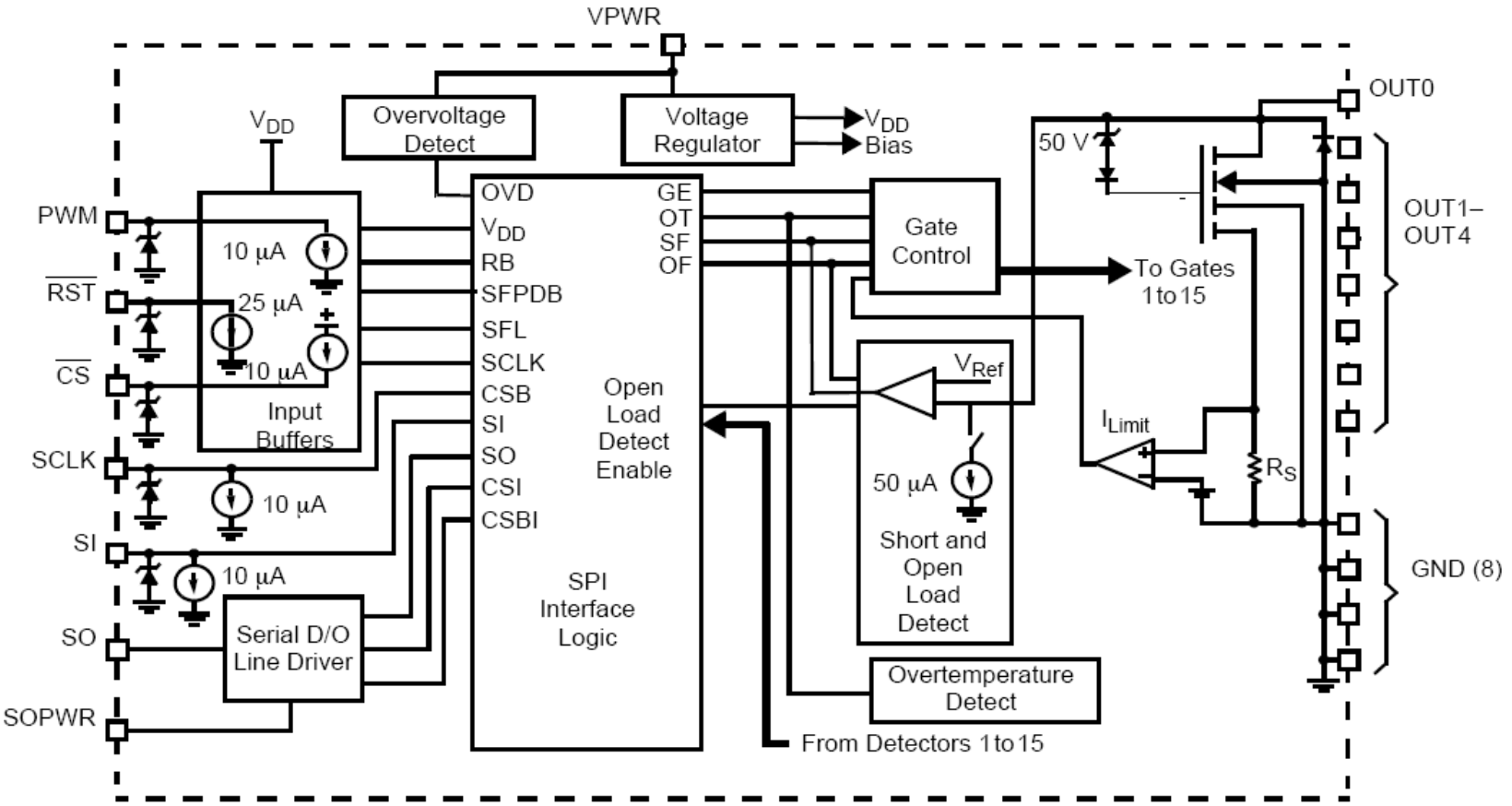
§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

●集成驱动芯片4/ Integrated Smart MOS–MC33996



◆特点：串行输入和反馈， 16路低边

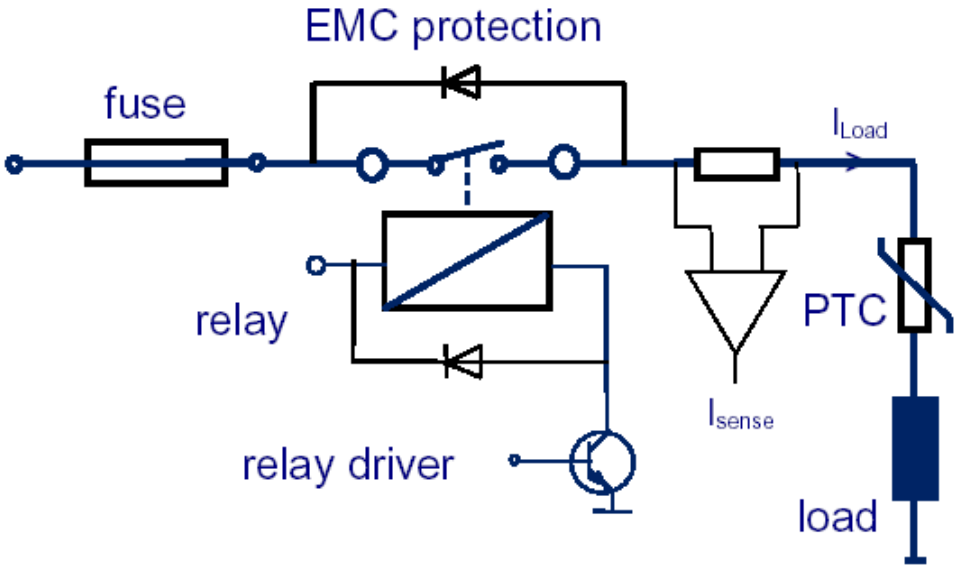
◆Features：Serial input and feedback



●集成驱动芯片5/ Integrated Smart MOS–BTS6144

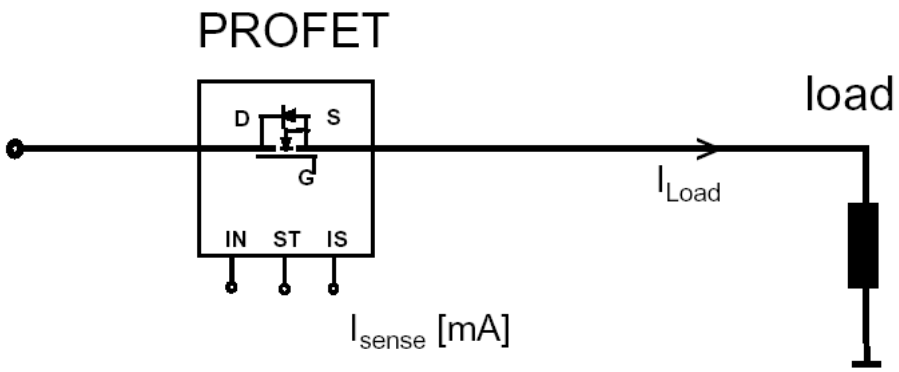
特点：

过载保护
限制电流
短路保护
过温保护
过压保护
负压钳位
反向低阻
电流反馈诊断
电流开路诊断
电源掉电保护
静电保护





+ *electronic fuse function*
+ *diagnostic / current sense*
+ *fully protection*
+ *PWM possibility*
+ *less wires and connectors*

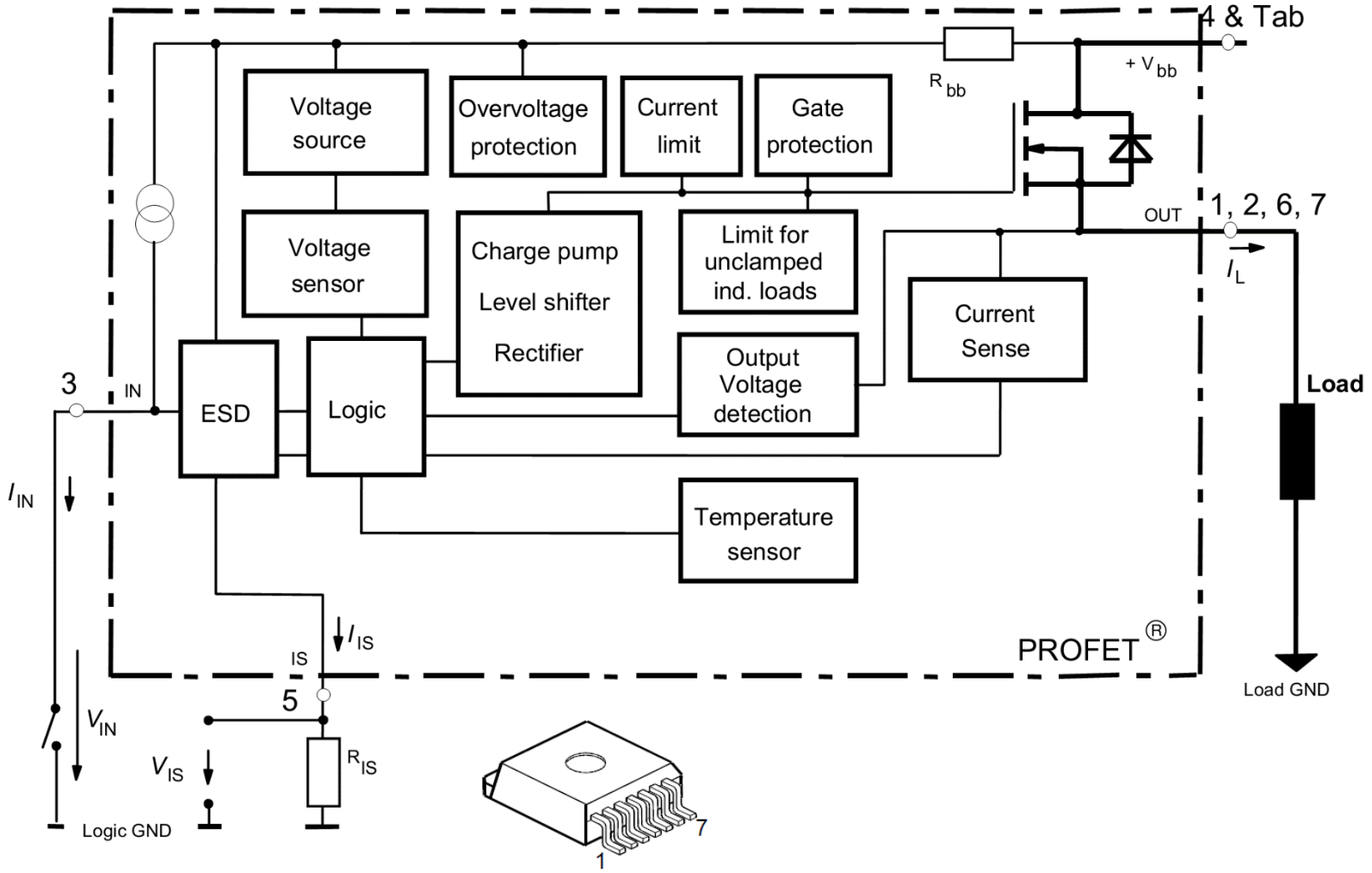


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§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

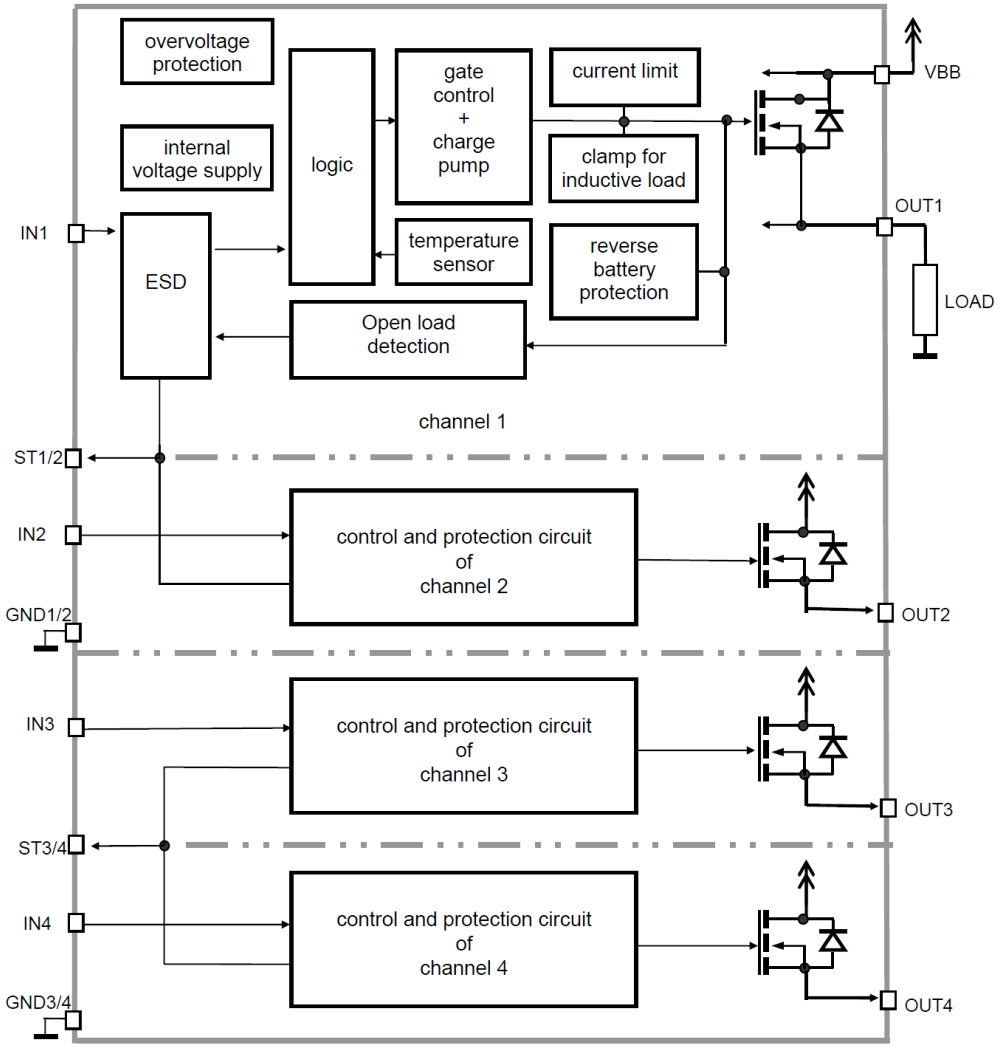
●集成驱动芯片5/ Integrated Smart MOS–BTS6144



§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

●集成驱动芯片6/ Integrated Smart MOS–BTS724





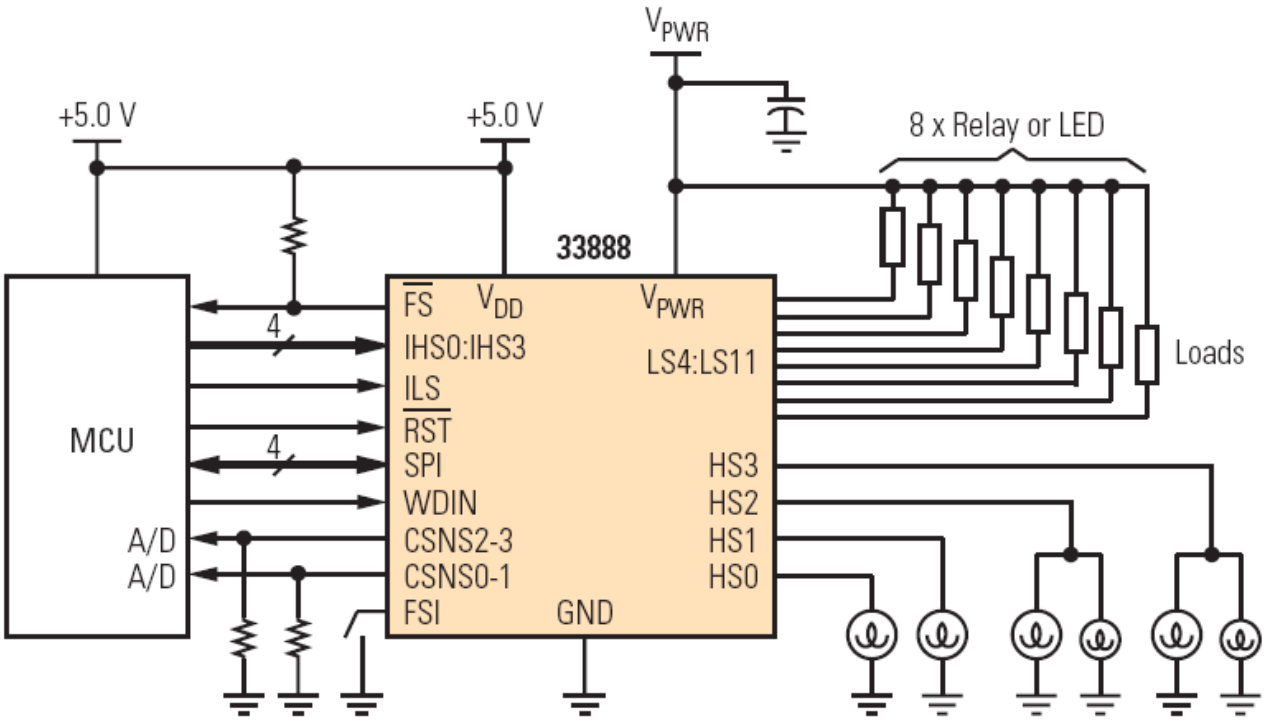
并行4路高边驱动

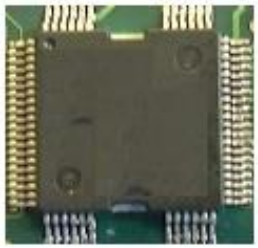
/ Parallel 4 HS

§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

●集成驱动芯片7/ Integrated Smart MOS–MC33888

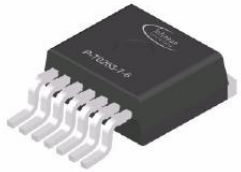
◆ 集成了8路低边和4路高边驱动/ 4 HS+8LS

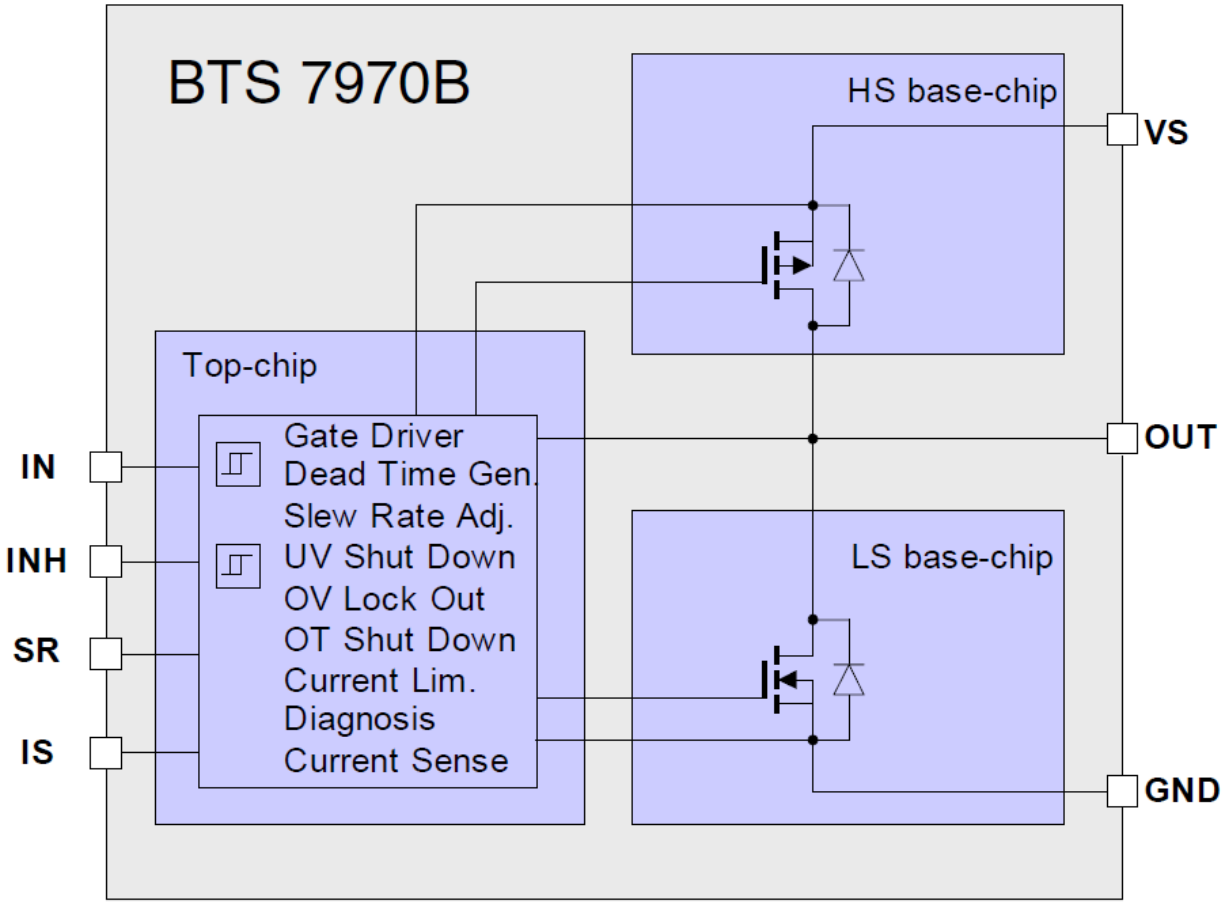




●集成驱动芯片8/ Integrated Smart MOS– BTS7970

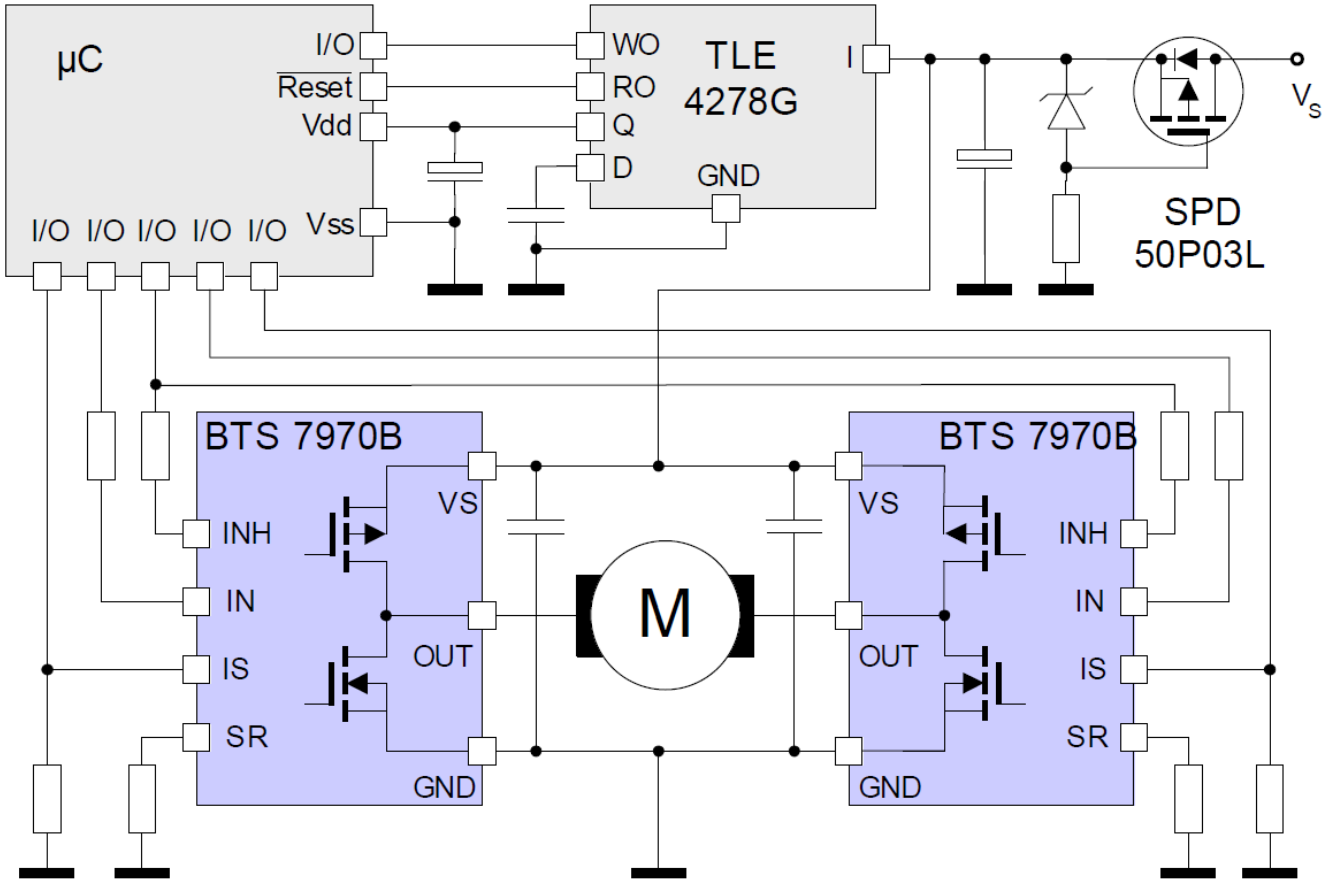
◆ 半桥驱动器/ Half bridge driver





●集成驱动芯片8/ Integrated Smart MOS– BTS7970

◆ 两片BTS7970组成的全桥驱动器/Full bridge driver consists of 2 BTS7970





§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

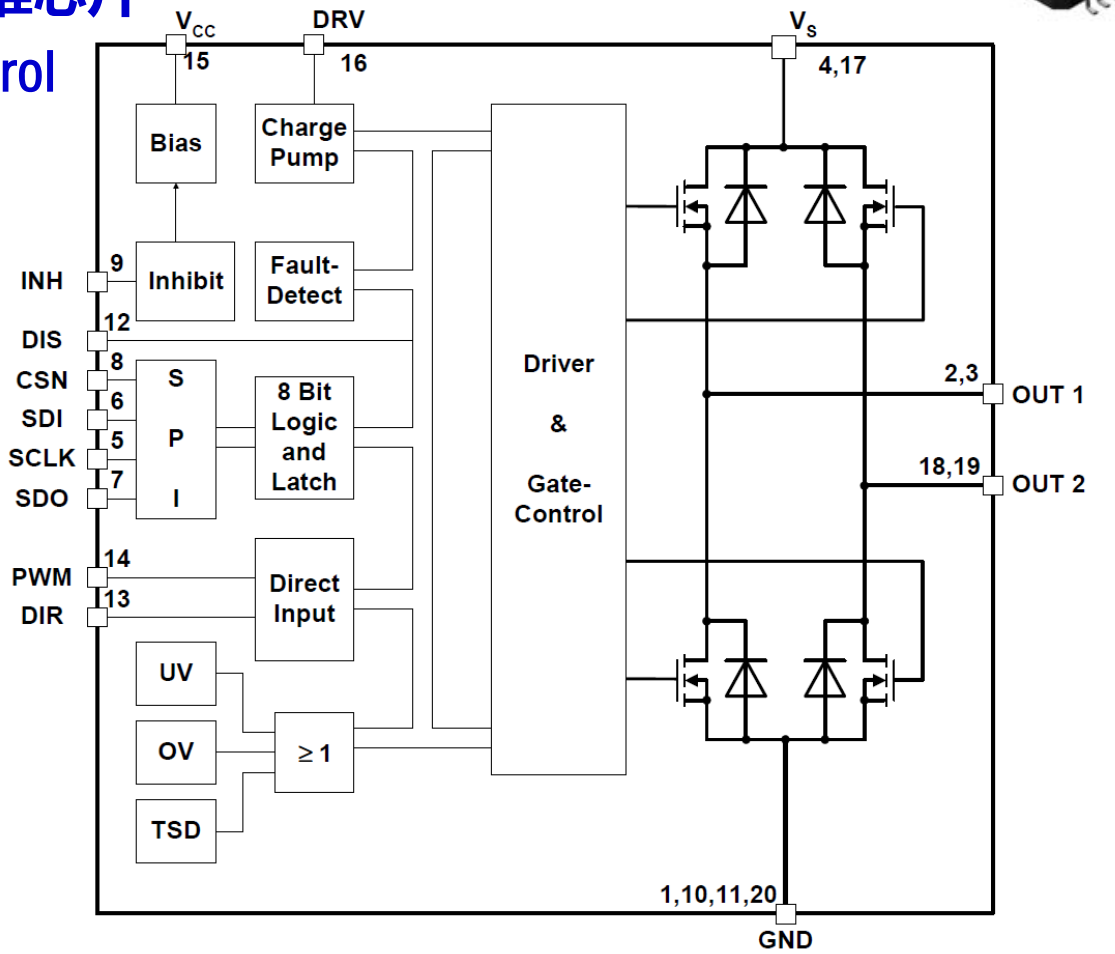
●集成驱动芯片9/ Integrated Smart MOS– TLE6209



◆全桥驱动器/Full bridge driver TLE6209

◆电子节气门控制标准芯片

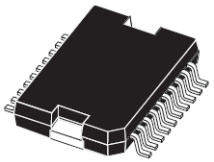
Electronic Throttle Control

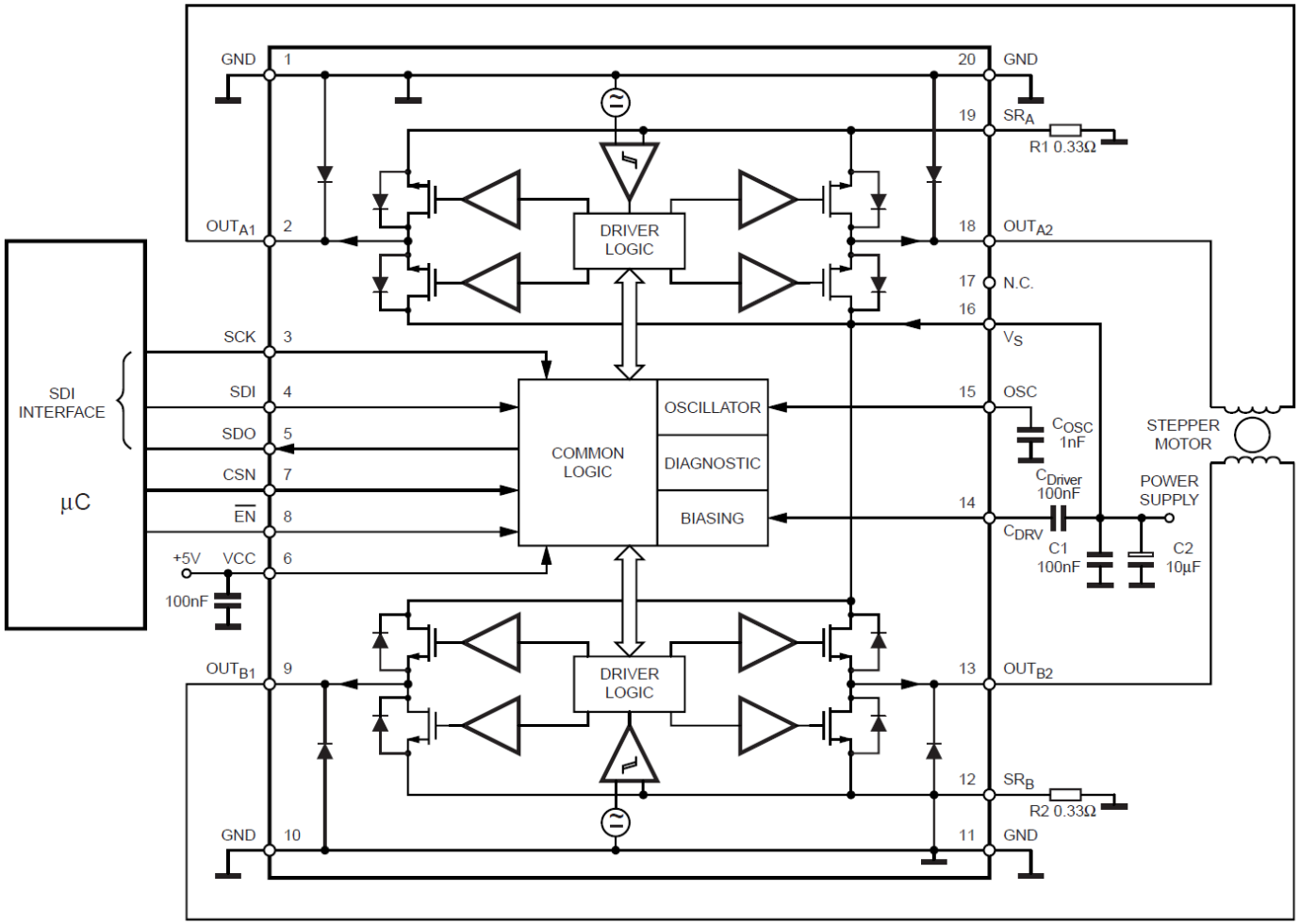


§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

●集成驱动芯片10/ Integrated Smart MOS– L9935

◆两相全桥驱动器/Two phase full bridge driver





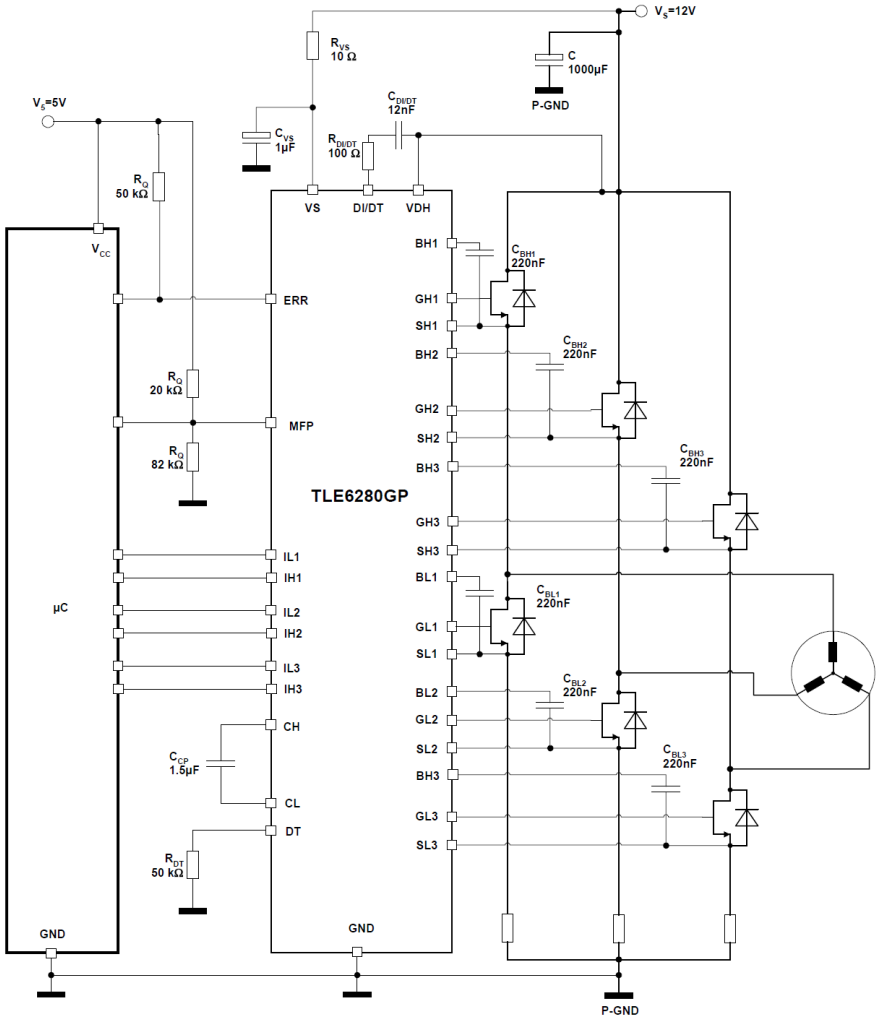
●集成驱动芯片11/ Integrated Smart MOS– TLE6280

◆三相全桥预驱动器/Three phase full bridge pre–driver



该芯片只是预驱动器，MOS管并没有集成在芯片中，设计电路需要单独的MOS管

/The chip is only a pre–drive, so the MOSFETs are not included in the chip! It is necessary to place the alone MOSFETs in circuit design.



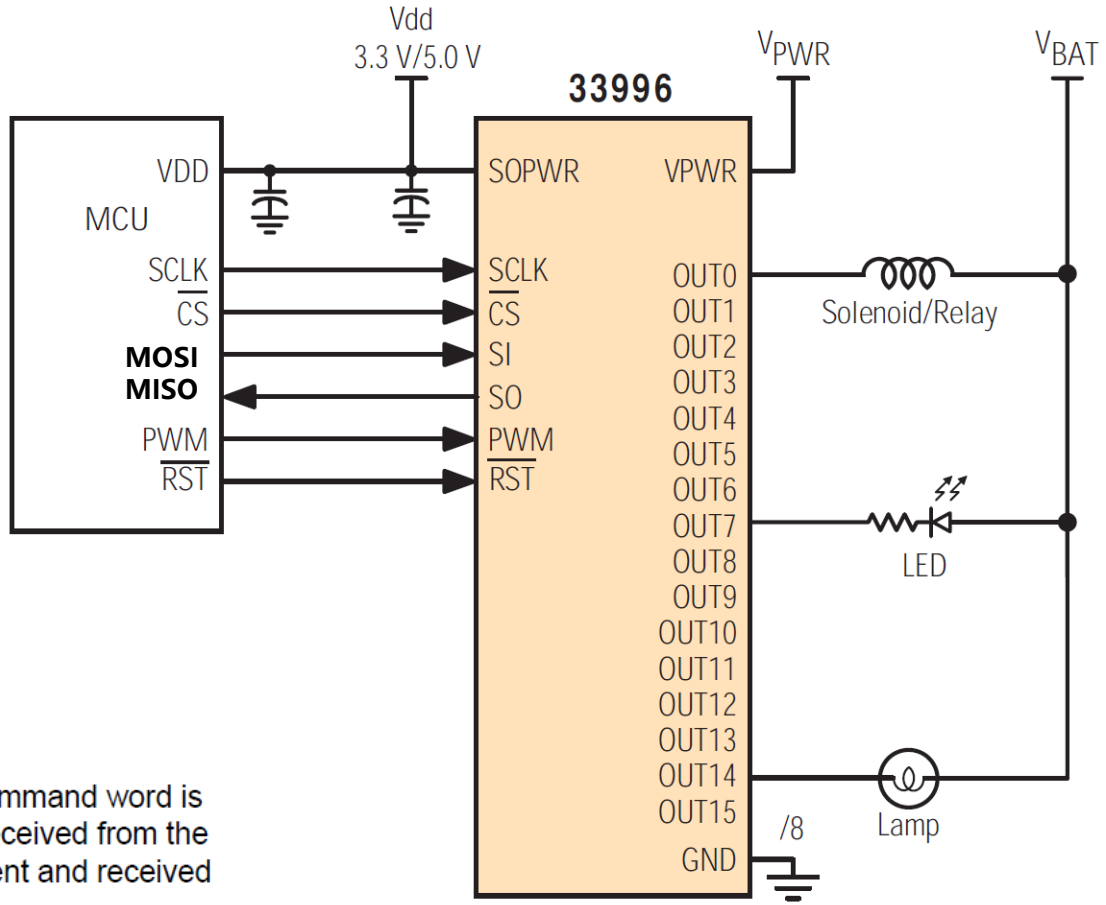
●集成驱动芯片调试举例/ Example of Integrated Drive Chip

SPI一共4路信号：

- CS片选
- MISO为主入从出
- MOSI为主出从入
- SCLK为时钟信号

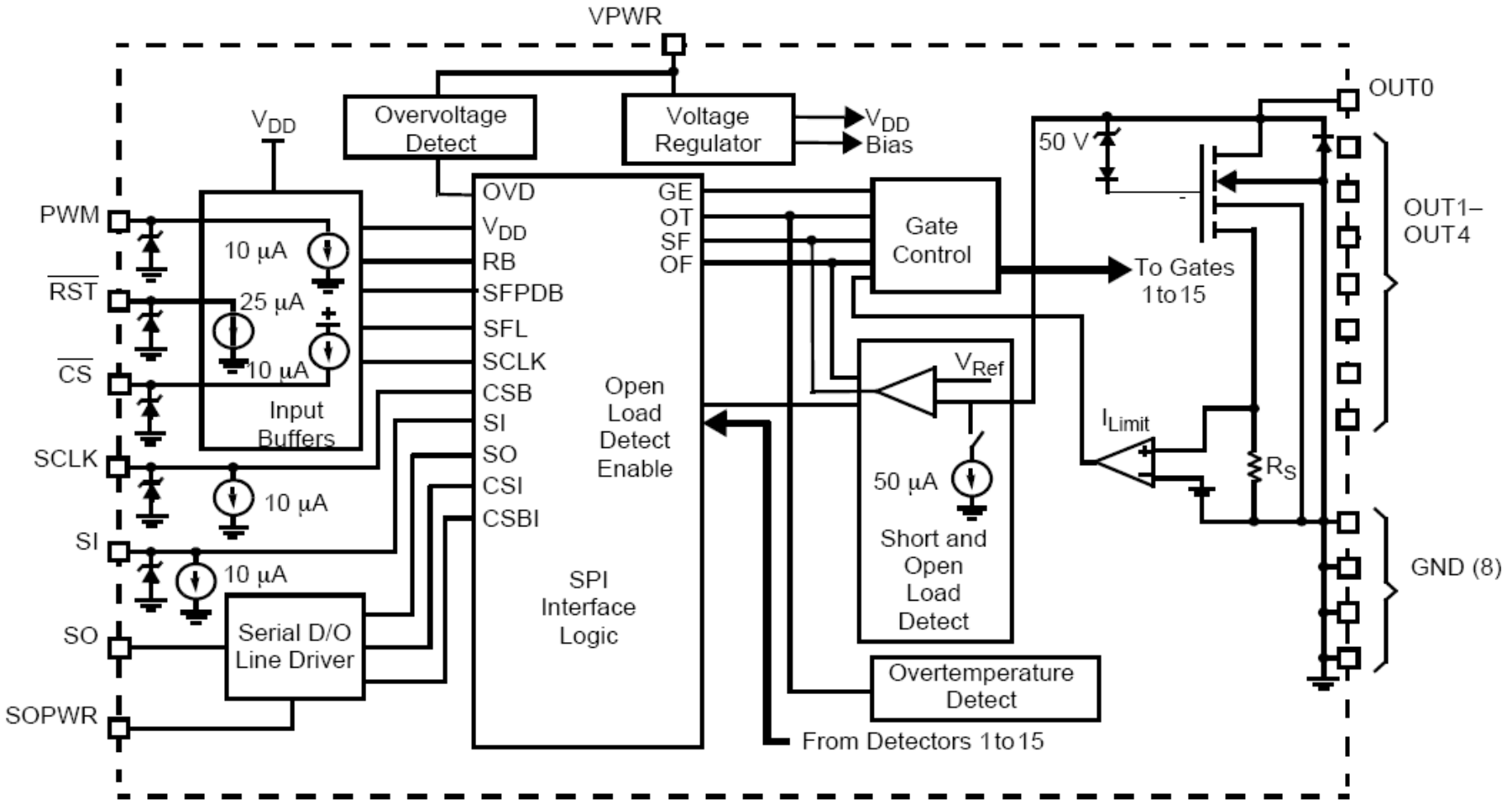
RST为复位信号

PWM信号

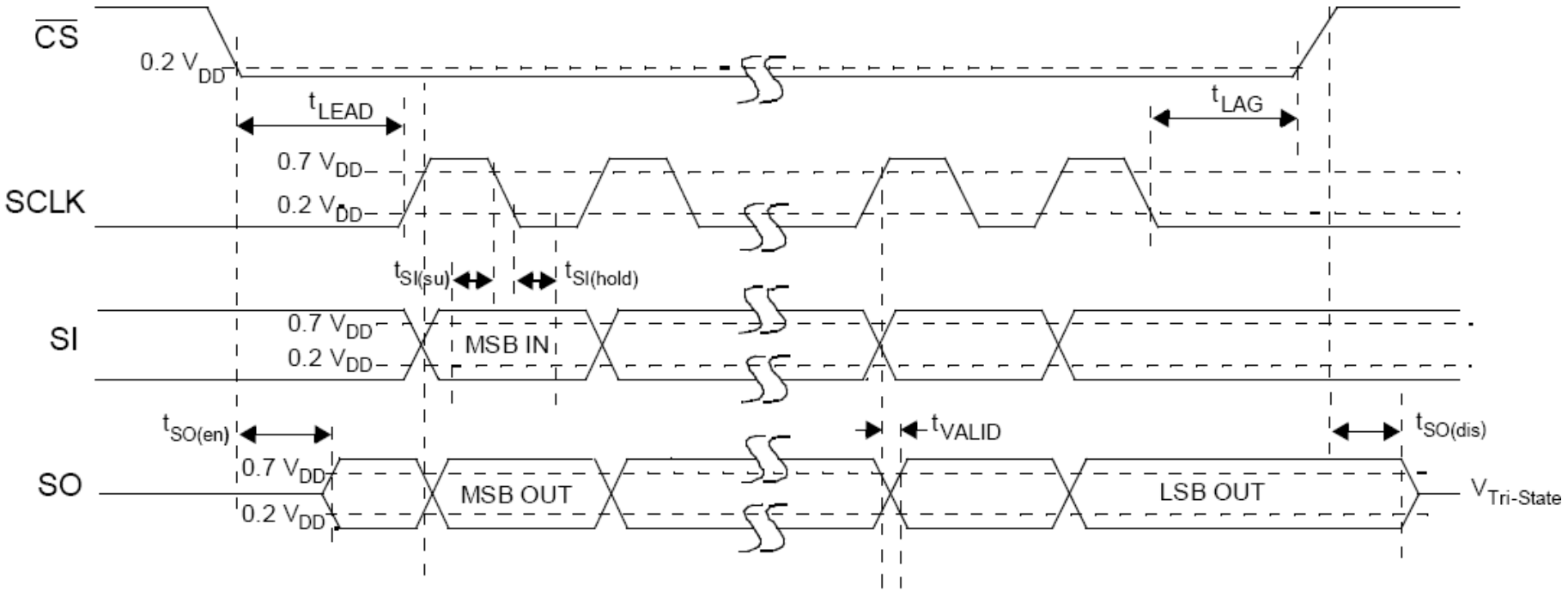


On each SPI communication, a 24 bit command word is sent to the 33996 and 24 bit fault word is received from the 33996. The Most Significant Bit (MSB) is sent and received first.

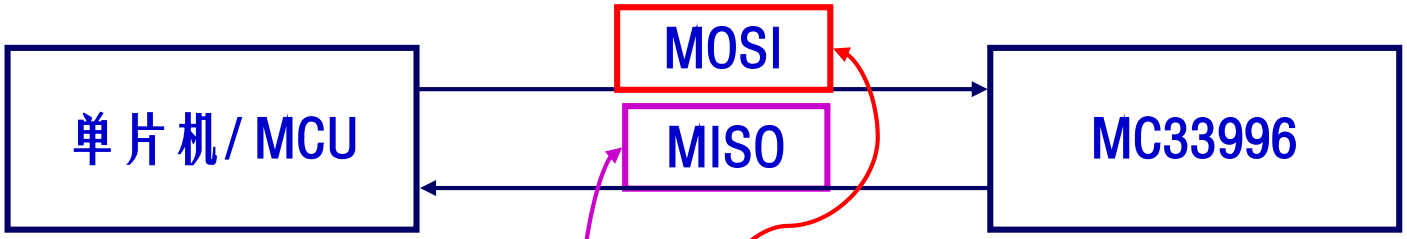
- 驱动逻辑/Logic circuit of MC33996



- SPI接口定义/ SPI interface definition of MC33996
  - CS=0表示传输有效
  - SLCK的下降沿锁存数据（有的芯片上升沿锁存）



● SPI数据定义/ SPI data format of MC33996

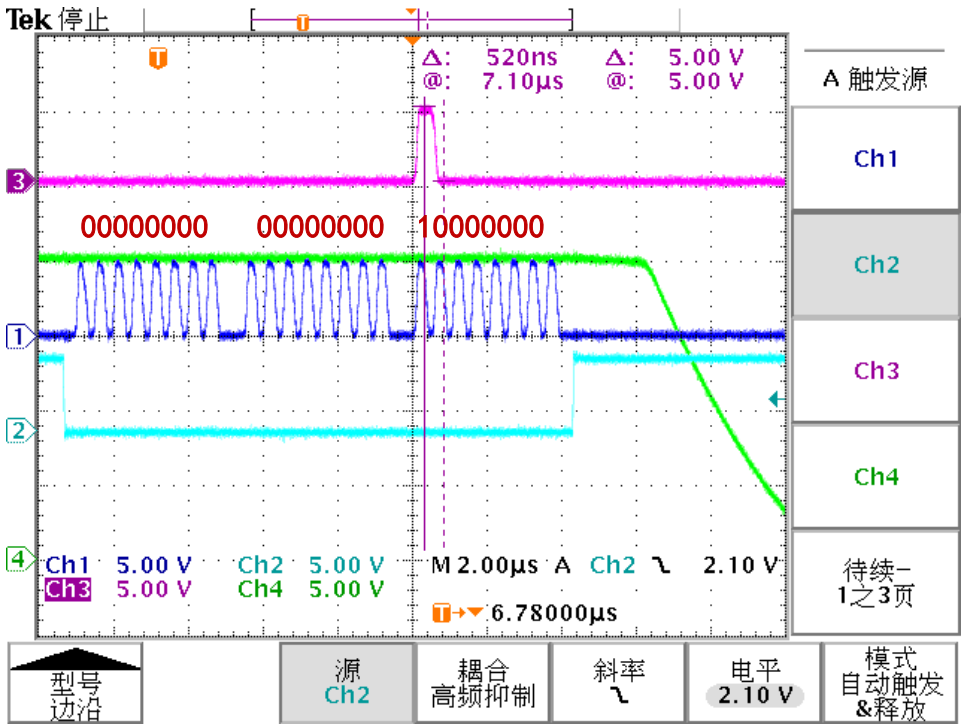


MSB Bits LSB

Commands	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ON/OFF Control 0=off, 1=on	0	0	0	0	0	0	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Open Load Current Enable 0=disable, 1=enable	0	0	0	0	0	1	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Global Shutdown/Retry Control 0=shutdown, 1=retry	0	0	0	0	1	0	Thermal Bit 0	Over-voltage 0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SFPD Control 1=therm only, 0=V_{DS}	0	0	0	0	1	1	X	X	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1
PWM Enable 0=SPI only, 1=PWM	0	0	0	1	0	0	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
AND/OR Control 0=PWM terminal AND with SPI 1=PWM terminal OR with SPI	0	0	0	1	0	1	X	X	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Reset	0	0	0	1	1	0	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
SO Response 0=No Fault, 1=Fault	Any Fault	over-volt-age	0	0	0	0	0	0	OUT 15	OUT 14	OUT 13	OUT 12	OUT 11	OUT 10	OUT 9	OUT 8	OUT 7	OUT 6	OUT 5	OUT 4	OUT 3	OUT 2	OUT 1	OUT 0

● 实际调试信号/ Tested Signals of MC33996

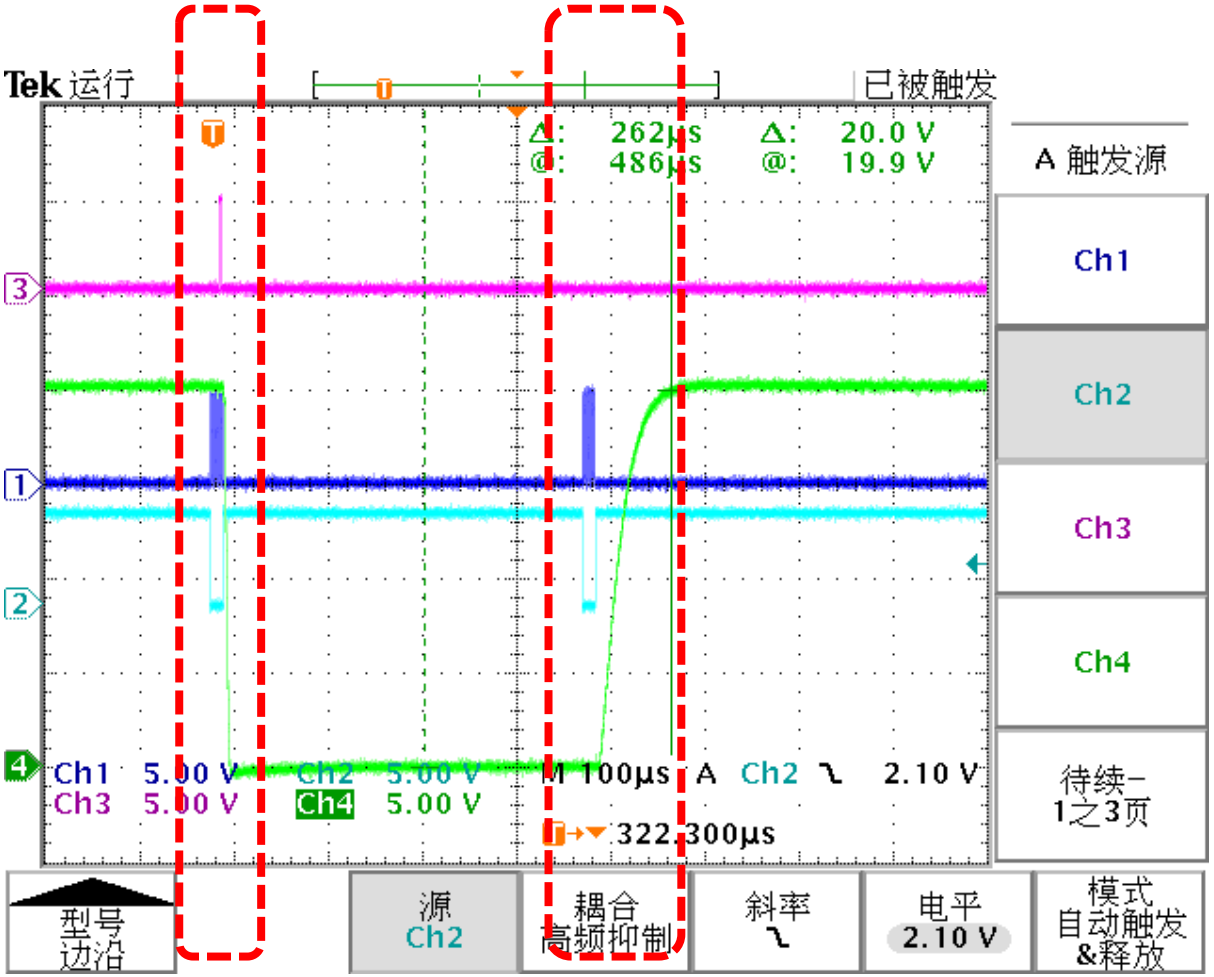
MOSI	SCK	PCS2	V_{LSOUT1}



The SI / SO shifting of data follows a first-in, first-out protocol, with both input and output words transferring the Most Significant Bit (MSB) first.

MOSI: 00000000 00000000 10000000

- 实际调试信号/ Tested Signals of MC33996



MOSI

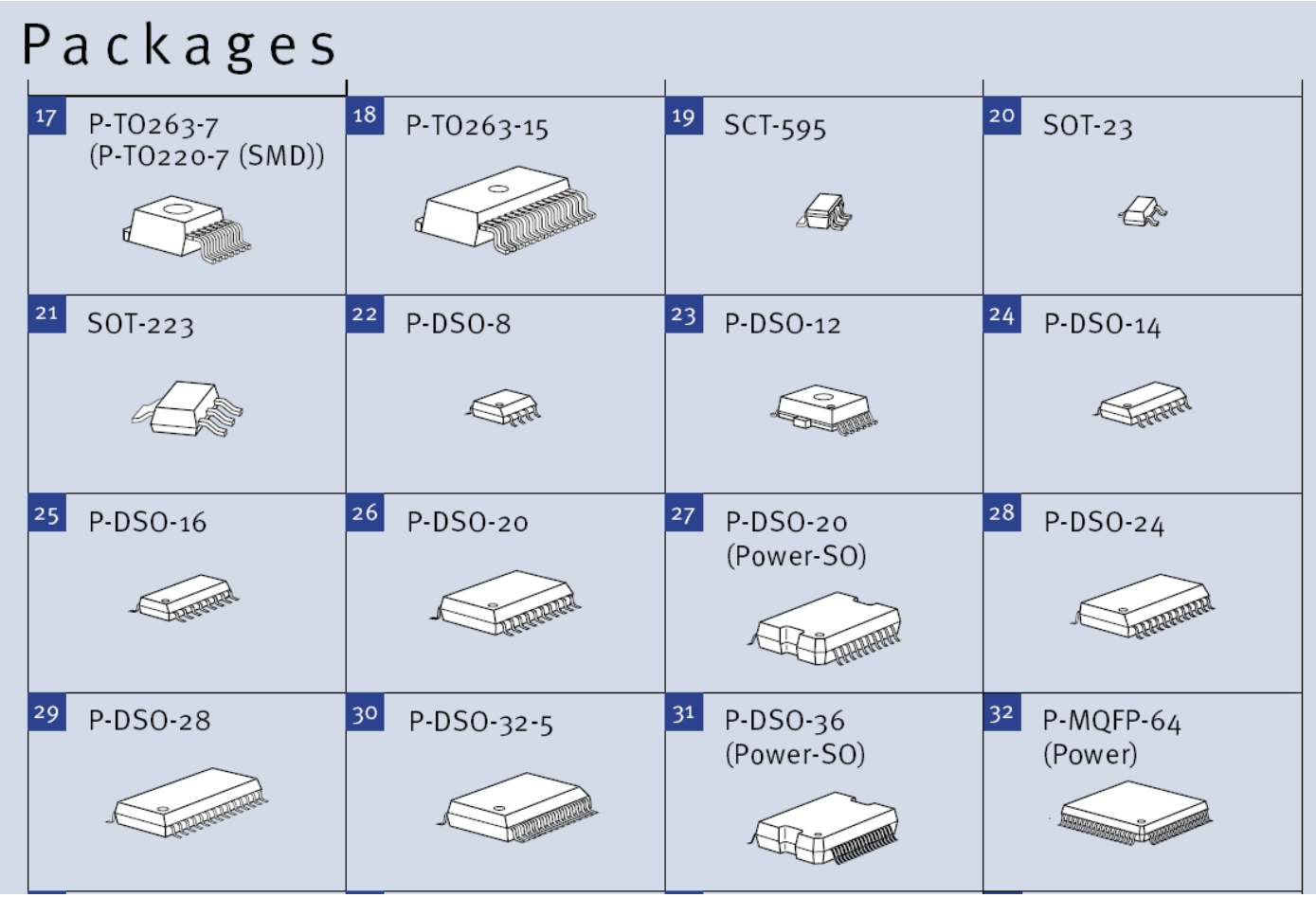
SCK

PCS2

V_{LSOUT1}

§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

- 更多的智能驱动器件/ More smart drive MOS
- Infineon
- Freescale
- ST
- BOSCH
- ……



§5 汽车专用功率驱动集成芯片
Integrated Power Chip for Automotive

- 并不是所有的地方都可以用集成MOS管/ The smart MOS cannot be suitable in some applications
- 当电压和电流较大时，集成驱动往往无法满足要求/ The smart MOS cannot fulfill requirement when the control voltage or current is too high.
- 如何MOS管选择

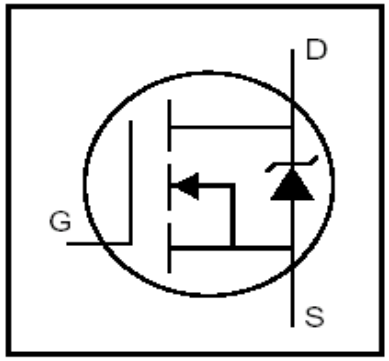
/How to select a MOS?

- 耐压值/ VDSS
- 最大电流/Rated current
- 内阻的大小/ Ron
- 开关速度/Switching speed
- 封装/散热要求

/ Package and dissipation

- 成本/Cost

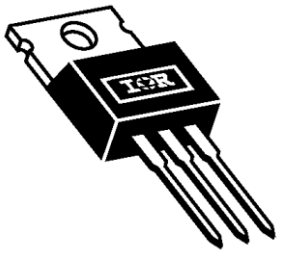
IRF3415S/L

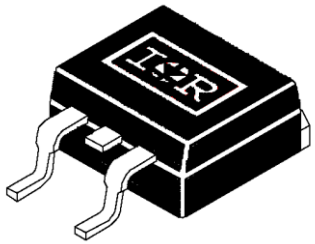


V_{DSS} = 150V

$R_{DS(on)}$ = 0.042Ω

I_D = 43A


TO-220

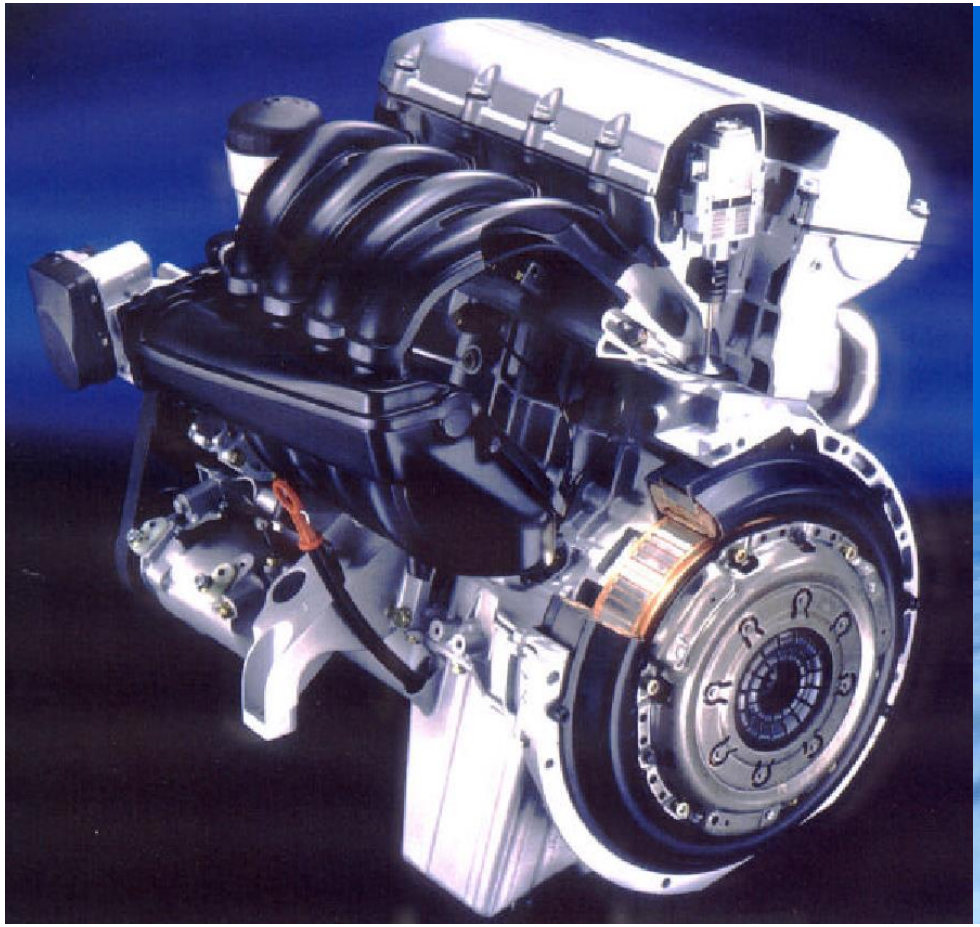

D^2Pak

- 选择参数列表/ Selection parameter table of MOSFETs

55 V

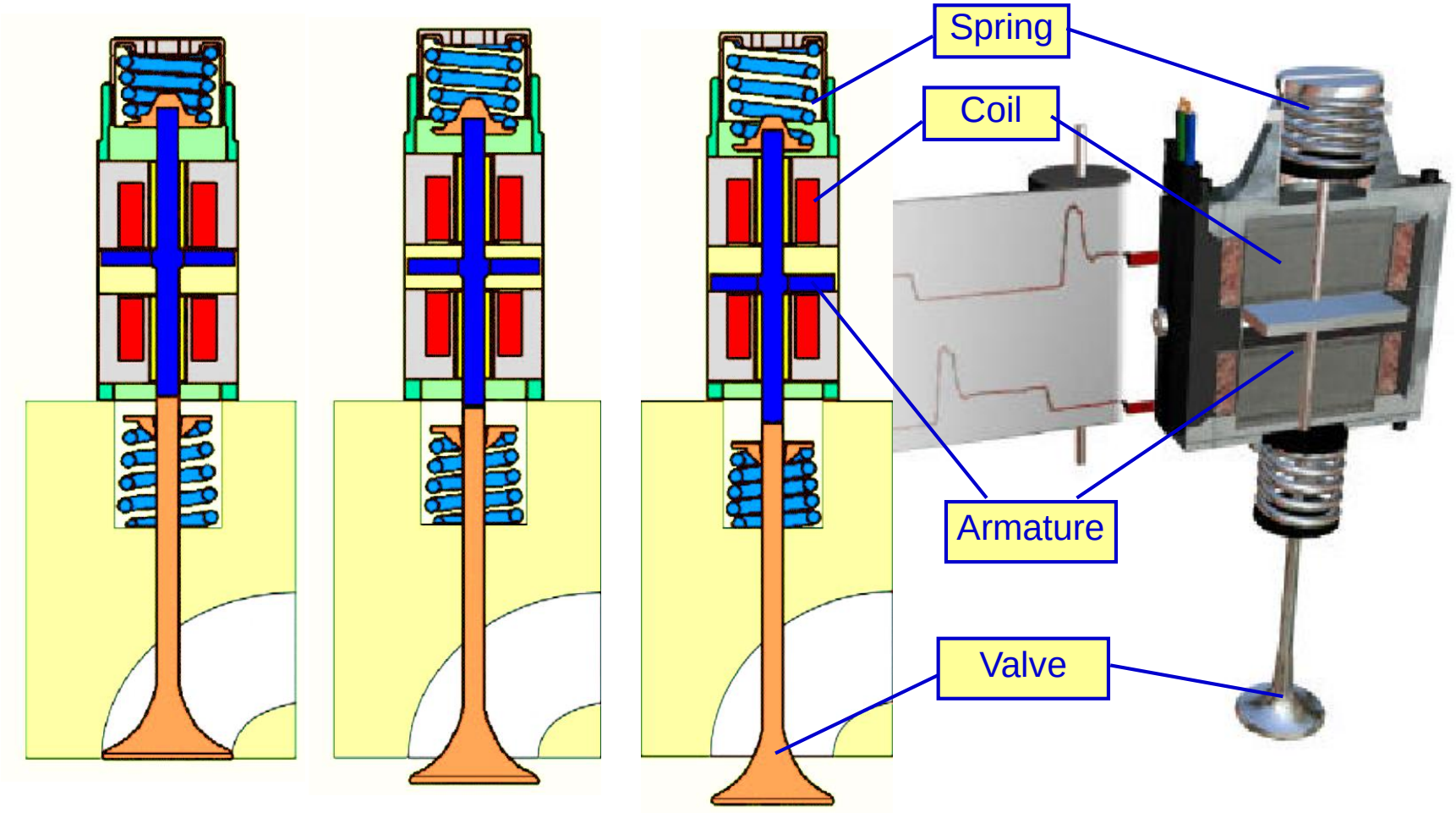
OptiMOS-T 55 V	$R_{DS(on)\ max.}$ [mΩ]	I_D [A]	$R_{thJC\ max.}$ [K/W]	Package*
IPD 90N06S3L-05	5.2	90	1.1	⑩
IPD 90N06S3-06	6.1	90	1.1	⑩
IPD 50N06S3L-06	6	50	1.1	⑩
IPD 50N06S3-07	6.9	50	1.1	⑩
IPD 90N06S3L-07	7.2	90	1.4	⑩
IPD 80N06S3-09	8.5	80	1.4	⑩
IPD 50N06S3L-08	7.8	50	1.4	⑩
IPD 50N06S3-09	9.1	50	1.4	⑩
IPD 50N06S3L-13	12.5	50	2	⑩
IPD 50N06S3-15	15	50	2	⑩
IPD 30N06S3L-20	20	30	2.8	⑩
IPD 30N06S3-23	23.4	30	2.8	⑩

- 驱动电路设计举例/ Examples of Driver circuit design
  - 电磁气门驱动/ Electric–Magnetic Valve Control



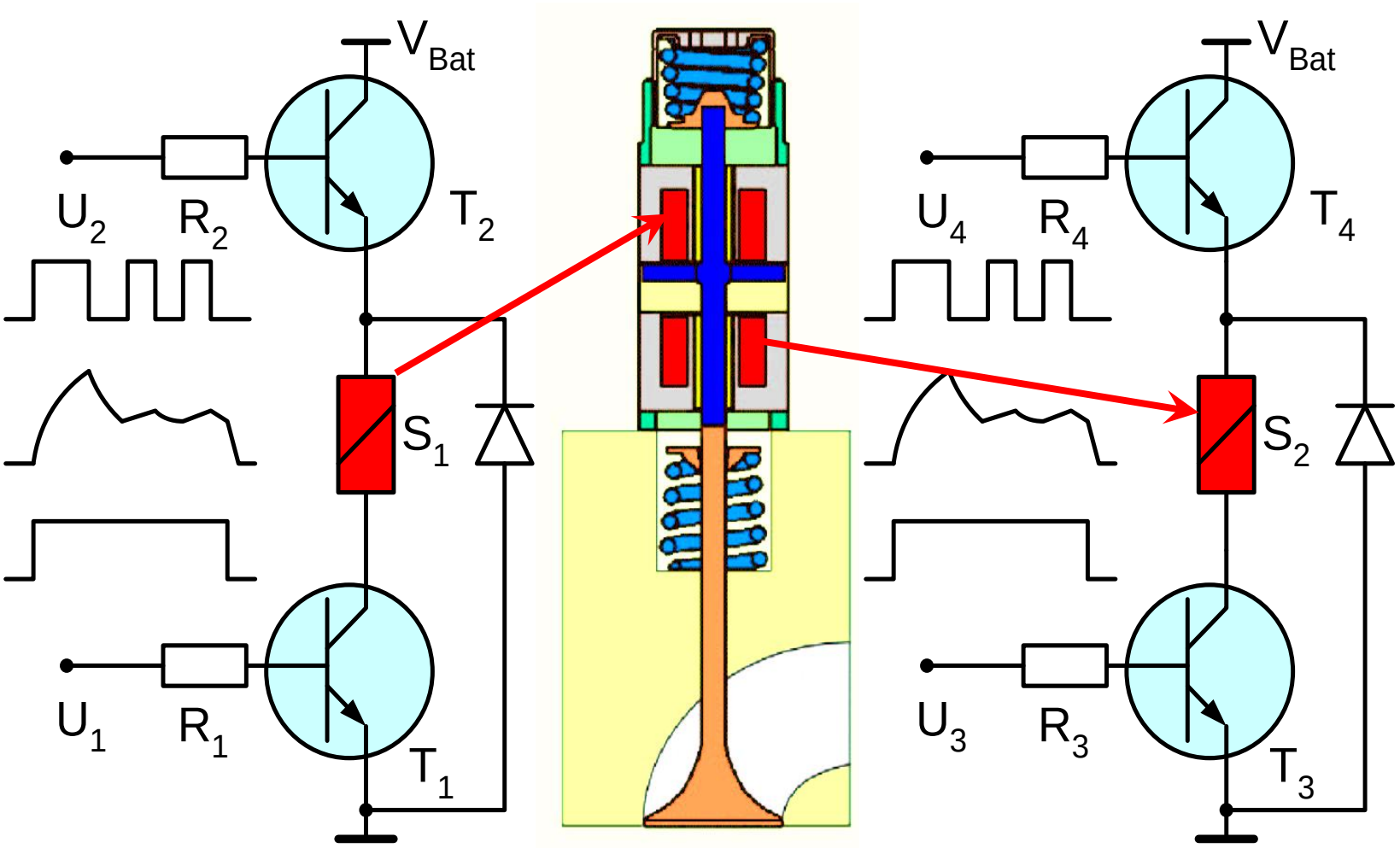


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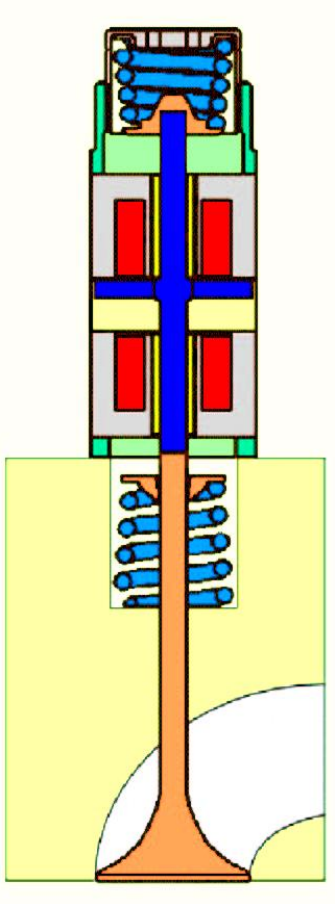
● 驱动电路设计举例/ Examples of Driver circuit design

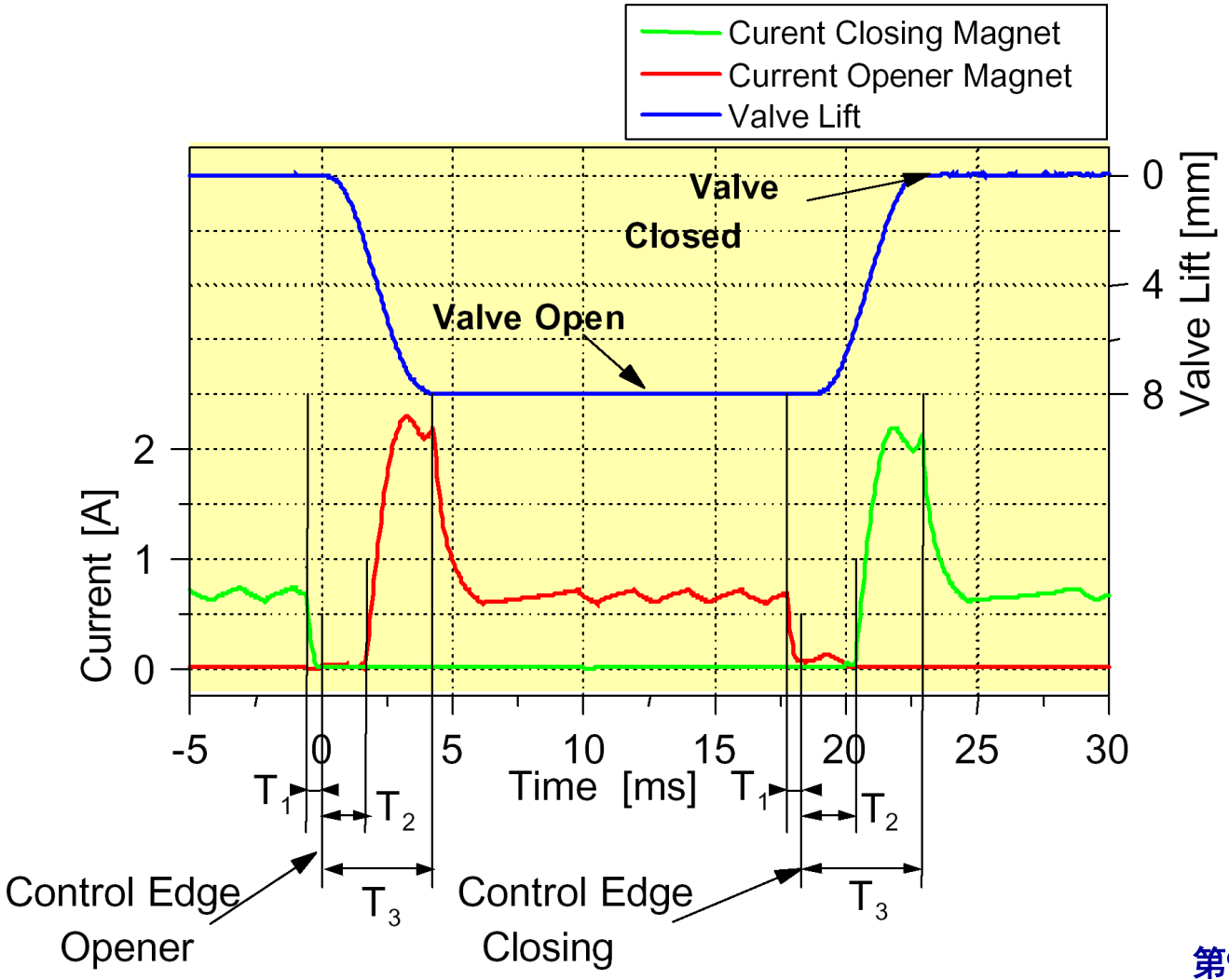
◆ 电磁气门驱动/ Electric–Magnetic Valve Control



● 驱动电路设计举例/ Examples of Driver circuit design

◆ 电磁气门驱动/ Electric-Magnetic Valve Control





小结
/Summary

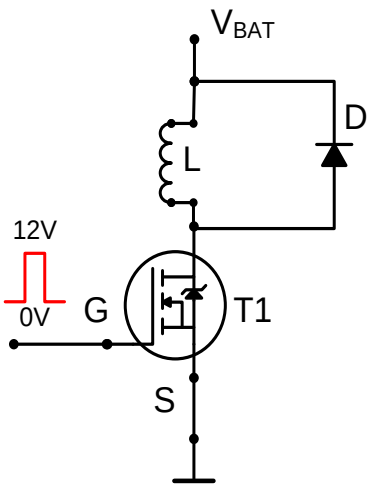
- 本节我们学习的主要知识：
  - ◆驱动电路的执行器类型和功率开关器件的类型；
  - ◆四种主要驱动电路拓扑结构
  - ◆四种执行器续流电路
  - ◆驱动电路和执行器的故障诊断和容错控制的方法
- 本节给予我们的启示：
  - （1）水涨船高—高边驱动中的栅极自举电容；
  - （2）续流是为了快速断流—续流回路分析；
  - （3）安全备份—输出通道冗余和容错；
  - （4）前馈和反馈—输出和输入相互配合；

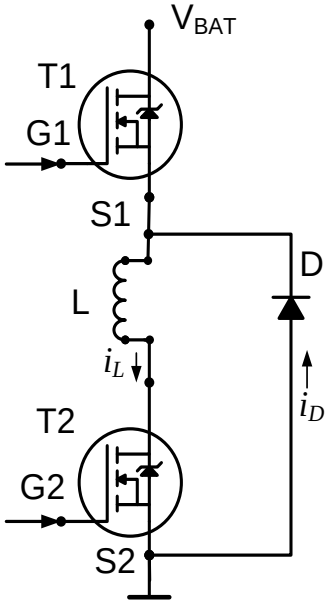
小结
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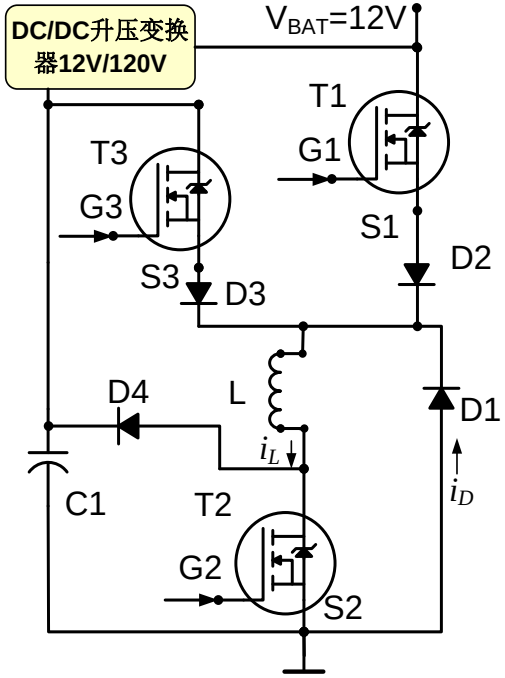
- 驱动电路四种拓扑结构：低边、高边、半桥和全桥/ Four topologies of drive circuit: low side, high side, half bridge and full bridge.
- 掌握MOSFET的导通和关断的稳/动态过程分析/ Understand and master the analysis of turn on and turn off process of MOSFETs.
- 理解高边驱动为什么需要浮动栅极驱动，以及自举电容“水涨船高”的原理/ Understand why does the high side drive need the float gates pre–drivers, and the principle of boost capacitor just like a boat rises up together with the water in a river.
- 学会利用MOS管、继电器等功率器件搭建四种拓扑结构的驱动电路/Familiar to construct a drive circuit by using MOSFET or relay.

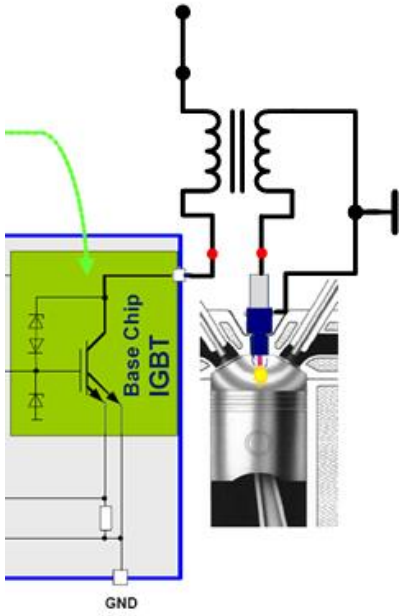
小结
/Summary

- 四种续流回路的工作特点/ Features of four freewheel circuit
- 估计执行器初始释能功率/Estimation of initial energy release rate of actuators
- 高压驱动：加快执行器充能和释能的速度/ High voltage drive will both faster energy storage and release.









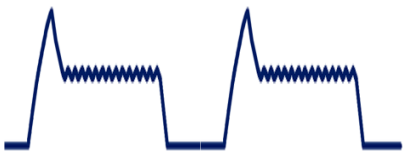
小结
/Summary

- 四种续流电路给我们的启示/ Inspiration of the freewheel circuits
  - ① 能量是守恒的，能量只会转移；“哪里有压迫，哪里就有反抗”！压制越高，释能功率越大！不要压制能量，要引导能量，引导好了可以做正功！
  - ② 遵守自然波动法则：电流和能量波动，涨落交替，春秋轮回；
  - ③ 处理人生态度：充放结合→一张一弛，才能长久工作，才能不断地产生火花，才能动力源源不断！
- 给自己和别人施压的同时，是否配置了释能泄压回路/Do you configure the freewheel circuit when you make the stress to yourself or others?

$P_{\mathrm{L}} = I_{\mathrm{Lmax}} \times 0.7\mathrm{V};$



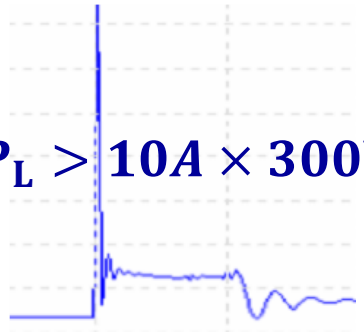
$P_{\mathrm{L}} \approx I_{\mathrm{L}} \times 70.7\mathrm{V};$



$P_{\mathrm{L}} \approx I_{\mathrm{L}} \times 121.4\mathrm{V};$



$P_{\mathrm{L}} > 10A \times 300\mathrm{V}$

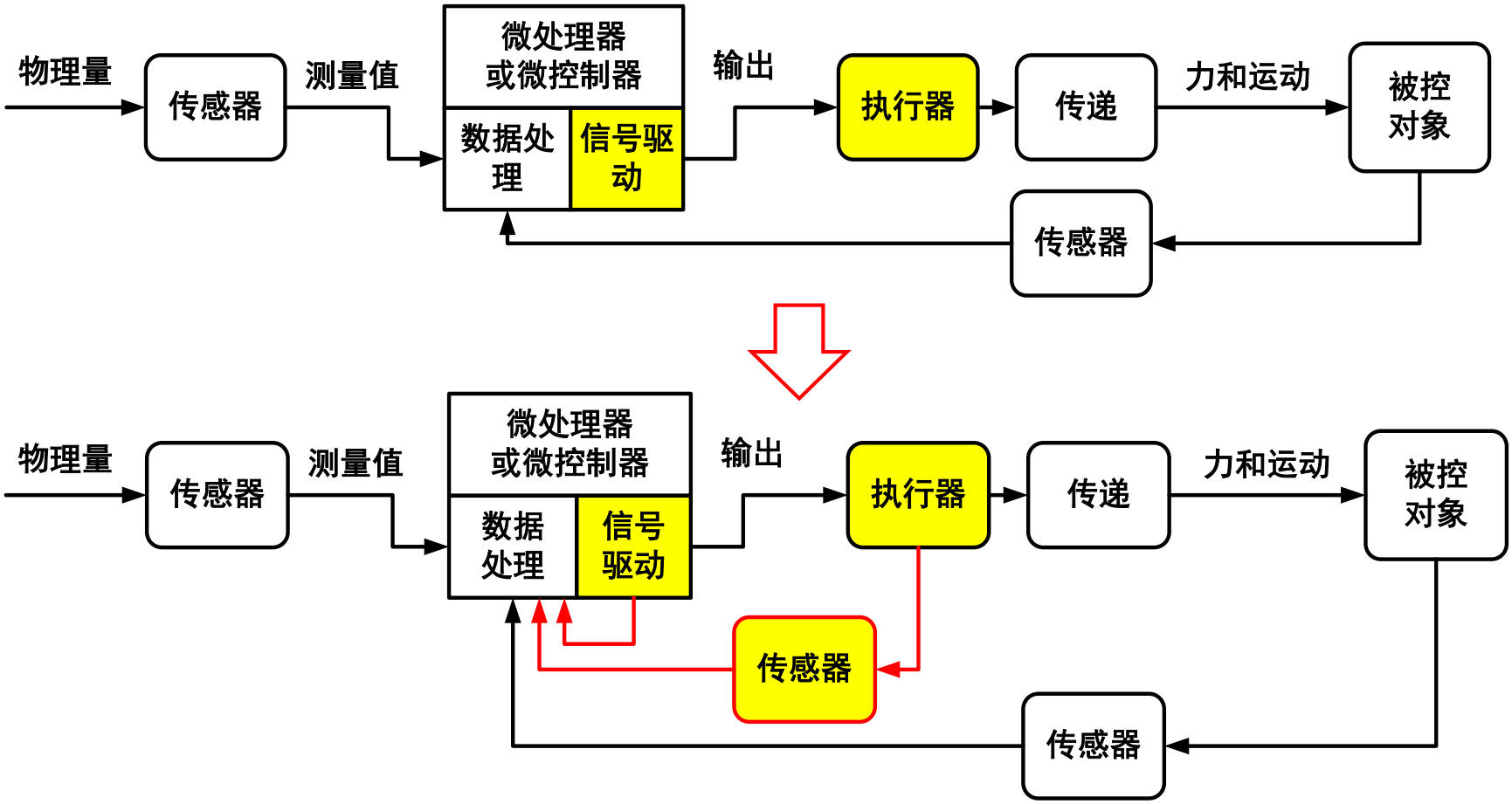


小结
/Summary

- 故障诊断知识小结/ Summary of knowledge:
  - ①电压和电流相结合的故障诊断方法/Diagnosis by using voltage and current.
  - ②通道冗余提高可靠性和功能安全性/Improve Reliability and function safety
  - ③驱动专用芯片采用并串结合实现控制、诊断与容错的集成/Drive IC integration of control, diagnosis and fault tolerant by parallel and serial.
- 故障诊断方法小结：
  - ① 重要的系统要设计好冗余备份，重要的事情要多考虑几种预案（容错控制）/ Back–up or redundancy for important system, multi–solution (fault tolerant) for important things.
  - ② 做事情要串联（省资源）和并联（效率高）相结合/Combination of Parallel(high efficiency) and Serial (low cost) when dealing problems.
  - ③ 正确处理前馈（输出）和反馈（输入）之间的关系/Right relationship between feed forward (output) and feedback (input).

小结
/Summary

- 输出驱动也有信号输入，前馈环节也有反馈/ Output drive might combine input signal, the feed forward channel sometime also needs feedback channel.

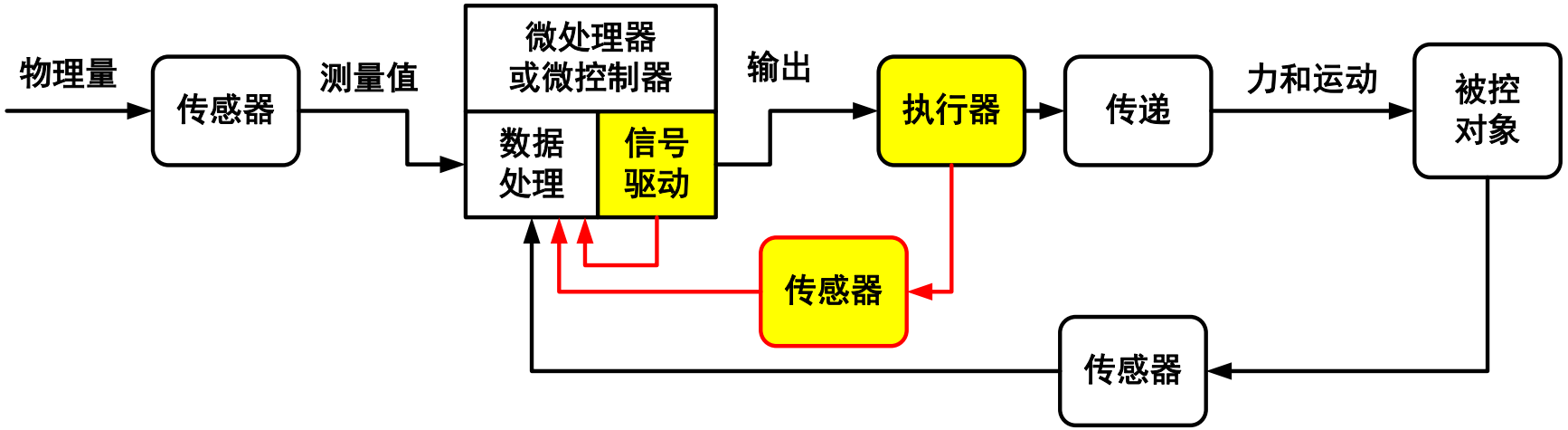


小结
/Summary

启示：在做事情的（前馈输出）过程中，一定要不断反省和获得他人的建议（反馈输入），才能把事情做得精准又可靠！

Inspirations: When you are doing something (feed forward channel), you need to make it more precise and reliable by continuously self-check and get others suggestions (feedback).

在你的成长道路上，是否接纳了别人的建议和反馈？是否开展了批评和自我批评？你配置了自己成长的反馈和诊断的通道了吗？ Do you build up the feedback channel for yourself?





本章结束，谢谢！

Thank you！